Aalto University

Spring 2020

Linear Algebra

Antti Hannukainen
Nuutti Hyvönen
Jarmo Malinen

November 3, 2025

Preface

These lecture notes cover basics of linear algebra from the standpoint of matrix computations: general vector spaces and linear maps are not in the focus of attention, but the primary objects of interest are matrices and (column) vectors that can directly be given as inputs to a computer. The students are expected to have basic knowledge about matrices; for example, the material presented during any of the courses MS-A00XX Matrix Algebra is sufficient. Many of the required fundamental tools are also reviewed in these notes.

The three Chapters 1, 2 and 3 correspond, respectively, to the three main themes of the lectures:

1. Solvability and stability of linear systems $A\boldsymbol{x} = \boldsymbol{b}$.

2. Eigenvalue problems $A\boldsymbol{v} = \lambda\boldsymbol{v}$ and their fundamental applications.

3. Least squares problems, their geometric interpretation and related matrix decompositions.

For simplicity and notational convenience, Chapters 1 and 3 consider real vectors and matrices, even though the extension to the complex case would be straightforward. However, Chapter 2 touches upon the complex extension that cannot be avoided in the treatment of eigenvalue problems.

Matrices and subspaces are denoted by capital letters (e.g., A or E), vectors by bolded lower case letters (e.g., $\boldsymbol{x}$ or $\boldsymbol{b}$) and scalars by standard lowercase letters (e.g., α or a_{ij}). In particular, the components of a vector $\boldsymbol{x} \in \mathbb{R}^n$ are x_j, $j = 1, \ldots, n$. However, the zero scalar and the zero vector of $\mathbb{R}^n$ are both denoted simply as 0. The Cartesian basis vectors for $\mathbb{R}^n$ are $\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_n$ and the identity matrix is denoted as

$$I := \big[\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_n\big] \in \mathbb{R}^{n\times n}.$$

We will frequently use such a notation to write matrices in terms of their column vectors. The positive integers n and m are reserved for spatial dimensions and $i, j, k, l \in \mathbb{N}_0$ are used as generic indices.

Lecture 1.

Chapter 1

Linear systems

Let $A \in \mathbb{R}^{m \times n}$ and $\boldsymbol{b} \in \mathbb{R}^m$. We focus on the following problem: Find $\boldsymbol{x} \in \mathbb{R}^n$ such that

$$A\boldsymbol{x} = \boldsymbol{b}. \tag{1.1}$$

Consider the following:

1. Is the problem (1.1) solvable? Is the solution unique?

2. How much does the solution $\boldsymbol{x}$ change if A or $\boldsymbol{b}$ is perturbed slightly?

3. How accurately can (1.1) be solved using a computer?

Answers to these questions are the subject of these lectures.

Definition 1.0.1. *The* **linear span** *of the vectors* $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k \in \mathbb{R}^n$ *is the set*

$$\operatorname{span}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k) := \left\{ \boldsymbol{z} \in \mathbb{R}^n \;\middle|\; \boldsymbol{z} = \sum_{j=1}^{k} \alpha_j \boldsymbol{q}_j \text{ for some } \boldsymbol{\alpha} \in \mathbb{R}^k \right\}.$$

Moreover, $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k \in \mathbb{R}^n$ *are called* **linearly independent** *if*

$$\sum_{j=1}^{k} \alpha_j \boldsymbol{q}_j = 0 \in \mathbb{R}^n$$

is equivalent to $\boldsymbol{\alpha} = 0 \in \mathbb{R}^k$.

If one stacks the vectors $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k \in \mathbb{R}^n$ as the columns of a matrix (as we will do frequently),

$$Q = \big[\boldsymbol{q}_1, \boldsymbol{q}_2, \ldots, \boldsymbol{q}_k\big] \in \mathbb{R}^{n \times k}, \tag{1.2}$$

then obviously

$$\operatorname{span}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k) = \big\{ \boldsymbol{z} \in \mathbb{R}^n \;\big|\; \boldsymbol{z} = Q\boldsymbol{\alpha} \text{ for some } \boldsymbol{\alpha} \in \mathbb{R}^k \big\}. \tag{1.3}$$

Moreover, $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k$ are linearly independent if and only if

$$Q\boldsymbol{\alpha} = 0 \quad \Longleftrightarrow \quad \boldsymbol{\alpha} = 0, \tag{1.4}$$

i.e., $Q\boldsymbol{\alpha} \in \mathbb{R}^n$ vanishes if and only if $\boldsymbol{\alpha} \in \mathbb{R}^k$ does.

1.1 Solvability of linear systems

The main aim of this section is to study the existence and uniqueness of a solution to (1.1). These properties can be characterised in terms of the **nullspace** and the **range** of the matrix A. In the following, we will denote the columns of A by $\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_n \in \mathbb{R}^m$ (cf. (1.2)).

Definition 1.1.1. *Let $A \in \mathbb{R}^{m\times n}$. The* **range** *of A is the set*

$$\begin{aligned} \mathrm{R}(A) &= \left\{ \boldsymbol{y} \in \mathbb{R}^m \mid \boldsymbol{y} = A\boldsymbol{x} \ \text{ for some } \boldsymbol{x} \in \mathbb{R}^n \right\} \\ &= \mathrm{span}(\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n) \subset \mathbb{R}^m \end{aligned} \tag{1.5}$$

and the **nullspace** *of A is the set*

$$\mathrm{N}(A) = \{ \boldsymbol{x} \in \mathbb{R}^n \mid A\boldsymbol{x} = 0 \} \subset \mathbb{R}^n.$$

The sets $\mathrm{R}(A)$ and $\mathrm{N}(A)$ are closed under addition and scalar multiplication. Indeed, if $\boldsymbol{x}, \boldsymbol{y} \in \mathrm{N}(A)$, then

$$A(\alpha\boldsymbol{x} + \beta\boldsymbol{y}) = \alpha A\boldsymbol{x} + \beta A\boldsymbol{y} = 0,$$

meaning that also $\alpha\boldsymbol{x} + \beta\boldsymbol{y} \in \mathrm{N}(A)$ for any $\alpha, \beta \in \mathbb{R}$.

Similarly, if $\boldsymbol{x}, \boldsymbol{y} \in \mathrm{R}(A)$, i.e. $\boldsymbol{x} = A\boldsymbol{z}$ and $\boldsymbol{y} = A\boldsymbol{w}$ for some $\boldsymbol{z}, \boldsymbol{w} \in \mathbb{R}^n$, then

$$A(\alpha\boldsymbol{z} + \beta\boldsymbol{w}) = \alpha A\boldsymbol{z} + \beta A\boldsymbol{w} = \alpha\boldsymbol{x} + \beta\boldsymbol{y},$$

i.e., $\alpha\boldsymbol{x} + \beta\boldsymbol{y} \in \mathrm{R}(A)$ for any $\alpha, \beta \in \mathbb{R}$.

This means that the subsets $\mathrm{R}(A) \subset \mathbb{R}^m$ and $\mathrm{N}(A) \subset \mathbb{R}^n$ are actually **subspaces**.

Definition 1.1.2. *Let $E \subset \mathbb{R}^n$ be such that for any $\boldsymbol{x}, \boldsymbol{y} \in E$ and $\alpha \in \mathbb{R}$*

$$\boldsymbol{x} + \boldsymbol{y} \in E \quad \textit{and} \quad \alpha\boldsymbol{x} \in E.$$

Then E is called a **subspace** *of $\mathbb{R}^n$. (In particular, $\mathbb{R}^n$ itself as well the* **trivial subspace** *$\{0\}$ are subspaces of $\mathbb{R}^n$.)*

The range of the matrix A induces a condition for the existence of a solution to the equation (1.1). By definition, $\boldsymbol{b} \in \mathrm{R}(A)$ if and only if there exists $\boldsymbol{z} \in \mathbb{R}^n$ such that

$$\boldsymbol{b} = A\boldsymbol{z}.$$

To sum up, there exists a solution to the linear system (1.1) if and only if $\boldsymbol{b} \in \mathrm{R}(A) = \mathrm{span}(\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n)$. Recall how matrix-vector products are computed.

Brush up your vector space understanding by checking the eight vector space axioms from, e.g., Wikipedia.

The uniqueness of a solution is studied via the homogeneous problem: Find $\boldsymbol{y} \in \mathbb{R}^n$ such that

$$A\boldsymbol{y} = 0 \quad \Longleftrightarrow \quad \boldsymbol{y} \in \mathrm{N}(A).$$

If the homogeneous equation has only the trivial solution $\boldsymbol{y} = 0$, a solution to the linear system (1.1) is unique (if one exists).

Lemma 1.1.1. *Assume that* (1.1) *has a solution. The solution is unique if and only if* $\mathrm{N}(A) = \{0\}$.

Proof. Let $\boldsymbol{x} \in \mathbb{R}^n$ be a solution of (1.1). Assume that $\mathrm{N}(A)$ is nontrivial, i.e., there exists $0 \neq \boldsymbol{z} \in \mathrm{N}(A)$. Then,

$$A(\boldsymbol{x} + \boldsymbol{z}) = A\boldsymbol{x} + A\boldsymbol{z} = A\boldsymbol{x} = \boldsymbol{b},$$

meaning that we have constructed a second, distinct solution $\boldsymbol{x}+\boldsymbol{z} \neq \boldsymbol{x}$.

This proves the 'only if' part of the claim.

In order to prove the 'if part', assume that $\mathrm{N}(A) = \{0\}$ and let both $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathbb{R}^n$ be solutions to (1.1), i.e.,

$$A\boldsymbol{x}_1 = \boldsymbol{b} \quad \text{and} \quad A\boldsymbol{x}_2 = \boldsymbol{b}.$$

Subtracting the two equations and linearity of A give

$$A(\boldsymbol{x}_1 - \boldsymbol{x}_2) = 0.$$

Hence, $\boldsymbol{x}_1 - \boldsymbol{x}_2 \in \mathrm{N}(A)$, which means by assumption that $\boldsymbol{x}_1 - \boldsymbol{x}_2 = 0$, and thus all solutions to (1.1) must be the same. □

To summarise:

The problem

$$A\boldsymbol{x} = \boldsymbol{b}. \tag{1.6}$$

has a unique solution if and only if

$$\boldsymbol{b} \in \mathrm{R}(A) \quad \text{and} \quad \mathrm{N}(A) = \{0\}. \tag{1.7}$$

Moreover, the proof of Lemma 1.1.1 characterises the possible nonuniqueness in solving (1.6):

Lemma 1.1.2. *Let $\boldsymbol{x}_p \in \mathbb{R}^n$ be a solution of (1.6). Then $\boldsymbol{x} \in \mathbb{R}^n$ is a solution of (1.6) if and only if $\boldsymbol{x} = \boldsymbol{x}_p + \boldsymbol{z}$ for some $\boldsymbol{z} \in \mathrm{N}(A)$.*

It turns out that the combined condition (1.7) **can be** true for all $\boldsymbol{b} \in \mathbb{R}^m$ only if $n = m$, i.e., only if A is a square matrix. This claim can be proved by relating the **dimensions** of $\mathrm{N}(A)$ and $\mathrm{R}(A)$ to each other and to n. To this end, we need the concept of **basis**.

Definition 1.1.3. *Let $E \subset \mathbb{R}^n$ be a subspace. A* **basis** *of E is a set of linearly independent vectors $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k\} \subset \mathbb{R}^n$ such that*

$$E = \mathrm{span}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k).$$

With the notation of the above definition, for each $\boldsymbol{x} \in E$ there exists a **coordinate vector** $\boldsymbol{\alpha} \in \mathbb{R}^k$ such that

$$\boldsymbol{x} = \sum_{i=1}^{k} \boldsymbol{q}_i \alpha_i \quad \Longleftrightarrow \quad \boldsymbol{x} = Q\boldsymbol{\alpha}, \tag{1.8}$$

where $Q = [\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k] \in \mathbb{R}^{n \times k}$ is a matrix with the basis vectors as its columns.

As the basis vectors are by definition linearly independent, such a coordinate vector $\boldsymbol{\alpha}$ is unique for any given $\boldsymbol{x}$.

Indeed, if (1.8) holds for two coordinate vectors $\boldsymbol{\alpha}, \tilde{\boldsymbol{\alpha}} \in \mathbb{R}^k$, then subtracting the corresponding representations yields

$$Q(\boldsymbol{\alpha} - \tilde{\boldsymbol{\alpha}}) = 0 \quad \Longrightarrow \quad \boldsymbol{\alpha} - \tilde{\boldsymbol{\alpha}} = 0$$

by virtue of (1.4).

There exists infinitely many different bases for any nontrivial subspace $\{0\} \neq E \subset \mathbb{R}^n$.

However, it can be proven that each basis of a subspace E has the same number of basis vectors (the proof is omitted).

Definition 1.1.4. *Let $E \subset \mathbb{R}^n$ be a subspace and $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k\} \subset \mathbb{R}^n$ its basis. Then the* **dimension** *of E is defined to be*

$$\dim(E) = k = \#\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k\}.$$

For a matrix $A \in \mathbb{R}^{m \times n}$, $\dim \mathrm{R}(A)$ is called the **rank** *of A.*

The dimension of $\mathbb{R}^n$ itself is n since the natural basis vectors $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n$ form its basis. All other subspace of $\mathbb{R}^n$ have dimension lower than n.

In the following example two different bases are introduced for the same subspace.

Example 1.1.1. *Let*

$$A = \begin{bmatrix} 1 & 1 \\ 0 & -1 \\ 2 & 2 \end{bmatrix}.$$

The column vectors of A are clearly linearly independent and so they form a basis for $\mathrm{R}(A) \subset \mathbb{R}^3$; see (1.5). Let

$$\boldsymbol{w}_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \qquad \boldsymbol{w}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

be another pair of vectors in $\mathbb{R}^3$ and denote $W = [\boldsymbol{w}_1, \boldsymbol{w}_2] \in \mathbb{R}^{3\times 2}$.

To study if $\boldsymbol{w}_1$ and $\boldsymbol{w}_2$ also form a basis for $\mathrm{R}(A)$, one has to check two conditions:

- *Are the vectors $\boldsymbol{w}_1$ and $\boldsymbol{w}_2$ linearly independent? (This is obviously true in the considered setting.)*

- *Can any $\boldsymbol{y} \in \mathrm{R}(A)$ be represented as a linear combination of $\boldsymbol{w}_1$ and $\boldsymbol{w}_2$? (This is not quite obvious.)*

The latter question can be answered by first introducing a generic element of $\mathrm{R}(A)$*, i.e.* $\boldsymbol{y} = A\boldsymbol{z}$ *for some* $\boldsymbol{z} \in \mathbb{R}^2$*, and then trying to express it as a linear combination of* $\boldsymbol{w}_1$ *and* $\boldsymbol{w}_2$*. In other words, one must check if for any* $\boldsymbol{z} \in \mathbb{R}^2$*, there exists* $\boldsymbol{x} \in \mathbb{R}^2$ *such that*

$$W\boldsymbol{x} = x_1\boldsymbol{w}_1 + x_2\boldsymbol{w}_2 = A\boldsymbol{z},$$

that is,

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 2 & 0 \end{bmatrix} \boldsymbol{x} = \begin{bmatrix} 1 & 1 \\ 0 & -1 \\ 2 & 2 \end{bmatrix} \boldsymbol{z}.$$

Gaussian elimination gives

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \boldsymbol{x} = \begin{bmatrix} 1 & 1 \\ 0 & -1 \\ 0 & 0 \end{bmatrix} \boldsymbol{z},$$

and thus $x_1 = z_1 + z_2$ *and* $x_2 = -z_2$ *provides the needed solution. In consequence,* $\{\boldsymbol{w}_1, \boldsymbol{w}_2\}$ *is a basis of* $\mathrm{R}(A)$*. (Note that we did not need to separately check whether* $\boldsymbol{w}_1$ *and* $\boldsymbol{w}_2$ *belong to* $\mathrm{R}(A)$ *because we knew to begin with that* $\dim \mathrm{R}(A) = 2$*.)*

We start the actual discussion on the relationship between the dimensions of $\mathrm{R}(A)$ and $\mathrm{N}(A)$ with another example in $\mathbb{R}^3$.

Example 1.1.2. *Let*

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 2 \\ -1 & 1 & 0 \end{bmatrix}.$$

The nullspace of A can be computed by solving the equation

$$A\boldsymbol{x} = 0. \tag{1.9}$$

Using Gaussian elimination (Homework!), the solutions to (1.9) are the scalar multiples of

$$\boldsymbol{x} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix},$$

i.e.,

$$\mathrm{N}(A) = \left\{ \boldsymbol{x} \in \mathbb{R}^3 \mid \boldsymbol{x} = s[-1, -1, 1]^T \text{ for some } s \in \mathbb{R} \right\}.$$

Hence, the dimension of $\mathrm{N}(A)$ is one.

Recall that

$$\mathrm{R}(A) = \{\boldsymbol{y} \in \mathbb{R}^3 \mid \boldsymbol{y} = A\boldsymbol{x} \text{ for some } \boldsymbol{x} \in \mathbb{R}^3\},$$

which is spanned by the column vectors of A.

To deduce $\dim\mathrm{R}(A)$, *we introduce a cleverly chosen basis for* $\mathbb{R}^3$:

$$\boldsymbol{v}_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \qquad \boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \qquad \boldsymbol{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

The first vector $\boldsymbol{v}_1$ *is a basis for* $\mathrm{N}(A)$, *and the other two are chosen so that they together form a basis for* $\mathbb{R}^3$. *(Homework!)*

Denote $V = [\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3]$. *Since the columns of* $V \in \mathbb{R}^{3\times 3}$ *form a basis for* $\mathbb{R}^3$, *any vector* $\boldsymbol{x} \in \mathbb{R}^3$ *can be given as their linear combination, i.e., as* $\boldsymbol{x} = V\boldsymbol{\alpha}$ *for some* $\boldsymbol{\alpha} \in \mathbb{R}^3$.

Hence, the range of A *can be alternatively expressed as*

$$\mathrm{R}(A) = \{\boldsymbol{y} \in \mathbb{R}^3 \mid \boldsymbol{y} = A(V\boldsymbol{\alpha}) \text{ for some } \boldsymbol{\alpha} \in \mathbb{R}^3\}.$$

By construction $\boldsymbol{v}_1 \in \mathrm{N}(A)$, *and so*

$$\begin{aligned} A(V\boldsymbol{\alpha}) = (AV)\boldsymbol{\alpha} &= [A\boldsymbol{v}_1, A\boldsymbol{v}_2, A\boldsymbol{v}_3]\boldsymbol{\alpha} \\ &= [0, A\boldsymbol{v}_2, A\boldsymbol{v}_3]\boldsymbol{\alpha} = \alpha_2 A\boldsymbol{v}_2 + \alpha_3 A\boldsymbol{v}_3. \end{aligned}$$

Hence, $\mathrm{R}(A)$ *is spanned by* $A\boldsymbol{v}_2$ *and* $A\boldsymbol{v}_3$, *which are linearly independent because they are by construction the (obviously linearly independent) first and second columns of* A.

Thus, $\dim \mathrm{R}(A) = 2$.

Lecture 2.

In Example 1.1.2,

$$\dim \mathrm{N}(A) + \dim \mathrm{R}(A) = 3,$$

which is the dimension of the considered space $\mathbb{R}^3$. The basic idea of Example 1.1.2 is also valid for a general $A \in \mathbb{R}^{m\times n}$, and then the result is known as the Dimension Theorem.

Theorem 1.1.1. *For any* $A \in \mathbb{R}^{m\times n}$,

$$\dim \mathrm{R}(A) + \dim \mathrm{N}(A) = n. \tag{1.10}$$

Proof. Let $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$ be a basis for $\mathrm{N}(A)$ with $\dim \mathrm{N}(A) = k$.

We introduce additional vectors $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_{n-k}$ such that

$$\{\boldsymbol{v}_1, \ldots \boldsymbol{v}_k, \boldsymbol{w}_1, \ldots, \boldsymbol{w}_{n-k}\}$$

is a basis for the whole $\mathbb{R}^n$.[1]

Define $V \in \mathbb{R}^{n\times k}$ and $W \in \mathbb{R}^{n\times(n-k)}$ via

$$V = \begin{bmatrix}\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\end{bmatrix} \quad \text{and} \quad W = \begin{bmatrix}\boldsymbol{w}_1, \ldots, \boldsymbol{w}_{n-k}\end{bmatrix}.$$

[1] Any linearly independent set can be completed to a basis in any vector space. In finite dimensions, this can be carried out but Gram–Schmidt process introduced later.

Because the columns of the composite matrix $[V,\, W] \in \mathbb{R}^{n\times n}$ form a basis for $\mathbb{R}^n$, any $\boldsymbol{x} \in \mathbb{R}^n$ can be written as $\boldsymbol{x} = [V,\, W]\boldsymbol{\alpha}$ for some $\boldsymbol{\alpha} \in \mathbb{R}^n$.

Further,

$$\mathrm{R}(A) = \big\{\boldsymbol{y} \in \mathbb{R}^n \mid \boldsymbol{y} = A[V,\, W]\boldsymbol{\alpha} \text{ for some } \boldsymbol{\alpha} \in \mathbb{R}^n\big\},$$

where

$$\begin{aligned} A[V,W]\boldsymbol{\alpha} &= \big[A\boldsymbol{v}_1, \ldots, A\boldsymbol{v}_k, A\boldsymbol{w}_1, \ldots, A\boldsymbol{w}_{n-k}\big]\boldsymbol{\alpha} \\ &= \big[0, \ldots, 0, A\boldsymbol{w}_1, \ldots, A\boldsymbol{w}_{n-k}]\boldsymbol{\alpha} \\ &= \sum_{i=1}^{n-k} \alpha_{i+k} A\boldsymbol{w}_i. \end{aligned}$$

Hence,

$$\mathrm{R}(A) = \mathrm{span}(A\boldsymbol{w}_1, \ldots, A\boldsymbol{w}_{n-k})$$

and $\dim \mathrm{R}(A) \leq n - k$.

We proceed to show that $\dim \mathrm{R}(A) = n - k$ by showing that the set

$$\{A\boldsymbol{w}_1, A\boldsymbol{w}_2, \ldots, A\boldsymbol{w}_{n-k}\}$$

is linearly independent.

Let $\boldsymbol{\alpha} = [\alpha_1, \ldots, \alpha_{n-k}]^T \in \mathbb{R}^{n-k}$ be such that

$$\sum_{i=1}^{n-k} \alpha_i (A\boldsymbol{w}_i) = \left[A\boldsymbol{w}_1, \ldots, A\boldsymbol{w}_{n-k}\right]\boldsymbol{\alpha} = AW\boldsymbol{\alpha} = 0.$$

Thus, $W\boldsymbol{\alpha} \in \mathrm{N}(A)$.

Since the columns of $V \in \mathbb{R}^{n\times k}$ form, by assumption, a basis for $\mathrm{N}(A)$, there exists (a unique) $\boldsymbol{\beta} \in \mathbb{R}^k$ such that $W\boldsymbol{\alpha} = V\boldsymbol{\beta}$, or equivalently:

$$[V,\ W] \begin{bmatrix} \boldsymbol{\beta} \\ -\boldsymbol{\alpha} \end{bmatrix} = 0.$$

Since the columns of $[V,\, W] \in \mathbb{R}^{n\times n}$ form a basis for $\mathbb{R}^n$ by our construction, it must hold that $\boldsymbol{\beta} = 0$ and $-\boldsymbol{\alpha} = 0$. Since $\boldsymbol{\alpha} = 0$, the vectors $A\boldsymbol{w}_1, \ldots, A\boldsymbol{w}_{n-k}$ are linearly independent. □

Let us then return to the unique solvability of

$$A\boldsymbol{x} = \boldsymbol{b}, \quad \text{where} \quad \boldsymbol{b} \in \mathbb{R}^m \quad \text{and} \quad \boldsymbol{x} \in \mathbb{R}^n.$$

In order for the 'unique solvability conditions' (1.7) to hold for all $\boldsymbol{b} \in \mathbb{R}^m$, one must obviously have $\mathrm{R}(A) = \mathbb{R}^m$ and $\mathrm{N}(A) = \{0\}$.

Thus,

$$\dim \mathrm{R}(A) + \dim \mathrm{N}(A) = m + 0 = m.$$

Considering the Dimension Theorem, this implies $m = n$. Thus A must necessarily be a square matrix for unique solvability for for all $\boldsymbol{b} \in \mathbb{R}^m$.

Obviously, the equation $A\boldsymbol{x} = \boldsymbol{b}$ may fail to be (uniquely) solvable even if A is square.

If A is not square, then at least one of the following conditions hold:

1. The equation $A\boldsymbol{x} = \boldsymbol{b}$ does not have any solution for some $\boldsymbol{b} \in \mathbb{R}^m$.

2. The equation $A\boldsymbol{x} = \boldsymbol{b}$ has many solutions for some $\boldsymbol{b} \in \mathbb{R}^m$.

Please, make sure that you understand the Aristotelian syllogisms. This is logic, not linear algebra any more.

Let us rephrase our conclusions for square matrices, i.e., $m = n$.

By the definition of $\mathrm{R}(A)$, the problem

$$A\boldsymbol{x} = \boldsymbol{b}. \tag{1.11}$$

has at least one solution for all $\boldsymbol{b} \in \mathbb{R}^n$ if and only if $\mathrm{R}(A) = \mathbb{R}^n$ if and only if $\operatorname{rank}(A) = \dim \mathrm{R}(A) = n$.

According to Dimension Theorem,

$$\dim \mathrm{N}(A) = n - \dim \mathrm{R}(A) = n - n = 0.$$

Thus, $\mathrm{N}(A) = \{0\}$, and the solution of (1.11) is also unique.

Conversely, the solution to (1.11) is unique if and only if $\dim \mathrm{N}(A) = 0$, which implies $\dim \mathrm{R}(A) = n$ by the Dimension Theorem 1.1.1. Then (1.11) is solvable for all $\boldsymbol{b} \in \mathbb{R}^n$.

Conclusion for square matrices A: The solvability of (1.11) for all $\boldsymbol{b} \in \mathbb{R}^n$ **and** the uniqueness of a solution to (1.11) are equivalent conditions.

Consider again the linear problem

$$A\boldsymbol{x} = \boldsymbol{b}. \tag{1.12}$$

Corollary 1.1.1. *Let A be $n \times n$ in (1.12).*

There exists a unique solution $\boldsymbol{x} \in \mathbb{R}^n$ for any $\boldsymbol{b} \in \mathbb{R}^n$ if and only if one of the following equivalent conditions holds:

1. $\mathrm{R}(A) = \mathbb{R}^n$,
2. *A has linearly independent columns,*
3. *A has linearly independent rows,*
4. $\mathrm{N}(A) = \{0\}$,
5. $\det A \neq 0$, *or*
6. *0 is not an eigenvalue of A.*

Proof. Conditions (1) and (2) are equivalent due to definition of $rangeA$.

The equivalence of (1) and (4) was already covered above.

The condition (5) is equivalent to both (2) and (3) by fundamental properties of the determinant (not covered in this course).

Finally, conditions (4) and (6) are equivalent since having 0 as an eigenvalue is equivalent to having a nontrivial nullspace. (Eigenvalues will be considered in detail in Chapter 2). □

When the equivalent conditions of Corollary 1.1.1 hold, we say that A is **non-singular** or **invertible**.

Assume that $A \in \mathbb{R}^{n\times n}$ is invertible.

Let $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathbb{R}^n$ and $\boldsymbol{b}_1, \boldsymbol{b}_2 \in \mathbb{R}^n$ satisfy

$$A\boldsymbol{x}_1 = \boldsymbol{b}_1 \quad \text{and} \quad A\boldsymbol{x}_2 = \boldsymbol{b}_2.$$

Then for any $\alpha, \beta \in \mathbb{R}$,

$$A(\alpha\boldsymbol{x}_1 + \beta\boldsymbol{x}_2) = \alpha A\boldsymbol{x}_1 + \beta A\boldsymbol{x}_2 = \alpha\boldsymbol{b}_1 + \beta\boldsymbol{b}_2.$$

So, $\boldsymbol{x} := \alpha\boldsymbol{x}_1 + \beta\boldsymbol{x}_2$ is the solution of $A\boldsymbol{x} = \boldsymbol{b}$ corresponding to $\boldsymbol{b} := \alpha\boldsymbol{b}_1 + \beta\boldsymbol{b}_2$. We conclude that the inverse mapping

$$\boldsymbol{b} \mapsto \boldsymbol{x} \quad \text{associated to the problem } A\boldsymbol{x} = \boldsymbol{b}$$

is linear.

The inverse mapping of A exists, since A represents a bijection from $\mathbb{R}^n$ onto itself.

Obviously, there cannot be two different inverse mappings for the same A.

(We go on assuming that $A \in \mathbb{R}^{n\times n}$ is invertible.)

Because any linear map can be represented as a matrix (in a given basis), it follows that the solution to $A\boldsymbol{x} = \boldsymbol{b}$ can be given as

$$\boldsymbol{x} = A^{-1}\boldsymbol{b}, \tag{1.13}$$

for some matrix $A^{-1} \in \mathbb{R}^{n\times n}$.

The matrix A^{-1} is called the **inverse** of A.

The inverse of a square matrix (when it exists) is unique.

Denoting $A = [\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n]$, we have

$$A^{-1}A = \left[A^{-1}\boldsymbol{a}_1, \ldots, A^{-1}\boldsymbol{a}_n\right] = [\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n] = I$$

since the unique solution of $A\boldsymbol{x} = \boldsymbol{a}_j$, $j = 1, \ldots, n$, is obviously the jth Cartesian basis vector $\boldsymbol{e}_j \in \mathbb{R}^n$.

Similarly, $AA^{-1} = I$ and $(A^{-1})^{-1} = A$.

Lecture 3.

1.2 Norms and inner products

To study the stability of the solution to (1.1) with respect to perturbations, inaccuracies or uncertainties in $\boldsymbol{b}$ and A, we need tools for measuring the 'size' of a vector or a matrix. We use (vector) norms for this purpose.

Further, orthogonality notions are studied in terms of inner products. Inner products may be used for defining norms, but not all norms can be produced this way. Finally, all inner products in $\mathbb{R}^n$ are characterised in terms of symmetric $n \times n$ matrices.

1.2.1 Vector norm

The size (length) of a vector $\boldsymbol{x} \in \mathbb{R}^n$ is measured by a (vector) norm.

There are many different vector norms.

Definition 1.2.1. *A function* $\|\cdot\| : \mathbb{R}^n \to \mathbb{R}$ *is a* **norm** *if it satisfies*

i. $\|\boldsymbol{x}\| \geq 0$ *for all* $\boldsymbol{x} \in \mathbb{R}^n$ *and* $\|\boldsymbol{x}\| = 0$ *if and only if* $\boldsymbol{x} = 0$,

ii. $\|\boldsymbol{x} + \boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$ *for all* $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$,

iii. $\|\alpha\boldsymbol{x}\| = |\alpha|\|\boldsymbol{x}\|$ *for all* $\boldsymbol{x} \in \mathbb{R}^n$ *and* $\alpha \in \mathbb{R}$.

Definition 1.2.1 characterises a reasonable measure of length.

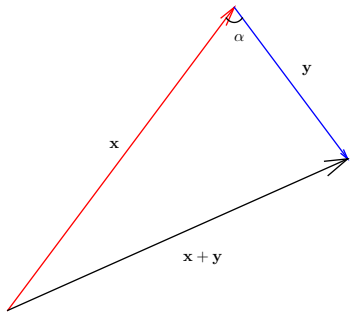


Figure 1.1: The triangle inequality.

As an example, consider the space $\mathbb{R}^2$ and associate to each $\boldsymbol{x} \in \mathbb{R}^2$ a geometric vector $x_1\boldsymbol{i} + x_2\boldsymbol{j}$:

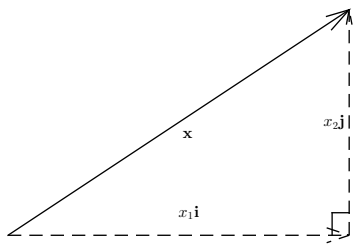


Figure 1.2: The Euclidian length of a vector in $\mathbb{R}^2$.

The perhaps most natural way to measure the length of a vector $\boldsymbol{x}$ is via the Pythagorean theorem.

This amounts to introducing the **Euclidean norm** $\|\cdot\|_2 : \mathbb{R}^2 \to \mathbb{R}$, given by

$$\|\boldsymbol{x}\|_2 := \left(x_1^2 + x_2^2\right)^{1/2}.$$

This function has the following properties:

(i) **Positivity:** For all $\boldsymbol{x} \in \mathbb{R}^2$,

$$\|\boldsymbol{x}\|_2 \geq 0,$$

with the equality holding only if and only if $\boldsymbol{x} = 0$.

(ii) **Scaling:** The vector length measured by $\|\cdot\|_2$ scales linearly with the multiplying scalar; i.e.,

$$\|\alpha\boldsymbol{x}\|_2 = |\alpha|\|\boldsymbol{x}\|_2$$

for all $\alpha \in \mathbb{R}$.

(iii) **Triangle inequality:** The three vectors $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{x}+\boldsymbol{y} \in \mathbb{R}^2$ form a triangle in the plane as shown in Fig. 1.1. For every triangle, the sum of the lengths of two sides is larger than the length of the third remaining side:

$$\|\boldsymbol{x}+\boldsymbol{y}\|_2 \leq \|\boldsymbol{x}\|_2 + \|\boldsymbol{y}\|_2.$$

Hence, $\|\cdot\|_2 : \mathbb{R}^2 \to \mathbb{R}$ is a norm in accordance with Definition 1.2.1.

The triangle inequality for the Euclidean norms will be proved, but surely we know it already?

In fact, the algebraic form of the Euclidean norm is a motivation for more general norms satisfying Definition 1.2.1.

The Euclidean norm generalises to an arbitrary spatial dimension by defining $\|\cdot\|_2 : \mathbb{R}^n \to \mathbb{R}$ through

$$\|\boldsymbol{x}\|_2 := \Big(\sum_{i=1}^{n} |x_i|^2 \Big)^{1/2}.$$

Further norms on $\mathbb{R}^n$ include the p-norms

$$\|\boldsymbol{x}\|_p := \Big(\sum_{i=1}^{n} |x_i|^p \Big)^{1/p}, \qquad 1 \leq p < \infty.$$

Clearly, the Euclidean norm is the same as the 2-norm.

The ∞-norm is given by

$$\|\boldsymbol{x}\|_\infty := \max_i |x_i|.$$

Homework: Show that $\lim_{p\to\infty} \|\boldsymbol{x}\|_p = \|\boldsymbol{x}\|_\infty$ for all $\boldsymbol{x}$. Can you give a short verbal explanation – a one your grandmother would like to hear – why this claim reasonably evident?

Proving that $\|\cdot\|_1$ and $\|\cdot\|_\infty$ (or even $\|\cdot\|_p$) are norms is left as an exercise, too.

A vector space equipped with a norm is called **normed vector space**.

The distance between two points (vectors) $\boldsymbol{x}$ and $\boldsymbol{y}$ is given by the **metric**

$$d(\boldsymbol{x}, \boldsymbol{y}) = \|\boldsymbol{x} - \boldsymbol{y}\|.$$

The norms used in linear algebra can be divided into **generic norms** and **problem specific norms**.

Much of the linear algebra theory relates to generic norms that plainly satisfy Definition 1.2.1, nothing more.

Problem specific norms are used when some practical application requires a particular way of measuring vector length.

Many times a given problem becomes much easier to solve if "the right kind of norm" is used in its solution. It often takes a certain amount of cleverness and experience to come up with "the right kind of norm", though.

But then again, one should not expect that mathematics is always easy. As it is not, and we have to deal with it.

Example 1.2.1. (a problem specific norm) *Consider a first order polynomial $p(x) = \alpha_0 + \alpha_1 x$ with real coefficients. The size of such a polynomial on $[0, 1]$ can be measured by the square integral*

$$\int_0^1 p^2(x)\,dx = \int_0^1 (\alpha_0^2 + 2\alpha_0\alpha_1 x + \alpha_1^2 x^2)\,dx = \alpha_0^2 + \alpha_0\alpha_1 + \frac{1}{3}\alpha_1^2.$$

On the other hand, any vector in $\mathbb{R}^2$ can be uniquely identified with a first order polynomial via

$$\begin{bmatrix} \alpha_0 \\ \alpha_1 \end{bmatrix} \quad \longleftrightarrow \quad \alpha_0 + \alpha_1 x.$$

Measuring the length of a vector defining the coefficients of a first order polynomial in, say, the 2-norm does not have any immediate interpretation. Instead it is more natural to measure the size of the coefficient vector using the function

$$\left\| \begin{bmatrix} \alpha_0 \\ \alpha_1 \end{bmatrix} \right\| := \left(\alpha_0^2 + \alpha_0\alpha_1 + \frac{1}{3}\alpha_1^2 \right)^{1/2}, \tag{1.14}$$

which is the square root of the square integral of the first order polynomial $\alpha_0 + \alpha_1 x$ introduced above.

We will later prove that (1.14), indeed, satisfies the conditions of Definition 1.2.1.

1.2.2 Inner product

Let $n = 2$ and consider two nonzero geometric vectors $x_1\boldsymbol{i} + x_2\boldsymbol{j}$ and $y_1\boldsymbol{i} + y_2\boldsymbol{j}$.

It follows from the **cosine theorem** in trigonometry that the angle θ between the two vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ satisfies

$$\begin{aligned}&\|\boldsymbol{x}\|_2^2 + \|\boldsymbol{y}\|_2^2 - 2\|\boldsymbol{x}\|_2\|\boldsymbol{y}\|_2 \cos\theta = \|\boldsymbol{x} - \boldsymbol{y}\|_2^2 \\ &= \|\boldsymbol{x}\|_2^2 + \|\boldsymbol{y}\|_2^2 - 2(x_1y_1 + x_2y_2),\end{aligned}$$

or, in other words,

$$\cos\theta = \frac{x_1y_1 + x_2y_2}{\|\boldsymbol{x}\|_2\|\boldsymbol{y}\|_2}.$$

Check the "law of cosines" in Wikipedia for an elementary proof using Pythagorean theorem and definitions of trigonometric functions.

Motivated by this formula, we define the **dot product** as

$$\boldsymbol{x} \cdot \boldsymbol{y} = x_1 y_1 + x_2 y_2,$$

yielding the familiar

$$\cos\theta = \frac{\boldsymbol{x} \cdot \boldsymbol{y}}{\|\boldsymbol{x}\|_2 \|\boldsymbol{y}\|_2}.$$

If $\boldsymbol{x} \cdot \boldsymbol{y} = 0$, the two vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ are in a right angle (or at least one of them vanishes).

Thus, thee dot product introduces the concept of orthogonality in the space $\mathbb{R}^2$.

The dot product in $\mathbb{R}^n$ is given by

$$\boldsymbol{x} \cdot \boldsymbol{y} := \sum_{i=1}^{n} x_i y_i = \boldsymbol{x}^T \boldsymbol{y}$$

for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$.

The vectors $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ are **orthogonal** (in the ordinary geometric sense, e.g., when $n = 3$) if $\boldsymbol{x} \cdot \boldsymbol{y} = 0$.

The dot product is a special case of a general **inner product**:

Definition 1.2.2. *A mapping* $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ *is an* **inner product** *if the following properties hold:*

i. Positive definiteness: $\langle \boldsymbol{x}, \boldsymbol{x} \rangle \geq 0$ *for all* $\boldsymbol{x} \in \mathbb{R}^n$ *and* $\langle \boldsymbol{x}, \boldsymbol{x} \rangle = 0$ *if and only if* $\boldsymbol{x} = 0$.

ii. Symmetry: $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \langle \boldsymbol{y}, \boldsymbol{x} \rangle$ *for all* $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$.

iii. Linearity w.r.t. the first variable position: $\langle \alpha\boldsymbol{x} + \beta\boldsymbol{y}, \boldsymbol{z} \rangle = \alpha\langle \boldsymbol{x}, \boldsymbol{z} \rangle + \beta\langle \boldsymbol{y}, \boldsymbol{z} \rangle$ *for all* $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}^n$ *and* $\alpha, \beta \in \mathbb{R}$.

The conditions (ii) and (iii) together imply that also $\langle \boldsymbol{z}, \alpha\boldsymbol{x} + \beta\boldsymbol{y} \rangle = \alpha\langle \boldsymbol{z}, \boldsymbol{x} \rangle + \beta\langle \boldsymbol{z}, \boldsymbol{y} \rangle$. Thus, we call the inner product **bilinear**.

A vector space equipped with an inner product is called **inner product space**.

If an inner product space is **complete** (meaning that all Cauchy sequences converge to some vector), it is called **Hilbert space**.

Fortunately, all finite-dimensional inner product spaces are complete (as far as the field of coefficients is $\mathbb{R}$ or $\mathbb{C}$).
(We do not concentrate on completeness in this course.)

Definition 1.2.3. *A set of vectors* $\{\boldsymbol{q}_1, \boldsymbol{q}_2, \ldots, \boldsymbol{q}_k\} \subset \mathbb{R}^n$ *is called* **orthogonal** *with respect to an inner product* $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ *if*

$$\langle \boldsymbol{q}_i, \boldsymbol{q}_j \rangle = 0$$

for all $i \neq j$.

Take note that different inner products give rise to different concepts of orthogonality in $\mathbb{R}^n$ through

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = 0.$$

It turns out that an orthogonal set of vectors is always linearly independent, unless one of the vectors is the zero vector.

Lemma 1.2.1. *If the nonzero vectors $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k \in \mathbb{R}^n$ compose a set of orthogonal vectors (with respect to any inner product of $\mathbb{R}^n$), then they are linearly independent.*

Proof. Let $\boldsymbol{\alpha} \in \mathbb{R}^k$ be such that

$$\sum_{i=1}^{k} \alpha_i \boldsymbol{q}_i = \mathbf{0}. \tag{1.15}$$

Take the inner product of LHS with $\boldsymbol{q}_j$, $j = 1, \ldots, k$ and use (iii) of Definition 1.2.2:

$$\left\langle \sum_{i=1}^{k} \alpha_i \boldsymbol{q}_i, \boldsymbol{q}_j \right\rangle = \sum_{i=1}^{k} \alpha_i \langle \boldsymbol{q}_i, \boldsymbol{q}_j \rangle = \alpha_j \langle \boldsymbol{q}_j, \boldsymbol{q}_j \rangle.$$

The second equality holds by orthogonality.

Take the inner product of RHS with $\boldsymbol{q}_j$, $j = 1, \ldots, k$:

$$\langle \mathbf{0}, \boldsymbol{q}_j \rangle \;=\; 0 \, \langle \mathbf{0}, \boldsymbol{q}_j \rangle = 0$$

due to (iii) of Definition 1.2.2 and the fact $\mathbf{0} = 0\mathbf{0}$.

Putting things together:

$$\alpha_j \langle \boldsymbol{q}_j, \boldsymbol{q}_j \rangle = \left\langle \sum_{i=1}^{k} \alpha_i \boldsymbol{q}_i, \boldsymbol{q}_j \right\rangle = \langle \boldsymbol{0}, \boldsymbol{q}_j \rangle = 0. \tag{1.16}$$

for any $j = 1, \ldots, k$.

Since $\langle \boldsymbol{q}_j, \boldsymbol{q}_j \rangle > 0$ by (i) of Definition 1.2.2 and our assumption on the orthogonal vectors, it must hold that $\alpha_j = 0$ for all $j = 1, \ldots, k$. Thus the vectors $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k$ are linearly independent. □

Recall that any set of n linearly independent vectors must be a basis for an n-dimensional (sub)space.

An immediate Corollary of Lemma 1.2.1 is the following: Any set of n non-zero orthogonal vectors $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n\} \subset \mathbb{R}^n$ is a basis for $\mathbb{R}^n$.

Suppose that the orthogonal vectors $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n\} \subset \mathbb{R}^n$ are given. If, in addition, $\langle \boldsymbol{q}_j, \boldsymbol{q}_j \rangle = 1$ for all $j = 1, \ldots, n$, these vectors are an **orthonormal basis** for $\mathbb{R}^n$.

Orthonormal = **ortho**gonal and **normal**ised.

How to make orthonormal vectors from non-zero orthogonal vectors?

Hence, for any $\boldsymbol{x} \in \mathbb{R}^n$, there exists a representation

$$\boldsymbol{x} = \sum_{i=1}^{n} \alpha_i \boldsymbol{q}_i \tag{1.17}$$

with some $\boldsymbol{\alpha} \in \mathbb{R}^n$.

The benefit of orthogonality (compared to a more general basis of $\mathbb{R}^n$) is that the coefficients α_i in (1.17) can be easily computed for any $\boldsymbol{x}$.

By taking the inner products of (1.17) with orthogonal vectors $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$ one at a time,

$$\alpha_j = \frac{\langle \boldsymbol{x}, \boldsymbol{q}_j \rangle}{\langle \boldsymbol{q}_j, \boldsymbol{q}_j \rangle}, \qquad j = 1, \ldots, n. \tag{1.18}$$

Eqs. (1.17) and (1.18) reduce to

$$\boldsymbol{x} = \sum_{i=1}^{n} \langle \boldsymbol{x}, \boldsymbol{q}_j \rangle \boldsymbol{q}_i$$

if $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n\} \subset \mathbb{R}^n$ is an orthonormal basis.

To motivate the connection between inner products and norms, observe that the dot product has a special connection to the Euclidean norm:

$$\|\boldsymbol{x}\|_2 = (\boldsymbol{x} \cdot \boldsymbol{x})^{1/2}$$

for all $\boldsymbol{x} \in \mathbb{R}^n$. More generally: any inner product induces a norm.

Lemma 1.2.2. *Let $\langle \cdot, \cdot \rangle$ be an inner product in $\mathbb{R}^n$. Then the function*

$$\|\boldsymbol{x}\| := \langle \boldsymbol{x}, \boldsymbol{x} \rangle^{1/2}, \qquad \boldsymbol{x} \in \mathbb{R}^n, \tag{1.19}$$

is a norm (positive square root chosen).

In addition, this norm satisfies the **Cauchy–Schwarz inequality**

$$|\langle \boldsymbol{x}, \boldsymbol{y} \rangle| \leq \|\boldsymbol{x}\| \|\boldsymbol{y}\| \tag{1.20}$$

for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$.

Proof. We need to check that $\|\cdot\| : \mathbb{R}^n \to \mathbb{R}$ satisfies the conditions (i-iii) of Definition 1.2.1 and that (1.20) holds.

By the positive definiteness of an inner product,

$$\|\boldsymbol{x}\| := \langle \boldsymbol{x}, \boldsymbol{x} \rangle^{1/2}$$

is well defined and equals 0 if and only if $\boldsymbol{x} = 0$. Since we use the non-negative branch of the square root function, $\|\boldsymbol{x}\| \geq 0$.

Furthermore, (bi)linearity of the inner product gives

$$\|\alpha \boldsymbol{x}\| = \langle \alpha \boldsymbol{x}, \alpha \boldsymbol{x} \rangle^{1/2} = \left(\alpha^2 \langle \boldsymbol{x}, \boldsymbol{x} \rangle\right)^{1/2} = |\alpha| \|\boldsymbol{x}\|.$$

Before proving the triangle inequality for the norm, it is convenient to prove the Cauchy–Schwarz inequality (1.20).

Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ be non-zero vectors and split $\boldsymbol{x}$ as

$$\boldsymbol{x} = \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\|\boldsymbol{y}\|^2} \boldsymbol{y} + \left(\boldsymbol{x} - \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\|\boldsymbol{y}\|^2} \boldsymbol{y} \right).$$

Using (ii) and (iii) of Definition 1.2.2, we thus obtain that

$$\begin{aligned}
\|\boldsymbol{x}\|^2 &= \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle^2}{\|\boldsymbol{y}\|^4} \|\boldsymbol{y}\|^2 + 2 \left\langle \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\|\boldsymbol{y}\|^2} \boldsymbol{y}, \boldsymbol{x} - \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\|\boldsymbol{y}\|^2} \boldsymbol{y} \right\rangle + \left\| \boldsymbol{x} - \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\|\boldsymbol{y}\|^2} \boldsymbol{y} \right\|^2 \\
&= \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle^2}{\|\boldsymbol{y}\|^2} + 2 \left(\frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle^2}{\|\boldsymbol{y}\|^2} - \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle^2}{\|\boldsymbol{y}\|^4} \|\boldsymbol{y}\|^2 \right) + \left\| \boldsymbol{x} - \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\|\boldsymbol{y}\|^2} \boldsymbol{y} \right\|^2 \\
&= \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle^2}{\|\boldsymbol{y}\|^2} + \left\| \boldsymbol{x} - \frac{\langle \boldsymbol{x}, \boldsymbol{y} \rangle}{\|\boldsymbol{y}\|^2} \boldsymbol{y} \right\|^2 .
\end{aligned}$$

Dropping the second (non-negative) term on the right-hand side and multiplying the resulting inequality by $\|\boldsymbol{y}\|^2$ gives

$$\|\boldsymbol{x}\|^2\|\boldsymbol{y}\|^2 \geq \langle \boldsymbol{x}, \boldsymbol{y}\rangle^2 \qquad \text{or} \qquad |\langle \boldsymbol{x}, \boldsymbol{y}\rangle| \leq \|\boldsymbol{x}\|\|\boldsymbol{y}\|,$$

which is the Cauchy–Schwarz inequality.

Finally, the triangle inequality follow from

$$\begin{aligned}\|\boldsymbol{x}+\boldsymbol{y}\|^2 &= \|\boldsymbol{x}\|^2 + 2\langle \boldsymbol{x}, \boldsymbol{y}\rangle + \|\boldsymbol{y}\|^2 \\ &\leq \|\boldsymbol{x}\|^2 + 2\|\boldsymbol{x}\|\|\boldsymbol{y}\| + \|\boldsymbol{y}\|^2 = (\|\boldsymbol{x}\| + \|\boldsymbol{y}\|)^2\end{aligned}$$

where the inequality is just (1.20). □

We complete this section by pointing out a one-to-one correspondence between positive definite matrices and inner products.

Definition 1.2.4. *A matrix $A \in \mathbb{R}^{n\times n}$ is called* **positive semidefinite** *if*

$$\boldsymbol{x}^T A\boldsymbol{x} = \boldsymbol{x} \cdot A\boldsymbol{x} \geq 0 \tag{1.21}$$

for all $\boldsymbol{x} \in \mathbb{R}^n$. If the second equality in (1.21) holds only for $\boldsymbol{x} = 0$, then A is called **positive definite**. *(Nota bene: Sometimes symmetricity of A is included in the definition of positive definiteness.)*

Any positive **definite** matrix $A \in \mathbb{R}^{n\times n}$ is invertible: If $\boldsymbol{x} \in \mathrm{N}(A)$, then

$$\boldsymbol{x}^T A\boldsymbol{x} = \boldsymbol{x}^T \mathbf{0} = 0.$$

Thus $\boldsymbol{x} = \mathbf{0}$ by definitenes, and $\mathrm{N}(A) = \{0\}$ follows.

The invertibility of A follows now from the equivalent conditions of Corollary 1.1.1.

Inner products of $\mathbb{R}^n$ and symmetric positive definite matrices $A \in \mathbb{R}^{n\times n}$ are in one-to-one correspondence:

Lemma 1.2.3. *For any inner product $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, there exists a symmetric positive definite matrix $A \in \mathbb{R}^{n\times n}$ such that*

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \boldsymbol{y}^T A \boldsymbol{x}. \tag{1.22}$$

for all $\boldsymbol{x} \in \mathbb{R}^n$. Conversely, the formula (1.22) defines an inner product for any symmetric positive definite $A \in \mathbb{R}^{n\times n}$.

Proof. Let $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ be an arbitrary inner product.

Expanding $\boldsymbol{x} = \sum_{i=1}^n x_i \boldsymbol{e}_i$ and $\boldsymbol{y} = \sum_{j=1}^n y_j \boldsymbol{e}_j$ in the natural unit basis vectors, it follows that

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \left\langle \sum_{i=1}^n x_i \boldsymbol{e}_i, \sum_{j=1}^n y_j \boldsymbol{e}_j \right\rangle = \sum_{i,j=1}^n x_i y_j \left\langle \boldsymbol{e}_i, \boldsymbol{e}_j \right\rangle,$$

where we repeatedly the linearity of the inner product in both arguments.

We now define the matrix $A \in \mathbb{R}^{n\times n}$ elementwise as $a_{ji} = \langle \boldsymbol{e}_i, \boldsymbol{e}_j \rangle$, $i, j = 1, \ldots, n$.

Because the inner product is symmetric, we have $a_{ij} = a_{ji}$. Thus A is symmetric, meaning that $A = A^T$.

Moreover, grouping terms in summation gives

$$\begin{aligned}\langle \boldsymbol{x}, \boldsymbol{y} \rangle &= \sum_{i,j=1}^{n} x_i y_j a_{ji} = \sum_{j=1}^{n} y_j \left(\sum_{i=1}^{n} a_{ji} x_i \right) \\ &= \sum_{j=1}^{n} y_j \left(A\boldsymbol{x} \right)_j = \boldsymbol{y}^T A \boldsymbol{x}\end{aligned}$$

by the definition of matrix-vector product. Here $\left(A\boldsymbol{x}\right)_j$ denotes the jth column of vector $A\boldsymbol{x}$.

Finally, the constructed A is positive definite since

$$\boldsymbol{x}^T A \boldsymbol{x} = \langle \boldsymbol{x}, \boldsymbol{x} \rangle > 0 \qquad \text{for all } 0 \neq \boldsymbol{x} \in \mathbb{R}^n$$

because $\langle \cdot, \cdot \rangle$ satisfies the corresponding axiom of the inner product in Definition 1.2.2.

The converse part is a straightforward computation, and can be either privately carried out as a fruitful and recommendable training of matrix manipulations... or just read it in the lecture notes if you already can. □

Lecture 4.

1.2.3 Matrix norm

Multiplication of a vector $\boldsymbol{x}$ by a matrix A produces another vector $\boldsymbol{y} := A\boldsymbol{x}$.

Typically, $\boldsymbol{y}$ differs from $\boldsymbol{x}$ both in length and direction. We don't care about the direction for now. Just the change of length.

Definition 1.2.5. *Let $\|\cdot\|$ be some vector norm. The* **operator norm** *induced by $\|\cdot\|$ is defined as*

$$\|A\|_{\text{op}} := \max_{0 \neq \boldsymbol{x} \in \mathbb{R}^n} \frac{\|A\boldsymbol{x}\|}{\|\boldsymbol{x}\|} = \max_{\|\boldsymbol{x}\|=1} \|A\boldsymbol{x}\|$$

where $A \in \mathbb{R}^{m \times n}$.

The operator norm $\|A\|_{\text{op}}$ gives the maximum amplification of the vector length under multiplication by A.

Different vector norms $\|\cdot\|$ induce different operator norms $\|\cdot\|_{\text{op}}$.

It is customary to drop the subscript from $\|\cdot\|_{\text{op}}$, and denote both a vector norm and the corresponding operator norm in the same way.

The operator norm measures the maximal stretching of the unit sphere

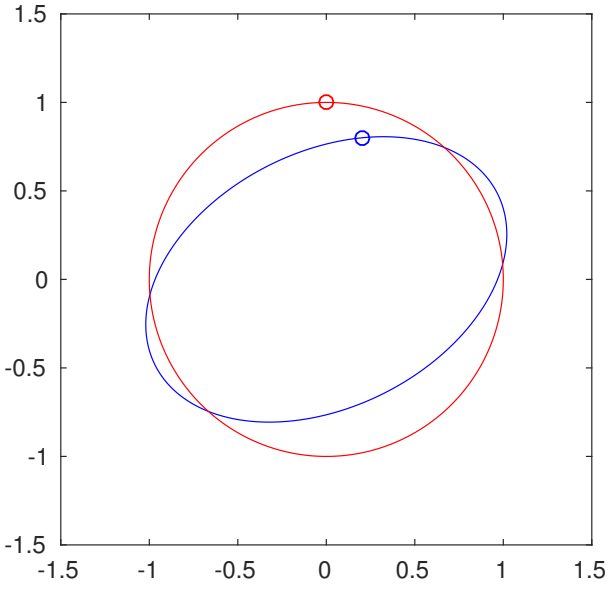
$$S := \{\boldsymbol{x} \in \mathbb{R}^n \mid \|\boldsymbol{x}\| = 1\}$$

under the action of a matrix A

The image of the unit sphere under the action of the matrix

$$A = \begin{bmatrix} 1 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}$$

with $\|A\| \approx 1.0817$.



The red circle is the unit sphere and the blue ellipse its image. Euclidean vector norm and the corresponding matrix norm is used.

Lemma 1.2.4. *Any operator norm satisfies,*

$$\|AB\| \leq \|A\|\|B\|\,, \qquad A \in \mathbb{R}^{l\times m},\ B \in \mathbb{R}^{m\times n} \tag{1.23}$$

and

$$\|A\boldsymbol{x}\| \leq \|A\|\|\boldsymbol{x}\|\,, \qquad A \in \mathbb{R}^{m\times n},\ \boldsymbol{x} \in \mathbb{R}^{n}. \tag{1.24}$$

Proof. For $\boldsymbol{x} \neq \mathbf{0}$:

$$\begin{aligned}\|A\boldsymbol{x}\| &= \frac{\|A\boldsymbol{x}\|}{\|\boldsymbol{x}\|}\cdot\|\boldsymbol{x}\| \\ &\leq \max_{\boldsymbol{z}\neq\mathbf{0}}\left(\frac{\|A\boldsymbol{z}\|}{\|\boldsymbol{z}\|}\cdot\|\boldsymbol{x}\|\right) = \max_{\boldsymbol{z}\neq\mathbf{0}}\left(\frac{\|A\boldsymbol{z}\|}{\|\boldsymbol{z}\|}\right)\cdot\|\boldsymbol{x}\| \\ &= \|A\|\,\|\boldsymbol{x}\|.\end{aligned}$$

Got it?

The other one is similar homework. □

Given a vector norm, an explicit formula for the corresponding operator norm is difficult — if not impossible — to give.

However, the operator norms induced by the vector norms $\|\cdot\|_1$, $\|\cdot\|_2$ and $\|\cdot\|_\infty$ can be described succinctly.

We leave it without proof that the operator norms of $A \in \mathbb{R}^{m\times n}$ corresponding to the 1 and ∞ vector norms are

$$\|A\|_1 = \max_{1\le j\le n} \sum_{i=1}^{m} |a_{ij}| \quad \text{and} \quad \|A\|_\infty = \max_{1\le i\le m} \sum_{j=1}^{n} |a_{ij}|.$$

These are maximum absolute column and row sums of A, respectively.

The 2-norm of a matrix can also be computed. We do that next using the eigenvalue theory of symmetric matrices.

Any symmetric matrix $B = B^T \in \mathbb{R}^{n\times n}$ has n (pairwise) orthonormal **real** eigenvectors so that

$$B\boldsymbol{q}_j = \lambda_j \boldsymbol{q}_j, \qquad j = 1, \ldots, n \tag{1.25}$$

where also all eigenvalues λ_j's are **real**.

Orthonormality plainly means

$$\|\boldsymbol{q}_j\|_2 = 1 \quad \text{for all} \quad j = 1, \ldots, n,$$

and

$$\boldsymbol{q}_j \cdot \boldsymbol{q}_k = \boldsymbol{q}_j^T \boldsymbol{q}_k = 0 \quad \text{for all} \quad j \neq k$$

in the Euclidean inner product.

Eq. (1.25) can be written in the form

$$BQ = Q\Lambda$$

where $Q = [\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n] \in \mathbb{R}^{n\times n}$ has the eigenvectors as its columns and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n) \in \mathbb{R}^{n\times n}$ is a diagonal matrix composed of the eigenvalues (some of which may have same values).

By orthonormality, $Q^TQ = I$ and thus $Q^T = Q^{-1}$. We now get what we require:

$$B = Q\Lambda Q^T. \tag{1.26}$$

Lemma 1.2.5. *For any* $A \in \mathbb{R}^{m\times n}$,

$$\|A\|_2 = \left(\lambda_{\max}(A^TA)\right)^{1/2},$$

where $\lambda_{\max}(A^TA)$ *is the largest eigenvalue of the symmetric matrix* A^TA.

Proof. First

$$\begin{aligned}\|A\|_2^2 &:= \max_{0\neq \boldsymbol{x}\in\mathbb{R}^n} \frac{\|A\boldsymbol{x}\|_2^2}{\|\boldsymbol{x}\|_2^2} \\ &= \max_{0\neq \boldsymbol{x}\in\mathbb{R}^n} \frac{\boldsymbol{x}^TA^TA\boldsymbol{x}}{\|\boldsymbol{x}\|_2^2} = \max_{0\neq \boldsymbol{x}\in\mathbb{R}^n} \frac{\boldsymbol{x}^TQ\Lambda Q^T\boldsymbol{x}}{\|\boldsymbol{x}\|_2^2},\end{aligned}$$

where

$$Q\Lambda Q^T = A^TA, \quad Q^TQ = I, \quad (A^TA)^T = A^TA$$

and

$$\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n).$$

We make the coordinate transformation $\boldsymbol{y} = Q^T\boldsymbol{x}$ where Q is an orthogonal, and hence an invertible matrix. The correspondence between $\boldsymbol{x}$ and $\boldsymbol{y}$ is one-to-one on all of $\mathbb{R}^n$.

Writing $\boldsymbol{y} = [y_1, y_2, \ldots, y_n]^T$, we continue

$$\begin{aligned} \|A\|_2^2 &= \max_{0\neq\boldsymbol{y}\in\mathbb{R}^n} \frac{\boldsymbol{y}^T\Lambda\boldsymbol{y}}{\|Q\boldsymbol{y}\|_2^2} = \max_{0\neq\boldsymbol{y}\in\mathbb{R}^n} \frac{\sum_{i=1}^n \lambda_i y_i^2}{\boldsymbol{y}^T Q^T Q\boldsymbol{y}} \\ &= \max_{0\neq\boldsymbol{y}\in\mathbb{R}^n} \frac{\sum_{i=1}^n \lambda_i y_i^2}{\|\boldsymbol{y}\|_2^2} \\ &= \max_{0\neq\boldsymbol{y}\in\mathbb{R}^n} \sum_{i=1}^n \lambda_i \frac{y_i^2}{\|\boldsymbol{y}\|_2^2}. \end{aligned} \tag{1.27}$$

Define the **nonnegative** numbers

$$\alpha_i := \frac{y_i^2}{\|\boldsymbol{y}\|_2^2} \quad \text{satisfying} \quad \sum_{i=1}^n \alpha_i = 1$$

so that

$$\|A\|_2^2 = \max\left\{ \sum_{i=1}^n \lambda_i\alpha_i \mid \sum_{i=1}^n \alpha_i = 1, \alpha_i \geq 0 \right\}.$$

By common sense $\|A\|_2^2 = \max_i \lambda_i = \lambda_{\max}(A^TA)$. □

Corollary 1.2.1. *Lemma 1.2.5 remains valid if* $\lambda_{\max}(A^TA)$ *is replaced by* $\lambda_{\max}(AA^T)$.

Proof. It can be shown that the $\min\{n, m\}$ largest eigenvalues of the symmetric matrices $A^TA \in \mathbb{R}^{n\times n}$ and $AA^T \in \mathbb{R}^{m\times m}$ are the same when counted according to their algebraic multiplicities.

This is not very difficult to believe: $AA^T\boldsymbol{v} = \lambda\boldsymbol{v} \Rightarrow A^TA(A^T\boldsymbol{v}) = \lambda(A^T\boldsymbol{v})$ and $A^TA\boldsymbol{v} = \lambda\boldsymbol{v} \Rightarrow AA^T(A\boldsymbol{v}) = \lambda(A\boldsymbol{v})$.

A full proof will be presented in Chapter 3. □

...or we could argue directly that $\|A^T\|_2 = \|A\|_2$ by computing

$$\begin{aligned}\|A\|_2 &= \max_{\|\boldsymbol{x}\|_2=\|\boldsymbol{y}\|_2=1} \boldsymbol{y}^TA\boldsymbol{x} = \max_{\|\boldsymbol{x}\|_2=\|\boldsymbol{y}\|_2=1} (\boldsymbol{y}^TA\boldsymbol{x})^T \\ &= \max_{\|\boldsymbol{x}\|_2=\|\boldsymbol{y}\|_2=1} \boldsymbol{x}A^T\boldsymbol{y} = \|A^T\|_2\end{aligned}$$

where we used the observation that

$$\|\boldsymbol{z}\|_2 = \max_{\|\boldsymbol{y}\|_2=1} \boldsymbol{y}^T\boldsymbol{z}. \tag{1.28}$$

Homework: Convince yourself that (1.28) holds.

All $m \times n$ matrices A form a vector space $\mathbb{R}^{m\times n}$. You can sum such matrices and multiply them by scalars.

The dimension of $\mathbb{R}^{m\times n}$ is, of course, mn.

Any vector norm on $\mathbb{R}^{m\times n}$ is called, for clarity, **matrix norm**. Matrix norms are thus characterised as follows:

Definition 1.2.6. *A function* $\|\cdot\| : \mathbb{R}^{m\times n} \to \mathbb{R}$ *is a* **matrix norm** *if*

i. $\|A\| \geq 0$ *and* $\|A\| = 0$ *if and only if* $A = 0$ *is the zero matrix,*

ii. $\|A + B\| \leq \|A\| + \|B\|$,

iii. $\|\alpha A\| = |\alpha|\|A\|$

for all $A, B \in \mathbb{R}^{m\times n}$ *and* $\alpha \in \mathbb{R}$.

Do the operator norms defined above always satisfy the axioms of the (matrix) norm?

Yes, they do.

Lemma 1.2.6. *Let $\|\cdot\|$ denote a vector norm and also the corresponding operator norm. Then $\|\cdot\|$ is also a matrix norm in accordance with Definition 1.2.6.*

Proof. Verification of (i)–(iii) of Definition 1.2.6 follows.

(i) Obviously $\|A\| \geq 0$, and $\|A\| = 0$ if $A = 0$.

If $A \in \mathbb{R}^{m\times n}$ is non-zero, there is $\boldsymbol{x}_0 \in \mathbb{R}^n$ such that $A\boldsymbol{x}_0 \neq \mathbf{0}$. Thus

$$\|A\| := \max_{\boldsymbol{x}\neq\mathbf{0}} \frac{\|A\boldsymbol{x}\|}{\|\boldsymbol{x}\|} \geq \frac{\|A\boldsymbol{x_0}\|}{\|\boldsymbol{x_0}\|} > 0.$$

Since we all are good students of the ancient logician Aristotle (aren't we?), we see that the condition required is now proved.

(ii) Plainly, by the triangle inequality

$$\begin{aligned}\|A+B\| &= \max_{\boldsymbol{x}\neq\boldsymbol{0}} \frac{\|A\boldsymbol{x}+B\boldsymbol{x}\|}{\|\boldsymbol{x}\|} \\ &\leq \max_{\boldsymbol{x}\neq\boldsymbol{0}} \left(\frac{\|A\boldsymbol{x}\|}{\|\boldsymbol{x}\|} + \frac{\|B\boldsymbol{x}\|}{\|\boldsymbol{x}\|}\right) \\ &\leq \max_{\boldsymbol{x}\neq\boldsymbol{0}} \frac{\|A\boldsymbol{x}\|}{\|\boldsymbol{x}\|} + \max_{\boldsymbol{x}\neq\boldsymbol{0}} \frac{\|B\boldsymbol{x}\|}{\|\boldsymbol{x}\|} \\ &= \|A\| + \|B\|\end{aligned}$$

where the last inequality follows since "the maximum of a sum is less than the sum of the maxima of the summands".

(iii) Finally:

$$\begin{aligned}\|\alpha A\| &= \max_{\boldsymbol{x}\neq\boldsymbol{0}} \frac{\|\alpha A\boldsymbol{x}\|}{\|\boldsymbol{x}\|} \\ &= \max_{\boldsymbol{x}\neq\boldsymbol{0}} \frac{|\alpha|\|A\boldsymbol{x}\|}{\|\boldsymbol{x}\|} = |\alpha| \max_{\boldsymbol{x}\neq\boldsymbol{0}} \frac{\|A\boldsymbol{x}\|}{\|\boldsymbol{x}\|} = |\alpha|\|A\|.\end{aligned}$$

□

So, all operator norms are (matrix) norms.

However, not all matrix norms are operator norms.

Examples of matrix norms not coming from the definition of the operator norm:

The **Frobenius norm** given by

$$\|A\|_F := \left(\sum_{i=1}^{m} \sum_{j=1}^{n} |a_{ij}|^2 \right)^{1/2},$$

which is related to an inner product on $\mathbb{R}^{m\times n}$.

Homework: Write down the formula for the inner product.

The **max-norm** given by

$$\|A\|_{\max} := \max_{1\le i\le m} \max_{1\le j\le n} |a_{ij}|.$$

Obviously, these are $\|\cdot\|_2$ and $\|\cdot\|_\infty$ vector norms on the vector space $\mathbb{R}^{m\times n}$ of matrices.

Lecture 5.

1.3 Stability of the solution

We consider the equation

$$A\boldsymbol{x} = \boldsymbol{b}. \tag{1.29}$$

for an invertible $n \times n$ matrix A with its inverse inverse $A^{-1} \in \mathbb{R}^{n\times n}$.

Let $\delta A \in \mathbb{R}^{n\times n}$ and $\delta\boldsymbol{b} \in \mathbb{R}^n$. The perturbations δA and $\delta\boldsymbol{b}$ model (additive) uncertainties or inaccuracies in A and $\boldsymbol{b}$, respectively.

We consider the perturbed linear equation

$$(A + \delta A)\tilde{\boldsymbol{x}} = \boldsymbol{b} + \delta\boldsymbol{b}. \tag{1.30}$$

We ask how large $\tilde{\boldsymbol{x}} - \boldsymbol{x}$ is in terms of δA and $\delta\boldsymbol{b}$.

We employ the operator and vector norms to study how much the perturbations of A and $\boldsymbol{b}$ in (1.29) affect the corresponding solution $\tilde{\boldsymbol{x}}$ in (1.30).

We assume that $\|\delta A\| \ll \|A\|$ and $\|\delta\boldsymbol{x}\| \ll \|\boldsymbol{x}\|$.

Observe that δ is not here anything as such but only a part of the matrix/vector name in δA, $\delta\boldsymbol{x}$!

A sufficient condition for the solvability of

$$(A + \delta A)\tilde{\boldsymbol{x}} = \boldsymbol{b}$$

assuming that the unperturbed problem

$$A\boldsymbol{x} = \boldsymbol{b}$$

is solvable is given next:

Theorem 1.3.1. *Let $\|\cdot\|$ be an operator norm. If $A \in \mathbb{R}^{n\times n}$ is invertible and $\delta A \in \mathbb{R}^{n\times n}$ satisfies*

$$\|\delta A\| < \frac{1}{\|A^{-1}\|}, \tag{1.31}$$

then $A + \delta A$ is also invertible.

The topological interpretation: The invertible matrices form an open subset in the space of all matrices equipped with some operator topology.

In layman's terms: If a matrix is invertible, then so are all nearby matrices.

Before giving the proof, we must say some words about **Cauchy sequences** and **completeness**.

Let V be a vector space and $\|\cdot\|$ a norm on it. Recall that $(V, \|\cdot\|)$ is known as a **normed vector space**.

In a normed vector space, a sequence $\{x_j\}_{j\in\mathbb{N}}$ is called **Cauchy sequence** if for each $\epsilon > 0$ there exists $N = N_\epsilon \in \mathbb{N}$ such that

$$\min\{j, k\} > N_\epsilon \quad \Rightarrow \quad \|x_j - x_k\| < \epsilon.$$

In general normed vector spaces, some Cauchy sequences converge to a limit vector

$$x = \lim_{j\to\infty} x_j$$

and some others do not.

However, any convergent sequence is a Cauchy sequence in any normed vector space.

The convergence is here understood in the sense of Weierstrass: For all $\epsilon > 0$ there exists $M = M_\epsilon \in \mathbb{N}$ such that

$$j > M_\epsilon \quad \Rightarrow \quad \|x_j - x\| < \epsilon.$$

We say that the normed vector space V is **complete** if all Cauchy sequences converge to a limit. In other words, there are no non-convergent Cauchy sequences in a complete normed vector space.

Complete normed vector spaces are called **Banach spaces**. If their norms are defined in terms of inner products, they are called **Hilbert spaces**.

It can be shown that any **finite-dimensional** vector space V equipped with any vector norm $\|\cdot\|$ is a Banach space.

In particular, $(\mathbb{R}, |\cdot|)$ is a Banach space whereas $(\mathbb{Q}, |\cdot|)$ is not. This is known as the completeness of the real number system $\mathbb{R}$.

You must take some deeper course on analysis or functional analysis to learn details about completeness. Here we use it only as a tool.

Proof. We assume that A is invertible with inverse A^{-1}. Then the perturbed version of it satisfies

$$A + \delta A = A\left(I + A^{-1} \cdot \delta A\right)$$

If (1.31) holds, then

$$\|A^{-1} \cdot \delta A\| \leq \|A^{-1}\| \cdot \|\delta A\| < 1. \tag{1.32}$$

Since the product of two invertible matrices is invertible, it is enough to show that $I - B$ is invertible if $\|B\| < 1$. (Here $B := -A^{-1} \cdot \delta A$.)

For each k define the sum

$$s_k := \sum_{i=0}^{k} B^i = I + B + \ldots + B^{k-1} + B^k.$$

(We make it a notational agreement that always $B^0 = I$ even if $B = 0$.)

Now, the sequence $\{s_k\}$ is a Cauchy sequence in the vector space $\mathbb{R}^{n\times n}$ equipped with the operator norm $\|\cdot\|$. Indeed, for $j<k$ we have

$$\begin{aligned}\|s_k - s_j\| = \left\|\sum_{i=j+1}^{k} B^i\right\| &\leq \sum_{i=j+1}^{k} \left\|B^i\right\| \leq \\ &\leq \sum_{i=j+1}^{k} \|B\|^i = \|B\|^{j+1} \sum_{i=0}^{k-j-1} \|B\|^i \\ &\leq \|B\|^{j+1} \sum_{i=0}^{\infty} \|B\|^i = \frac{\|B\|^{j+1}}{1-\|B\|}\end{aligned}$$

where we used the sum formula of a geometric series $1+q+q^2+\ldots$ with $q=\|B\|<1$.

Now we see that if $\epsilon>0$ is arbitrary, choosing

$$N_\epsilon := \frac{\ln(1-\|B\|)\epsilon}{\ln\|B\|}$$

will quarantee that

$$\min\{j,k\} > N_\epsilon \quad \Rightarrow \quad \|s_j - s_k\| < \epsilon.$$

The vector space $\mathbb{R}^{n\times n}$ is finite-dimensional of dimension n^2. Thus, $(\mathbb{R}^{n\times n}, \|\cdot\|)$ is a complete Banach space, and hence

$$\lim_{k\to\infty} s_k =: s \in \mathbb{R}^{n\times n}.$$

exists for any B with $\|B\| < 1$.

We proceed to show that $(I - B)s = I$, which completes the proof since then $(I-B)$ is invertible and $(I-B)^{-1} = s$ by Corollary 1.1.1.

Using the usual rules of limit computations, we get

$$\begin{aligned}(I - B)s &= (I - B)\lim_{k\to\infty} s_k = \lim_{k\to\infty}(I - B)s_k \\ &= \lim_{k\to\infty}(I - B)\sum_{i=0}^{k} B^i \\ &= \lim_{k\to\infty}(I - B^{k+1}) = I - \lim_{k\to\infty} B^{k+1} = I\end{aligned}$$

since obviously

$$\|B^{k+1}\| \le \|B\|^{k+1} \to 0 \quad \text{as} \quad k \to \infty.$$

□

In fact, the above argument *almost* produces a norm estimate for the inverse of $A + \delta A$ as we shall see next.

We need the **inverse triangle inequality** in the proof of the next corollary.

For any vectors $\boldsymbol{a}, \boldsymbol{b}$, the usual triangle inequality gives

$$\|\boldsymbol{a}\| = \|\boldsymbol{a} - \boldsymbol{b} + \boldsymbol{b}\| \leq \|\boldsymbol{a} - \boldsymbol{b}\| + \|\boldsymbol{b}\|$$

and hence

$$\|\boldsymbol{a}\| - \|\boldsymbol{b}\| \leq \|\boldsymbol{a} - \boldsymbol{b}\|.$$

Similarly,

$$\|\boldsymbol{b}\| - \|\boldsymbol{a}\| \leq \|\boldsymbol{a} - \boldsymbol{b}\|.$$

Combining these two inequalities gives

$$|\|\boldsymbol{a}\| - \|\boldsymbol{b}\|| \leq \|\boldsymbol{a} - \boldsymbol{b}\|.$$

Corollary 1.3.1. *Make the same assumptions as in Theorem 1.3.1. Then*

$$\left\|(A+\delta A)^{-1}\right\| \leq \frac{\|A^{-1}\|}{1-\|\delta A\|\|A^{-1}\|}$$

for any operator norm $\|\cdot\|$.

Proof. We use the same splitting as in Theorem 1.3.1:

$$A+\delta A = A(I-B) \quad \text{with} \quad B := -A^{-1}\cdot\delta A$$

and $\|B\| < 1$.

The "dressing–undressing" principle says that

$$(A+\delta A)^{-1} = (I-B)^{-1}A^{-1}$$

and thus

$$\|\,(A+\delta A)^{-1}\,\| \leq \|(I-B)^{-1}\|\cdot\|A^{-1}\|.$$

So, we need to estimate $\|(I-B)^{-1}\|$.

Now, using the notation of Theorem 1.3.1:

$$(I - B)^{-1} = s = \lim_{k\to\infty} s_k.$$

We need to show first that $\|s\| = \lim_{k\to\infty} \|s_k\|$.

By the inverse triangle inequality

$$|\|s\| - \|s_k\|| \leq \|s - s_k\| \to 0 \quad \text{as} \quad k \to \infty.$$

Thus $\|s_k\| \to \|s\| = \|(I - B)^{-1}\|$ as claimed.

On the other hand, for each k we have

$$\|s_k\| = \|\sum_{i=0}^{k} B^i\| \leq \sum_{i=0}^{k} \|B\|^i \leq \sum_{i=0}^{\infty} \|B\|^i = \frac{1}{1 - \|B\|}.$$

We conclude that $\|(I - B)^{-1}\| \leq \frac{1}{1-\|B\|}$ and hence

$$\left\|(A + \delta A)^{-1}\right\| \leq \frac{\|A^{-1}\|}{1 - \|A^{-1} \cdot \delta A\|}.$$

The claim now follows by observing that

$$\|A^{-1} \cdot \delta A\| \leq \|A^{-1}\| \cdot \|\delta A\| < 1. \qquad \square$$

This is called a **Neumann series argument** due to German mathematician Carl Neumann in the 19th century. Not the same man as John von Neumann in the 20th century.

By subtracting the original linear problem

$$A\boldsymbol{x} = \boldsymbol{b}$$

from its perturbed version

$$(A + \delta A)\tilde{\boldsymbol{x}} = \boldsymbol{b} + \delta\boldsymbol{b}$$

we get an equation satisfied by the **error** $\tilde{\boldsymbol{x}} - \boldsymbol{x}$.

$$(A + \delta A)(\tilde{\boldsymbol{x}} - \boldsymbol{x}) = \delta\boldsymbol{b} - \delta A\boldsymbol{x}. \tag{1.33}$$

The size of the error can be estimated using (1.33).

From now on, we consider the **relative errors** given by

$$\frac{\|\delta\boldsymbol{b}\|}{\|\boldsymbol{b}\|}, \quad \frac{\|\delta A\|}{\|A\|}, \quad \text{and} \quad \frac{\|\tilde{\boldsymbol{x}} - \boldsymbol{x}\|}{\|\boldsymbol{x}\|}.$$

We also introduce the **condition number** of an invertible square matrix A, given by

$$\kappa(A) := \|A\|\|A^{-1}\|.$$

Of course, $\kappa(A)$ depends on the operator norm $\|\cdot\|$ used.

Theorem 1.3.2. *Make the same assumptions as in Theorem 1.3.1. Then*

$$\frac{\|\tilde{\boldsymbol{x}} - \boldsymbol{x}\|}{\|\boldsymbol{x}\|} \leq \frac{\kappa(A)}{1 - \frac{\|\delta A\|}{\|A\|}\kappa(A)} \left(\frac{\|\delta \boldsymbol{b}\|}{\|\boldsymbol{b}\|} + \frac{\|\delta A\|}{\|A\|} \right) \tag{1.34}$$

for any $\|\cdot\|$ operator norm and the corresponding condition number $\kappa(A)$.

Proof. By Theorem 1.3.1, the matrix $A + \delta A \in \mathbb{R}^{n \times n}$ is invertible.

The error $\tilde{\boldsymbol{x}} - \boldsymbol{x}$ can be solved from (1.33) as

$$\tilde{\boldsymbol{x}} - \boldsymbol{x} = (A + \delta A)^{-1}(\delta \boldsymbol{b} - \delta A \boldsymbol{x}).$$

Straightforward estimation

$$\begin{aligned} \|\tilde{\boldsymbol{x}} - \boldsymbol{x}\| &\leq \|(A + \delta A)^{-1}\| \|\delta \boldsymbol{b} - \delta A \boldsymbol{x}\| \\ &\leq \|(A + \delta A)^{-1}\| \left(\|\delta \boldsymbol{b}\| + \|\delta A \boldsymbol{x}\| \right) \end{aligned}$$

by what we have already learned about norm inequalities.

We attack the first term on RHS using Corollary 1.3.1:

$$\|\tilde{\boldsymbol{x}} - \boldsymbol{x}\| \leq \frac{\|A^{-1}\|}{1 - \|\delta A\| \|A^{-1}\|} \left(\|\delta \boldsymbol{b}\| + \|\delta A\| \, \|\boldsymbol{x}\| \right)$$

where the second term on RHS was treated by $\|\delta A \boldsymbol{x}\| \leq \|\delta A\| \, \|\boldsymbol{x}\|$.

Dividing by $\|\boldsymbol{x}\|$ and manipulating the terms horrendously:

$$\begin{aligned}\frac{\|\tilde{\boldsymbol{x}}-\boldsymbol{x}\|}{\|\boldsymbol{x}\|} &\leq \frac{\|A^{-1}\|}{1-\|\delta A\|\|A^{-1}\|}\left(\frac{\|\delta\boldsymbol{b}\|}{\|\boldsymbol{x}\|}+\|\delta A\|\right)\\ &= \frac{\|A\|\|A^{-1}\|}{1-\frac{\|\delta A\|}{\|A\|}\|A^{-1}\|\|A\|}\left(\frac{\|\delta\boldsymbol{b}\|}{\|A\|\|\boldsymbol{x}\|}+\frac{\|\delta A\|}{\|A\|}\right) \qquad (1.35)\\ &= \frac{\kappa(A)}{1-\frac{\|\delta A\|}{\|A\|}\kappa(A)}\left(\frac{\|\delta\boldsymbol{b}\|}{\|A\|\|\boldsymbol{x}\|}+\frac{\|\delta A\|}{\|A\|}\right).\end{aligned}$$

We are almost at the home base now. We just don't like that ugly denominator in our perfectly nice term

$$\frac{\|\delta\boldsymbol{b}\|}{\|A\|\|\boldsymbol{x}\|}$$

that we gun down from above by

$$\frac{\|\delta\boldsymbol{b}\|}{\|A\|\|\boldsymbol{x}\|} \leq \frac{\|\delta\boldsymbol{b}\|}{\|A\boldsymbol{x}\|} = \frac{\|\delta\boldsymbol{b}\|}{\|\boldsymbol{b}\|}$$

since the circuit court judge says $\|A\|\|\boldsymbol{x}\| \geq \|A\boldsymbol{x}\| = \|\boldsymbol{b}\|$.

Now you can get the marshall. □

1.4 Solving $Ax = b$ on a computer

The Lecture Notes contain a section for solving a linear system

$$A\boldsymbol{x} = \boldsymbol{b}$$

by computer using **floating point arithmetics**.

Computers are inherently unable to compute with general real numbers in $\mathbb{R}$ since they are finite (state) machines — not quite even **Turing machines** since the length of their "strip of tape" is bounded by the physical storage capacity.

The finite accuracy of numerical floating point processors is given in terms of the **machine epsilon** $\boldsymbol{u}$ which represents the size of the smallest positive number that can be represented at all.

One way of intuitively understanding floating point numbers is to consider each such number – not as a real number – but an interval of real numbers of length $\boldsymbol{u}$.

Floating point numbers are a number system of some kind. The rules of computation differ from those of $\mathbb{R}$, but unfortunately, not favourably so.

When considering perturbations δA and δb as above related to solving $A\boldsymbol{x} = \boldsymbol{b}$, each of these perturbations may be due to the finite accurary achieved by the floating point arithmetics.

Of course, the data A and $\boldsymbol{b}$ itself can be inherently inaccurate as well. Theorem 1.3.2 applies then, too.

Moreover, in the course of computing $\boldsymbol{x}$ from $A\boldsymbol{x} = \boldsymbol{b}$, e.g., elementary (Gaussian) row operators, by numerous additions and multiplications, roundings take place at each step.

So, it is far from evident that computing a solution of $A\boldsymbol{x} = \boldsymbol{b}$ is even **in principle** possible for reasonably large $n \times n$ matrices A. A practical example involving **Hilbert matrices** is given in the Lecture Notes, and it is a hopeless problem for $n \geq 15$ using the floating point precision of usual computers. At $n = 15$, we are far from running out of computer memory. We run out of accuracy.

The student is invited to read Sections 1.4-1.4.1 in the Lecture Notes independently. About the machine epsilon, a rather good article can be found in English Wikipedia.

This is not a trivial matter.

Lecture 6.

Chapter 2

Eigenvalue problems

We discuss the following themes:

1. Basic eigenvalue theory with emphasis on Hermitian matrices,
2. Similarity transformations,
3. Matrix exponential and its applications.

In Chapter 1, we used exclusively the real vector spaces $\mathbb{R}^n$ and $\mathbb{R}^{m\times n}$. The results given there generalise to the complex vector spaces $\mathbb{C}^n$ and $\mathbb{C}^{m\times n}$ with small modifications.

In complex vector spaces, the inner product is defined with an extra conjugation in it. Thus, the symmetry of the inner product is replaced by the **conjugate symmetry** condition

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \overline{\langle \boldsymbol{y}, \boldsymbol{x} \rangle}. \tag{2.1}$$

The inner product remains linear with respect to its first variable position

$$\langle \alpha \boldsymbol{x}_1 + \beta \boldsymbol{x}_2, \boldsymbol{y} \rangle = \alpha \langle \boldsymbol{x}_1, \boldsymbol{y} \rangle + \beta \langle \boldsymbol{x}_2, \boldsymbol{y} \rangle . \tag{2.2}$$

Instead of bilinearity of the inner product in $\mathbb{R}^n$, Eqs. (2.1)—(2.2) result into **sesquilinearity**: the inner product is **conjugate linear** with respect to the second variable position:

$$\langle \boldsymbol{x}, \alpha \boldsymbol{y}_1 + \beta \boldsymbol{y}_2 \rangle = \overline{\alpha} \langle \boldsymbol{x}, \boldsymbol{y}_1 \rangle + \overline{\beta} \langle \boldsymbol{x}, \boldsymbol{y}_2 \rangle.$$

For vectors and matrices, the symbol $\overline{}$ marks componentwise or elementwise complex conjugation: For example, if $\boldsymbol{x} = \boldsymbol{a} + \mathrm{i}\boldsymbol{b}$ for some $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^n$, then

$$\overline{\boldsymbol{x}} = \boldsymbol{a} - \mathrm{i}\boldsymbol{b}.$$

The usual inner product and 2-norm (corresponding the Euclidean inner product and norm in $\mathbb{R}^n$) are defined in $\mathbb{C}^n$ as

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle := \boldsymbol{x} \cdot \overline{\boldsymbol{y}} = \boldsymbol{y}^* \boldsymbol{x} \qquad \text{and} \qquad \|\boldsymbol{x}\|_2 := (\boldsymbol{x}^* \boldsymbol{x})^{1/2}.$$

Here $*$ denotes the conjugation and transposition combined:

$$\boldsymbol{x}^* := \overline{\boldsymbol{x}}^T \quad \text{or} \quad A^* := \overline{A}^T.$$

The operation $\boldsymbol{x} \mapsto \boldsymbol{x}^*$ or $A \mapsto A^*$ is called **adjoining**.

It is a consequence of the sesquilinear inner product that A^T is usually replaced by A^* in $\mathbb{C}^{m \times n}$.

Almost always Banach and Hilbert spaces are considered as complex vector spaces. Real Banach and Hilbert spaces have little motivation as such.

The eigenvalues and eigenvectors of a **square** matrix $A \in \mathbb{C}^{n\times n}$ are the solutions of the following problem: Find $(\lambda, \boldsymbol{v}) \in \mathbb{C} \times (\mathbb{C}^n \setminus \{0\})$ such that

$$A\boldsymbol{v} = \lambda\boldsymbol{v}. \tag{2.3}$$

We denote the set of all eigenvalues of A by $\sigma(A)$:

$$\sigma(A) := \{\lambda \in \mathbb{C} \mid A\boldsymbol{v} = \lambda\boldsymbol{v} \text{ for some } \boldsymbol{v} \neq \mathbf{0}\}.$$

The set $\sigma(A)$ is called the **spectrum** of A.

It is a part of the definition that **an eigenvector can never be** the zero vector $\mathbf{0}$. Otherwise, any number would be an eigenvalue corresponding to $\mathbf{0}$ which would be empty nonsense or nonsensical emptiness.

We don't do empty nonsense in mathematics, do we?

Why complex numbers?

Even $A \in \mathbb{R}^{2\times 2}$ given by $\left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ has eigenvalues $\pm i \in \mathbb{C}$. So there is no way we can do satisfactory eigenvalue theory without using complex numbers at all.

It is better develop the theory for complex valued vectors $\mathbb{C}^n$ and matrices $\mathbb{C}^{n\times n}$ throughout.

Eigenvalues of matrices are roots (zeroes) of a characteristic polynomial. If matrix A is complex, the coefficients of the characteristic polynomial $p_A(\lambda)$ are (typically) complex-valued. However, the complex numbers $\mathbb{C}$ are an **algebraically closed field** — roots of all polynomials with coefficients in $\mathbb{C}$ are in $\mathbb{C}$.

So, we do not need yet another *non plus ultra* further extension of $\mathbb{C}$ to deal with eigenvalues of $A \in \mathbb{C}^{n\times n}$.

2.1 Basic eigenvalue theory

Rewriting $A\boldsymbol{v} = \lambda\boldsymbol{v}$ for $A \in \mathbb{C}^{n\times n}$ gives

$$(A - \lambda I)\,\boldsymbol{v} = 0.$$

Thus,

$$\lambda \in \sigma(A) \quad \Leftrightarrow \quad \mathrm{N}(A - \lambda I) \neq \{\mathbf{0}\}.$$

By Corollary 1.1.1:

$$\mathrm{N}(A - \lambda I) \neq \{\mathbf{0}\} \quad \Leftrightarrow \quad \det(A - \lambda I) = 0.$$

Definition 2.1.1. *The* **characteristic polynomial** *of a matrix $A \in \mathbb{C}^{n\times n}$ is defined as*

$$p_A(\lambda) = \det(A - \lambda I).$$

Indeed, $p_A(\lambda)$ is a polynomial in λ of degree n. This will be shown in Lemma 2.1.1.

This approach requires solving the roots of a polynomial equation of degree n coming from the determinant condition. It could be quite a challenge if $n > 4$ without using a numerical approach.

When an eigenvalue $\lambda \in \mathbb{C}$ has been found, the related eigenvector(s) are computed by finding a basis for $\mathrm{N}(A - \lambda I)$.

There exists infinitely many eigenvectors for each eigenvalue: the eigenvectors corresponding to an eigenvalue $\lambda \in \sigma(A)$ — together with the null vector $\mathbf{0}$ which is **not** an eigenvector — form a non-trivial subspace of $\mathbb{C}^n$:

$$E_\lambda := \mathrm{N}(A - \lambda I), \quad \lambda \in \sigma(A).$$

The subspace E_λ is called the **eigenspace** of A associated to eigenvalue λ.

The number

$$\mu_G(\lambda) := \dim E_\lambda, \quad 1 \leq \mu_G(\lambda) \leq n$$

is called the **geometric multiplicity** of the eigenvalue λ.

Recall the **subdeterminant rule** that can be used for computing determinants by hand (for example, as a form of a particularly cruel punishment for $n \geq 4$).

The rule can be applied either column-wise,

$$\det A = (-1)^{j-1} \sum_{i=1}^{n} (-1)^{i-1} a_{ij} \det [A]_{i,j},$$

or row-wise,

$$\det A = (-1)^{j-1} \sum_{i=1}^{n} (-1)^{i-1} a_{ji} \det [A]_{j,i}.$$

In the above formulas, j is the index of the column or the row with respect to which the determinant is expanded. Moreover, $[A]_{i,j} \in \mathbb{C}^{(n-1)\times(n-1)}$ is the square matrix that is obtained by removing the ith row and the jth column from the matrix A.

This is obviously a recursive rule.

It suffices to say that not even computers are used for computing determinants this way. The consumption of time and electricity would become excessive.

A couple of examples are in order (not in the same order as in the Lecture Notes).

Example 2.1.1. *Let*

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Then the submatrices are as follows:

$$[A]_{1,1} = \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}, \quad [A]_{1,2} = \begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}, \quad [A]_{1,3} = \begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}.$$

Example 2.1.2. *For*

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 3 & 4 & 0 & 5 \\ 0 & 6 & 7 & 0 \\ 8 & 0 & 0 & 10 \end{bmatrix},$$

the subdeterminant rule with respect to the first row gives

$$\det A = \det \begin{bmatrix} 4 & 0 & 5 \\ 6 & 7 & 0 \\ 0 & 0 & 10 \end{bmatrix} - 2 \det \begin{bmatrix} 3 & 0 & 5 \\ 0 & 7 & 0 \\ 8 & 0 & 10 \end{bmatrix}.$$

Applying the subdeterminant rule next to $[A]_{1,1}$*, we obtain*

$$\det \begin{bmatrix} 4 & 0 & 5 \\ 6 & 7 & 0 \\ 0 & 0 & 10 \end{bmatrix} = 4 \det \begin{bmatrix} 7 & 0 \\ 0 & 10 \end{bmatrix} + 5 \det \begin{bmatrix} 6 & 7 \\ 0 & 0 \end{bmatrix} = 280.$$

The determinant of $[A]_{1,2}$ *is computed recursively according to the same procedure. This gives* $\det [A]_{1,2} = -70$ *and altogether* $\det A = 420$*.*

Example 2.1.3. *Let*

$$A = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}.$$

The eigenvalues of A are the solutions of the equation

$$\begin{aligned} \det(A - \lambda I) = \det \begin{bmatrix} 3-\lambda & -1 \\ 1 & 1-\lambda \end{bmatrix} &= (3-\lambda)(1-\lambda) + 1 \\ &= \lambda^2 - 4\lambda + 4 = (\lambda - 2)^2 = 0. \end{aligned}$$

Hence, there is only one eigenvalue $\lambda = 2$ that is a double root of the characteristic polynomial. The eigenspace corresponding to the eigenvalue $\lambda = 2$ is $E_2 = \mathrm{N}(A - 2I)$.

Because

$$A - 2I = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix},$$

it is easy to deduce that $E_2 = \mathrm{span}\{[1, 1]^T\}$ is a one-dimensional subspace of $\mathbb{R}^2$.

We conclude that $\mu_G(2) = 1$ and the algebraic multiplicity $\mu_A(2) = 2$ as defined below.

Lemma 2.1.1. *If $A \in \mathbb{C}^{n\times n}$, then $p_A(\lambda) = \det(A - \lambda I)$ is a polynomial of degree n. Moreover, the coefficient of the nth order term is $(-1)^n$.*

Proof. The proof is based on induction with respect to the dimension of a square matrix.

Base case: For a general $A \in \mathbb{C}^{2\times 2}$,

$$\begin{aligned}\det(A - \lambda I) &= \det\begin{bmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{bmatrix} \\ &= \lambda^2 - (a_{11} + a_{22})\lambda + (a_{11}a_{22} - a_{12}a_{21})\end{aligned}$$

is clearly a second order polynomial and the coefficient of the term λ^2 is $1 = (-1)^2$.

Induction assumption: For any $B \in \mathbb{C}^{k\times k}$, $\det(B - \lambda I)$ is a polynomial of degree k and the coefficient of the kth order term is $(-1)^k$.

Induction step: Let $A \in \mathbb{C}^{(k+1)\times(k+1)}$ be arbitrary. The subdeterminant rule yields

$$\begin{aligned}&\det(A-\lambda I)\\&=(a_{11}-\lambda)\det([A-\lambda I]_{1,1})\\&+\sum_{i=2}^{k}(-1)^{i-1}a_{i1}\det([A-\lambda I]_{i,1}).\end{aligned}\tag{2.4}$$

By the induction assumption, each subdeterminant in (2.4) is a polynomial of degree k having $(-1)^k$ as the coefficient of the kth order term.

Hence, the sum term on the right-hand side of (2.4) amounts to a polynomial of degree k in λ, and the first term on the right-hand side of (2.4) is a polynomial of degree $k+1$ in λ having $(-1)^{k+1}$ as the coefficient of the $(k+1)$th order term.

Thus, $\det(A-\lambda I)$ is altogether a polynomial of degree $k+1$ having $(-1)^{k+1}$ as the coefficient of the $(k+1)$th order term, which completes the proof. □

Lost with the induction proof? There is a good page in English Wikipedia about "mathematical induction". Look it up now.

By virtue of the **fundamental theorem of algebra**, the characteristic polynomial p_A can be factored as

$$p_A(\lambda) = (-1)^n \prod_{i=1}^{k} (\lambda - \lambda_i)^{\mu_A(\lambda_i)}, \tag{2.5}$$

where $\sum_{i=1}^{k} \mu_A(\lambda_i) = n$ since $p_A(\lambda)$ is of degree n.

The positive integer power $\mu_A(\lambda_i)$ is the **algebraic multiplicity** of the eigenvalue λ_i, i.e., its multiplicity as a root of the characteristic polynomial.

Obviously

$$1 \leq \mu_A(\lambda) \leq n \quad \text{for all} \quad \lambda \in \sigma(A).$$

Lemma 2.1.2. *A matrix $A \in \mathbb{C}^{n\times n}$ has at most n* **distinct** *eigenvalues.*

If counted according to their algebraic multiplicities (i.e., each $\lambda \in \sigma(A)$ counted $\mu_A(\lambda)$ times), then there are **exactly** *n eigenvalues.*

Proof. The eigenvalues of $A \in \mathbb{C}^{n\times n}$ are the roots of the characteristic polynomial p_A.

Because the polynomial p_A is exactly of the order n, it has exactly n roots in $\mathbb{C}$ by the fundamental theorem of algebra.

Some of the roots may coincide, and hence there are at most n distinct roots. □

The following inequality always holds for $A \in \mathbb{C}^{n\times n}$ and $\lambda \in \sigma(A)$:

$$1 \leq \mu_G(\lambda) \leq \mu_A(\lambda) \leq n.$$

Let's not prove it on this course.

However, if $\mu_A(\lambda_i) = 1$, then also $\mu_G(\lambda_i) = \mu_A(\lambda_i) = 1$ because each eigenvalue has at least one-dimensional eigenspace.

The eigenvectors related to different eigenvalues are linearly independent.

Theorem 2.1.1. *Any eigenvectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k \in \mathbb{C}^n$ corresponding to the distinct eigenvalues $\lambda_1, \ldots, \lambda_k \in \mathbb{C}$ of $A \in \mathbb{C}^{n \times n}$ are linearly independent.*

Proof. The proof is based on induction with respect to the number of considered eigenvalues.

Base case (i.e., any pair of eigenvectors are linearly independent):

Without loss of generality, it is enough consider the first two eigenvalues λ_1 and λ_2, $\lambda_1 \neq \lambda_2$, with $\lambda_1 \neq 0$.

(We can reorder the eigenvalues if necessary, so that any two may appear as λ_1 and λ_2.)

We argue by contradiction: Assume that the corresponding eigenvectors $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ are linearly dependent.

Then there exists $0 \neq \alpha \in \mathbb{C}$ such that

$$\boldsymbol{v}_1 = \alpha \boldsymbol{v}_2.$$

Multiplying this equation with A and using the definition of eigenvectors gives

$$\lambda_1 \boldsymbol{v}_1 = \alpha \lambda_2 \boldsymbol{v}_2, \qquad \text{i.e.,} \qquad \boldsymbol{v}_1 = \frac{\lambda_2}{\lambda_1} \alpha \boldsymbol{v}_2.$$

Subtracting the above identities leads to

$$\Big(1 - \frac{\lambda_2}{\lambda_1}\Big) \alpha \boldsymbol{v}_2 = 0.$$

As $\alpha \neq 0$ by assumption and $\boldsymbol{v}_2 \neq 0$ by definition, it must hold that $\lambda_2/\lambda_1 = 1$, which is a contradiction because $\lambda_1 \neq \lambda_2$.

Hence, $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ are linearly independent.

Induction assumption: Assume that any j vectors chosen from the set $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$ are linearly independent $(1 \leq j \leq k-1)$.

Induction step: Consider any $j+1$ vectors chosen from the set $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$. After renumbering, we may assume that they are $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{j+1}$ and that $\lambda_1 \neq 0$.

Let $\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \ldots, \alpha_{j+1}] \in \mathbb{C}^{j+1}$ be such that

$$\sum_{i=1}^{j+1} \alpha_i \boldsymbol{v}_i = 0. \tag{2.6}$$

Multiplying (2.6) by A gives

$$\sum_{i=1}^{j+1} \alpha_i \lambda_i \boldsymbol{v}_i = 0.$$

Hence,

$$\alpha_1 \boldsymbol{v}_1 + \sum_{i=2}^{j+1} \alpha_i \frac{\lambda_i}{\lambda_1} \boldsymbol{v}_i = 0.$$

Subtracting this from (2.6), we get

$$\sum_{i=2}^{j+1} \alpha_i \Big(1 - \frac{\lambda_i}{\lambda_1}\Big) \boldsymbol{v}_i = 0.$$

Now, the following holds with this equation:

1. The j eigenvectors $\boldsymbol{v}_2, \ldots, \boldsymbol{v}_{j+1}$ are linearly independent by the induction assumption.

2. We have $\lambda_i/\lambda_1 \neq 1$ since all these eigenvalues are distinct.

We conclude that $\alpha_i = 0$ for all $i = 2, \ldots, j+1$.

Using this information on α_i's with (2.6), we conclude that also $\alpha_1 = 0$.

We have now shown that $\boldsymbol{\alpha} = 0$. Thus $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{j+1}$ are linearly independent. □

Theorem 2.1.1 implies that there exists n linearly independent eigenvectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ for A if

$$\mu_A(\lambda) = \mu_G(\lambda) \quad \text{for all} \quad \lambda \in \sigma(A) = \{\lambda_1, \ldots, \lambda_k\},$$

When this condition is satisfied, the matrix A is said to be **non-defective**. Otherwise, A is **defective**

Indeed, one can find $\mu_G(\lambda_i)$ linearly independent eigenvectors from each eigenspace E_{λ_i} of A...

... and the eigenvectors corresponding to distinct eigenvalues $\lambda_i \neq \lambda j$, $1 \leq i < j \leq k$ are automatically linearly independent by Theorem 2.1.1...

...and also

$$\sum_{i=1}^{k} \mu_G(\lambda_i) = \sum_{i=1}^{k} \mu_A(\lambda_i) = n.$$

So we have n linearly independent eigenvectors of A as claimed.

In particular, this holds if $\mu_G(\lambda) = \mu_A(\lambda) = 1$ for all eigenvalues.

Assume then that $A \in \mathbb{R}^{n\times n}$ has n linearly independent eigenvectors and stack them as the columns of a matrix:

$$V := \left[\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\right] \in \mathbb{C}^{n\times n}.$$

The matrix product AV gives

$$AV = \left[A\boldsymbol{v}_1, \ldots, A\boldsymbol{v}_n\right] = \left[\lambda_1\boldsymbol{v}_1, \ldots, \lambda_n\boldsymbol{v}_n\right], \tag{2.7}$$

where the eigenvalues are repeated according to their algebraic/geometric multiplicity. If $\Lambda \in \mathbb{C}^{n\times n}$ is the diagonal matrix carrying the eigenvalues of A in the corresponding order, i.e.,

$$\Lambda = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix},$$

then (2.7) can be rewritten as

$$AV = V\Lambda.$$

As V has linearly independent columns, it is invertible (see (2) in Corollary 1.1.1) and we obtain

$$A = V\Lambda V^{-1}. \tag{2.8}$$

Let us recap what we have now reached:

If the matrix $A \in \mathbb{R}^{n\times n}$ is non-defective, i.e., $\mu_G(\lambda) = \mu_A(\lambda)$ for all $\lambda \in \sigma(A)$, then A has n linearly independent eigenvectors that give the **diagonalisation**

$$A = V\Lambda V^{-1}.$$

where $\Lambda \in \mathbb{R}^{n\times n}$ is a diagonal matrix.

In particular, diagonalisation is possible if $A \in \mathbb{R}^{n\times n}$ has n distinct eigenvalues when $\mu_G(\lambda) = \mu_A(\lambda) = 1$ for all $\lambda \in \sigma(A)$,.

If the matrix $A \in \mathbb{R}^{n\times n}$ is defective, i.e., $\mu_G(\lambda) < \mu_A(\lambda)$ for some $\lambda \in \sigma(A)$, then A cannot be represented in the form of (2.1) for any diagonal Λ and invertible V.

A diagonalisation of A into the form (2.1) is never unique. One can change the order of the eigenvalues, the length of the eigenvectors, or even pick different bases for the eigenspaces E_λ satisfying $\mu_G(\lambda) > 1$.

Lecture 7.

2.2 Hermitian matrices

Hermitian matrices are encountered in numerous applications.

Definition 2.2.1. *A matrix* $A \in \mathbb{C}^{n\times n}$ *is* **Hermitian** *if* $A^* = A$.

Obviously, the diagonal elements of a Hermitian matrix must be real.

A **real** matrix is Hermitian if and only if it is **symmetric**:

$$A^T = A.$$

Definition 2.2.2. *A matrix* $U \in \mathbb{C}^{n\times n}$ *is* **unitary** *if it is invertible and* $U^* = U^{-1}$.

A **real** matrix is unitary if and only if it is **orthogonal**:

$$U^T = U^{-1}.$$

The columns (or the rows) of a unitary matrix form an orthonormal basis for $\mathbb{C}^n$.

Indeed, if $U = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n] \in \mathbb{C}^{n\times n}$ is unitary, we have

$$U^*U = \begin{bmatrix} \boldsymbol{u}_1^*\boldsymbol{u}_1 & \boldsymbol{u}_1^*\boldsymbol{u}_2 & \cdots & \boldsymbol{u}_1^*\boldsymbol{u}_n \\ \boldsymbol{u}_2^*\boldsymbol{u}_1 & \boldsymbol{u}_2^*\boldsymbol{u}_2 & \cdots & \boldsymbol{u}_2^*\boldsymbol{u}_n \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{u}_n^*\boldsymbol{u}_1 & \boldsymbol{u}_n^*\boldsymbol{u}_2 & \cdots & \boldsymbol{u}_n^*\boldsymbol{u}_n \end{bmatrix} = I. \tag{2.9}$$

Recall that the eigenvalues of a Hermitian matrix are real, and eigenvectors corresponding to different eigenvalues are orthogonal.

Analogously to this, we show that Hermitian matrix $A \in \mathbb{C}^{n\times n}$ is unitarily diagonalisable:

$$A = Q\Lambda Q^*, \tag{2.10}$$

where $Q \in \mathbb{C}^{n\times n}$ is unitary $Q^* = Q^{-1}$, and $\Lambda \in \mathbb{R}^{n\times n}$ a diagonal matrix with **real** entries.

Lemma 2.2.1. *The eigenvalues of a Hermitian $A \in \mathbb{C}^{n\times n}$ are real.*

Proof. Let $\lambda \in \sigma(A)$ and $\boldsymbol{v} \neq \mathbf{0}$ such that $A\boldsymbol{v} = \lambda\boldsymbol{v}$.

Then

$$\boldsymbol{v}^* A\boldsymbol{v} = \boldsymbol{v}^*(\lambda\boldsymbol{v}) = \lambda\|\boldsymbol{v}\|_2^2.$$

On the other hand, since $A^* = A$,

$$\boldsymbol{v}^* A\boldsymbol{v} = \boldsymbol{v}^* A^*\boldsymbol{v} = (A\boldsymbol{v})^*\boldsymbol{v} = (\lambda\boldsymbol{v})^*\boldsymbol{v} = \overline{\lambda}\|\boldsymbol{v}\|_2^2.$$

Subtracting these identities leads to

$$(\lambda - \overline{\lambda})\|\boldsymbol{v}\|_2^2 = 2\,\mathrm{Im}\lambda\,\|\boldsymbol{v}\|_2^2 = 0.$$

Since $\boldsymbol{v} \neq \mathbf{0}$, we get $\mathrm{Im}\lambda = 0$. □

What an excellent exam question!

We have shown for a general matrix A that eigenvectors corresponding to different eigenvalues are linearly free.

Even more is true for Hermitian matrices:

Lemma 2.2.2. *Let $\boldsymbol{q}_1, \boldsymbol{q}_2 \in \mathbb{C}^n$ be eigenvectors corresponding to eigenvalues $\lambda_1, \lambda_2 \in \mathbb{R}$, $\lambda_1 \neq \lambda_2$, of a Hermitian matrix $A \in \mathbb{C}^{n\times n}$, respectively. Then $\boldsymbol{q}_1$ and $\boldsymbol{q}_2$ are orthogonal in the sense that $\boldsymbol{q}_1^* \boldsymbol{q}_2 = 0$.*

Proof. Since the eigenvalues of a Hermitian matrix are real

$$\lambda_1 \boldsymbol{q}_1^* \boldsymbol{q}_2 = (\lambda_1 \boldsymbol{q}_1)^* \boldsymbol{q}_2 = (A\boldsymbol{q}_1)^* \boldsymbol{q}_2 = \boldsymbol{q}_1^* A^* \boldsymbol{q}_2 = \boldsymbol{q}_1^* A \boldsymbol{q}_2.$$

On the other hand,

$$\lambda_2 \boldsymbol{q}_1^* \boldsymbol{q}_2 = \boldsymbol{q}_1^* (\lambda_2 \boldsymbol{q}_2) = \boldsymbol{q}_1^* A \boldsymbol{q}_2.$$

Subtracting these equalities gives

$$(\lambda_2 - \lambda_1) \boldsymbol{q}_1^* \boldsymbol{q}_2 = 0,$$

which proves the claim since $\lambda_1 \neq \lambda_2$. □

Another perfectly fine exam question!

Now we can prove the unitary diagonalisation of Hermitian matrices again by mathematical induction.

Theorem 2.2.1. *For any Hermitian matrix $A \in \mathbb{C}^{n\times n}$, there exists a unitary $Q \in \mathbb{C}^{n\times n}$ and a diagonal $\Lambda \in \mathbb{R}^{n\times n}$ satisfying*

$$A = Q\Lambda Q^*.$$

Proof. The proof is based on induction with respect to the dimension n of a Hermitian matrix.

Base case ($n = 1$): A Hermitian 1×1 matrix is a single real number, say, $a \in \mathbb{R}$ that can be decomposed as $a = 1\, a\, 1$, i.e., $\Lambda = a$ itself and $Q = 1$.

Induction assumption: The claim is true for $n = k$.

Induction step: Let $A \in \mathbb{C}^{(k+1)\times(k+1)}$ be Hermitian.

It has at least one eigenvalue $\lambda_1 \in \mathbb{R}$ with a corresponding eigenvector $\boldsymbol{q}_1$ satisfying $\|\boldsymbol{q}_1\|_2 = 1$.

Introduce somehow k auxiliary unit vectors $\boldsymbol{q}_2, \ldots, \boldsymbol{q}_{k+1} \in \mathbb{C}^{k+1}$ that are mutually orthogonal as well as orthogonal to $\boldsymbol{q}_1$ so that

$$\boldsymbol{q}_j^* \boldsymbol{q}_k = \begin{cases} 0 \text{ if } j \neq k, \\ 1 \text{ if } j = k. \end{cases}$$

(This can always be done algorithmically by the Gram–Schmidt process considered later.)

Define

$$\widetilde{Q} := \big[\boldsymbol{q}_2, \ldots, \boldsymbol{q}_{k+1}\big] \in \mathbb{C}^{(k+1)\times k} \quad \text{and}$$
$$Q_0 := \big[\boldsymbol{q}_1, \widetilde{Q}\big] \in \mathbb{C}^{(k+1)\times(k+1)}.$$

Obviously, this Q_0 is unitary. Homework!

Now

$$
\begin{aligned}
Q_0^* A Q_0 &= \big[\boldsymbol{q}_1,\, \widetilde{Q}\big]^* A \big[\boldsymbol{q}_1,\, \widetilde{Q}\big] = \begin{bmatrix} \boldsymbol{q}_1^* \\ \widetilde{Q}^* \end{bmatrix} \big[A\boldsymbol{q}_1,\, A\widetilde{Q}\big] \\
&= \begin{bmatrix} \boldsymbol{q}_1^* A \boldsymbol{q}_1 & \boldsymbol{q}_1^* A \widetilde{Q} \\ \widetilde{Q}^* A \boldsymbol{q}_1 & \widetilde{Q}^* A \widetilde{Q} \end{bmatrix} = \begin{bmatrix} \lambda_1 \boldsymbol{q}_1^* \boldsymbol{q}_1 & (\lambda_1 \boldsymbol{q}_1)^* \widetilde{Q} \\ \lambda_1 \widetilde{Q}^* \boldsymbol{q}_1 & \widetilde{Q}^* A \widetilde{Q} \end{bmatrix} \\
&= \begin{bmatrix} \lambda_1 & 0 \\ 0 & \widetilde{Q}^* A \widetilde{Q} \end{bmatrix},
\end{aligned}
$$

where the last equality follows from the fact that $\boldsymbol{q}_1$ is of unit length and orthogonal to the columns of $\widetilde{Q}$.

Now we have to treat the lower right corner in last equation.

Observe that $\widetilde{Q}^* A \widetilde{Q}$ is Hermitian $k \times k$ matrix since A is Hermitian.

By the induction assumption, there exists a unitary $Q_1 \in \mathbb{C}^{k \times k}$ and a diagonal matrix $\Lambda_1 \in \mathbb{R}^{k \times k}$ such that

$$\widetilde{Q}^* A \widetilde{Q} = Q_1 \Lambda_1 Q_1^*.$$

Hence

$$Q_0^* A Q_0 = \begin{bmatrix} \lambda_1 & 0 \\ 0 & Q_1 \Lambda_1 Q_1^* \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & Q_1 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \Lambda_1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & Q_1^* \end{bmatrix},$$

where all factors are $k \times k$ matrices.

Due to the unitarity of Q_0, we have $Q_0^* Q_0 = Q_0 Q_0^* = I$ and

$$A = \left(Q_0 \begin{bmatrix} 1 & 0 \\ 0 & Q_1 \end{bmatrix} \right) \begin{bmatrix} \lambda_1 & 0 \\ 0 & \Lambda_1 \end{bmatrix} \left(Q_0 \begin{bmatrix} 1 & 0 \\ 0 & Q_1 \end{bmatrix} \right)^* \tag{2.11}$$

This completes the proof with $Q := Q_0 \left[\begin{smallmatrix} 1 & 0 \\ 0 & Q_1 \end{smallmatrix} \right]$ since the expression in the parenthesis is unitary:

$$\begin{aligned} \left(Q_0 \begin{bmatrix} 1 & 0 \\ 0 & Q_1 \end{bmatrix} \right)^* \left(Q_0 \begin{bmatrix} 1 & 0 \\ 0 & Q_1 \end{bmatrix} \right) &= \begin{bmatrix} 1 & 0 \\ 0 & Q_1^* \end{bmatrix} Q_0^* Q_0 \begin{bmatrix} 1 & 0 \\ 0 & Q_1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & Q_1^* \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & Q_1 \end{bmatrix} = I, \end{aligned}$$

where we used the unitarity of both Q_0 and Q_1. □

Learn to multiply **block matrices** following the same kind of matrix-matrix product rules as the usual matrix-matrix product.

Since $Q^{-1} = Q^*$, we have

$$\begin{aligned} A = Q\Lambda Q^* &\Leftrightarrow AQ = Q\Lambda \\ &\Leftrightarrow \big[A\boldsymbol{q}_1, \ldots, A\boldsymbol{q}_n\big] = \big[\lambda_1\boldsymbol{q}_1, \ldots, \lambda_n\boldsymbol{q}_n\big]. \end{aligned}$$

The columns of Q are orthonormal eigenvectors for A, and the diagonal elements of $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ are the corresponding eigenvalues repeated according to their algebraic multiplicity.

We just showed that A is diagonalisable. Hence, it is non-defective, and

$$\mu_A(\lambda) = \mu_G(\lambda)$$

for all $\lambda \in \sigma(A)$. So we can just speak about multiplicity.

The orthogonal eigenvectors $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$ form an eigenbasis for $\mathbb{C}^{n\times n}$ (cf. Lemma 1.2.1).

For a Hermitian matrix $A \in \mathbb{C}^{n\times n}$, the eigenvalues can also be characterised by an optimisation problem related to the **Rayleigh quotient**.

Definition 2.2.3. *For a Hermitian matrix $A \in \mathbb{C}^{n\times n}$, the corresponding Rayleigh quotient is defined as*

$$R(A, \boldsymbol{x}) = \frac{\boldsymbol{x}^* A \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}} = \frac{\boldsymbol{x}^* A \boldsymbol{x}}{\|\boldsymbol{x}\|_2^2}.$$

The Rayleigh quotient is real-valued $R(A, \boldsymbol{x}) \in \mathbb{R}$ because the numerator is real:

$$\boldsymbol{x}^* A \boldsymbol{x} = \boldsymbol{x}^* A^* \boldsymbol{x} = (\boldsymbol{x}^* A \boldsymbol{x})^* = \overline{\boldsymbol{x}^* A \boldsymbol{x}}.$$

Since they are real, all eigenvalues of a Hermitian $A \in \mathbb{C}^{n\times n}$ will be arranged in non-decreasing order

$$\lambda_1 \leq \lambda_2 \leq \ldots \leq \lambda_n,$$

where the eigenvalues are repeated according to their multiplicity.

Theorem 2.2.2. *For a Hermitian matrix* $A \in \mathbb{C}^{n\times n}$

$$\lambda_1 = \min_{0\neq \boldsymbol{x}\in\mathbb{C}^n} R(A,\boldsymbol{x}) \qquad \textit{and} \qquad \lambda_n = \max_{0\neq \boldsymbol{x}\in\mathbb{C}^n} R(A,\boldsymbol{x}),$$

where λ_1 *is the smallest eigenvalue of* A *and* λ_n *the largest.*

Proof. As before, let $A = Q\Lambda Q^*$ be a unitary diagonalisation of A where $Q = \big[\boldsymbol{q}_1, \boldsymbol{q}_2, \ldots, \boldsymbol{q}_n\big]$.

Expand an arbitrary $\boldsymbol{x} \in \mathbb{C}^n$ in the orthonormal basis as

$$\boldsymbol{x} = \sum_{i=1}^{n} \alpha_i \boldsymbol{q}_i, \quad \text{where} \quad \alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{C}.$$

By a direct computation

$$\boldsymbol{x}^* A \boldsymbol{x} = \sum_{i=1}^{n} |\alpha_i|^2 \lambda_i \quad \text{and} \quad \boldsymbol{x}^*\boldsymbol{x} = \sum_{j=1}^{n} |\alpha_j|^2.$$

Denote

$$\beta_i := \frac{|\alpha_i|^2}{\sum_{j=1}^{n} |\alpha_j|^2} \quad \text{for all} \quad i = 1, 2, \ldots, n.$$

Then $\beta_i \geq 0$ and $\sum_{i=1}^{n} \beta_i = 1$ for whatever we have as α_i's.

Values of the Rayleigh quotient can be represented as **convex combinations** of eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$

$$R(A, \boldsymbol{x}) = \frac{\boldsymbol{x}^* A \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}} = \sum_{i=1}^{n} \beta_i \lambda_i.$$

Since λ_1 is minimal of the eigenvalues,

$$\sum_{i=1}^{n} \beta_i \lambda_i \geq \sum_{i=1}^{n} \beta_i \lambda_1 \geq \lambda_1 \sum_{i=1}^{n} \beta_i = \lambda_1$$

and also choosing $\beta_1 = 1, \beta_i = 0$ for $i = 2, 3, \ldots, n$ shows that the value is actually attained:

$$\lambda_1 = \min_{0 \neq \boldsymbol{x} \in \mathbb{C}^n} R(A, \boldsymbol{x}).$$

The second part is left as an excercise. □

The other eigenvalues of a Hermitian matrix A can be characterised via

$$\lambda_k = \min_{\dim U = k} \max_{0 \neq \boldsymbol{x} \in U} \frac{\boldsymbol{x}^* A \boldsymbol{x}}{\boldsymbol{x}^* \boldsymbol{x}}, \tag{2.12}$$

where the outer minimisation is over subspaces of dimension k.

This is known as the Courant–Fisher min-max theorem...

...but we will not have it now even though we can anticipate its proof from the proof of Theorem 2.2.2.

Can you modify the proof of Theorem 2.2.2 in terms on convex combinations of λ_i's to prove the theorem of Courant and Fisher?

In particular, can you reason where the $\min_{\dim U = k}$ part in Eq. (2.12) comes from?

Lecture 8.

2.3 Similarity transformations

Recall that the following are equivalent for any matrix $n \times n$ matrix $A \in \mathbb{C}^{n \times n}$:

1. A is diagonalisable.
2. Matrix is **non-defective**, meaning that
$$\mu_A(\lambda) = \mu_G(\lambda) \quad \text{for all} \quad \lambda \in \sigma(A).$$
3. A has n linearly independent eigenvectors.

For **defective** matrices $A \in \mathbb{C}^{n \times n}$ one can still look for some other decomposition of the form

$$A = SBS^{-1}, \tag{2.13}$$

where B is a **more simple matrix** than the original A in some other sense.

These decompositions have great theoretical and practical value in carrying out computations on matrices A.

Definition 2.3.1. *The matrices* $A \in \mathbb{C}^{n\times n}$ *and* $B \in \mathbb{C}^{n\times n}$ *are called* **similar** *if there exists an invertible* $S \in \mathbb{C}^{n\times n}$ *such that*

$$A = SBS^{-1}.$$

Matrix similarity is an **equivalence relation** in the set of all $n \times n$ matrices $\mathbb{C}^{n\times n}$:

1. **Reflexivity**: A is similar with A itself.
2. **Symmetricity**: A is similar with B if and only if B is similar with A.
3. **Transitivity**: If A is similar with B and B is similar with C, then A is similar with C.

Check that the above three conditions hold.

Lemma 2.3.1. *If $A, B \in \mathbb{C}^{n\times n}$ are similar, then their characteristic polynomials are the same, i.e., $p_A = p_B$.*

Proof. If $A = SBS^{-1}$, we get

$$\begin{aligned} p_A(\lambda) &= \det A - \lambda I) = \det\big(SBS^{-1} - \lambda SS^{-1}\big) \\ &= \det\big(S(B-\lambda I)S^{-1}\big) = \det(S)\det(B-\lambda I)\det(S^{-1}) \\ &= \det(S)\det\left(B-\lambda I\right)\det(S)^{-1} \\ &= \det\left(B-\lambda I\right) = p_B(\lambda). \end{aligned}$$

□

An immediate consequence: If matrices A and B are similar, then they have the same eigenvalues $\sigma(A) = \sigma(B)$ with the same algebraic multiplicities.

Also the geometric multiplicities are the same.

Check this from $(A-\lambda I)S = S(B-\lambda I)$ with invertible S.

Definition 2.3.2. *Let* $A \in \mathbb{C}^{n\times n}$ *and*

$$p(s) = p_n s^n + p_{n-1}s^{n-1} + \dots p_1 s + p_0, \quad p_n \neq 0$$

be a polynomial with cofficients in $\mathbb{C}$*. Then the corresponding matrix polynomial is defined as*

$$p(A) = p_n A^n + p_{n-1}A^{n-1} + \dots p_1 A + p_0 I$$

where the powers $A^j = A \cdot A \cdot \dots \cdot A$ *are defined through matrix-matrix product.*

This is a beginning of a **functional calculus**: we can evaluate values of polynomials on square matrices. Later, we will do the exponential function, too.

If matrices A and B are similar, then so are the matrices $p(A)$ and $p(B)$ for any polynomial $p(t)$.

Check that this holds.

For two similar matrices A and B, the powers of B could be much easier to compute than the powers of A. This would happen if

1. B is diagonal.
2. B is upper or lower triangular; i.e., all elements below or above the main diagonal entries are zeroes.
3. B has otherwise a lot of zero elements.

Two $n \times n$ matrices A and B are similar if and only if *they represent the same linear mapping*, say $T : V \to V$, in two different bases of V with $\dim V = n$.

If $A = SBS^{-1}$ represents a linear mapping T in the natural Cartesian basis $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_n\}$, then B represents the same T in the basis defined by the columns of S^{-1}. Indeed, the columns of S^{-1} are the vectors $\{S^{-1}\boldsymbol{e}_1, S^{-1}\boldsymbol{e}_2, \ldots, S^{-1}\boldsymbol{e}_n\}$.

A diagonalisable matrix A is represented by a diagonal matrix Λ in its own eigenbasis.

Further, we have seen that the eigenbasis can be chosen to be orthonormal if A is Hermitian.

The **Schur decomposition** states that any square matrix is unitarily similar to an upper (or lower) triangular matrix.

Theorem 2.3.1. *For any $A \in \mathbb{C}^{n\times n}$, there exist a unitary $Q \in \mathbb{C}^{n\times n}$ and an upper triangular $T \in \mathbb{C}^{n\times n}$ such that*

$$A = QTQ^*. \tag{2.14}$$

Proof. The proof is based on induction with respect to the dimension n. It is very much like the proof of Theorem 2.2.1 above.

Base case ($n = 1$): A general 1×1 matrix is a single complex number, say, $a \in \mathbb{C}$ that can be decomposed as $a = 1\, a\, 1$, i.e., $T = a$ itself and $Q = 1$.

Induction assumption: The claim is true for $n = k$.

Induction step: Let $A \in \mathbb{C}^{(k+1)\times(k+1)}$.

It has at least one eigenvalue $\lambda_1 \in \mathbb{C}$ with a corresponding eigenvector $\boldsymbol{q}_1$ satisfying $\|\boldsymbol{q}_1\|_2 = 1$.

Introduce, again, by some means k auxiliary unit vectors $\boldsymbol{q}_2, \ldots, \boldsymbol{q}_{k+1} \in \mathbb{C}^{k+1}$ that are mutually orthogonal as well as orthogonal to $\boldsymbol{q}_1$.

Define

$$\widetilde{Q} := \left[\boldsymbol{q}_2, \ldots, \boldsymbol{q}_{k+1}\right] \in \mathbb{C}^{(k+1)\times k} \qquad \text{and}$$
$$Q_0 := \left[\boldsymbol{q}_1,\, \widetilde{Q}\right] \in \mathbb{C}^{(k+1)\times(k+1)},$$

Obviously, Q_0 is unitary, again.

Exactly as in the proof of Theorem 2.2.1, we get

$$Q_0^* A Q_0 = \begin{bmatrix} \boldsymbol{q}_1^* A \boldsymbol{q}_1 & \boldsymbol{q}_1^* A \widetilde{Q} \\ \widetilde{Q}^* A \boldsymbol{q}_1 & \widetilde{Q}^* A \widetilde{Q} \end{bmatrix} = \begin{bmatrix} \lambda_1 \boldsymbol{q}_1^* \boldsymbol{q}_1 & \boldsymbol{q}_1^* A \widetilde{Q} \\ \lambda_1 \widetilde{Q}^* \boldsymbol{q}_1 & \widetilde{Q}^* A \widetilde{Q} \end{bmatrix}$$
$$= \begin{bmatrix} \lambda_1 & \boldsymbol{q}_1^* A \widetilde{Q} \\ 0 & \widetilde{Q}^* A \widetilde{Q} \end{bmatrix}.$$

where we used the fact that $\boldsymbol{q}_1$ is a unit vector and orthogonal to the columns of $\widetilde{Q}$.

However, since A is not assumed to be Hermitian, we cannot write $\boldsymbol{q}_1^* A \widetilde{Q} = \mathbf{0}^T$ in the right top corner as we did in Theorem 2.2.1.

Now this is a major difference, bereaving us of all hopes of diagonalisation!

By the induction assumption, there exists an upper triangular $T_1 \in \mathbb{C}^{k \times k}$ and a unitary $Q_1 \in \mathbb{C}^{k \times k}$ such that

$$\widetilde{Q}^* A \widetilde{Q} = Q_1 T_1 Q_1^*$$

since $\widetilde{Q}^* A \widetilde{Q} \in \mathbb{C}^{k \times k}$.

Hence,

$$Q_0^* A Q_0 = \begin{bmatrix} \lambda_1 & \boldsymbol{q}_1^* A \widetilde{Q} \\ 0 & Q_1 T_1 Q_1^* \end{bmatrix} = \begin{bmatrix} \lambda_1 & \boldsymbol{q}_1^* A \widetilde{Q} (Q_1 Q_1^*) \\ 0 & Q_1 T_1 Q_1^* \end{bmatrix}$$
$$= \begin{bmatrix} 1 & 0 \\ 0 & Q_1 \end{bmatrix} \begin{bmatrix} \lambda_1 & \boldsymbol{q}_1^* A \widetilde{Q} Q_1 \\ 0 & T_1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & Q_1^* \end{bmatrix}.$$

Since $T_1 \in \mathbb{C}^{k \times k}$ is upper triangular, so is the middle (block) matrix above, too.

It would be very commendable action that we would now construct the block unitary matrix Q from Q_0 and $\left[\begin{smallmatrix} 1 & 0 \\ 0 & Q_1 \end{smallmatrix}\right]$ exactly as we did in the proof of Theorem 2.2.1.

But then again, commandability is overrated. □

By the Schur decomposition

$$A = QTQ^* = QTQ^{-1}$$

with upper triangular T for any square matrix A; here $Q^* = Q^{-1}$.

So, the Schur decomposition is also a **similarity transformation** in addition to a **congruence transformation** with an upper triangular matrix.

Hence, by Lemma 2.3.1

$$p_A(\lambda) = p_T(\lambda) = \det(T - \lambda I).$$

The determinant of the upper triangular matrix $T - \lambda I$ is the product of its diagonal elements, and the eigenvalues of A are on the diagonal of T repeated as many times as their algebraic multiplicity indicates.

To complete this section, we introduce the **Jordan normal (canonical) form** without proof.

Jordan form relates to Schur for in the following sense: If we do not require the congruence/similarity transformation Q in the Schur decomposition

$$A = QTQ^* = QTQ^{-1}$$

to be unitary (but only invertible), then we can introduce many more zeroes to the upper triangular T.

Given $A \in \mathbb{C}^{n\times n}$, its Jordan canonical form is the similarity transformation

$$A = SJS^{-1} \quad \text{where} \quad S \quad \text{is invertible,}$$

and the "almost diagonal" matrix J is some **Jordan matrix** as defined below.

A $n \times n$ Jordan matrix is any upper triangular, **block diagonal matrix** of the form

$$J := \begin{bmatrix} J_{n_1}(\lambda_1) & & & \\ & J_{n_2}(\lambda_2) & & \\ & & \ddots & \\ & & & J_{n_l}(\lambda_l) \end{bmatrix} \tag{2.15}$$

where the omitted elements (blocks) are all (full of) zeroes. We do **not** assume here that $\lambda_j \neq \lambda_k$ for $j \neq k$.

Take note that $n_1 + n_2 + \cdots + n_l = n$.

Here each $J_{n_j}(\lambda_j) \in \mathbb{C}^{n_j \times n_j}$ is a **Jordan block** related to an eigenvalue λ_j,

$$J_{n_j}(\lambda_j) := \begin{bmatrix} \lambda_j & 1 & & & \\ & \lambda_j & 1 & & \\ & & \ddots & \ddots & \\ & & & \lambda_j & 1 \\ & & & & \lambda_j \end{bmatrix},$$

where the omitted elements are all zeroes.

As in the Schur form, each eigenvalue is repeated according to its algebraic multiplicity in J.

Theorem 2.3.2. *Any $A \in \mathbb{C}^{n\times n}$ has a decomposition*

$$A = SJS^{-1}$$

where $J \in \mathbb{C}^{n\times n}$ is a Jordan matrix, and the matrix $S \in \mathbb{C}^{n\times n}$ is invertible.

Rather than trying to prove this by a technical argument (readily found elsewhere), let us try to understand what the claim means in terms of the algebraic and geometric multiplicities.

Considering a single $n \times n$ Jordan block $J_n(\lambda)$ as a matrix on its own, we have

$$\begin{aligned} &\sigma(J_n(\lambda)) = \{\lambda\}, \\ &\mu_A(\lambda) = n, \quad \text{and} \\ &\mu_G(\lambda) = 1. \quad \text{Easy to check! DIY!} \end{aligned}$$

So, $J_n(\lambda)$ is defective unless $n = 1$.

For $n > 1$, $J_n(\lambda)$ it actually is the **most defective** $n \times n$ **matrix** with the only eigenvalue λ there is since it satisfies

$$1 = \mu_G(\lambda) < \mu_A(\lambda) = n$$

with the maximum gap $n - 1$ possible between $\mu_A(\lambda)$ and $\mu_G(\lambda)$.

Given $A \in \mathbb{C}^{n\times n}$ and $\lambda \in \sigma(A)$, the required multiplicities $\mu_A(\lambda)$ and $\mu_G(\lambda)$ satisfying

$$1 \leq \mu_G(\lambda) \leq \mu_A(\lambda) \leq n$$

can be achieved for Jordan matrix J by playing with sufficiently many Jordan blocks $J_{n_j}(\lambda)$ of different sizes.

If all $n_j = 1$ in Jordan canonical form of A, then we get back the diagonalisation of A.

Given the Jordan form J of A and an eigenvalue $\lambda \in \sigma(A) = \sigma(J)$, we see that

1. $\mu_A(\lambda)$ is the number of times λ appears in the diagonal of J; and

2. $\mu_G(\lambda)$ is the number of distinct Jordan blocks associated to λ.

Homework: Write down all 4×4 Jordan matrices J satisfying $\sigma(J) = \{0\}$ and $\mu_G(0) = 2$. Are they all similar to each other?

2.4 Matrix exponential

Similarity transformations can be used to calculate matrix-valued functions such as polynomials, the square root, or the exponential of a matrix. This piece of work is known as **functional calculus**.

We consider the **matrix exponential** and its application to qualitative analysis of **systems of ordinary differential equations**.

For all $t \in \mathbb{R}$ and $a \in \mathbb{C}$, the exponential function can be defined through its convergent power series expansion

$$\mathrm{e}^{at} = \sum_{j=0}^{\infty} \frac{(at)^j}{j!} \tag{2.16}$$

where $t^0 = 1$ holds as a convention.

It can be shown that this power series can be differentiated termwise

$$\begin{aligned} \frac{d}{dt}\mathrm{e}^{at} &= \sum_{j=0}^{\infty} \frac{d}{dt}\frac{(at)^j}{j!} = \sum_{j=1}^{\infty} \frac{j a^j t^{(j-1)}}{j!} \\ &= a \sum_{j=1}^{\infty} \frac{(at)^{(j-1)}}{(j-1)!} = a \sum_{j=0}^{\infty} \frac{(at)^j}{j} = a\mathrm{e}^{at}. \end{aligned} \tag{2.17}$$

We conclude that the power series representation is consistent with the formula

$$\frac{d}{dt}\mathrm{e}^{at} = a\mathrm{e}^{at}$$

for any $a \in \mathbb{C}$.

It is further basic course knowledge that the initial value problem

$$\begin{cases} x'(t) &= ax(t) \quad \text{for all } t > 0, \\ x(0) &= x_0 \in \mathbb{C}, \end{cases} \tag{2.18}$$

has always a unique solution continuous on $[0, \infty)$ and differentiable on $(0, \infty)$, given by

$$x(t) = \mathrm{e}^{at}x_0 \quad \text{for all } t \geq 0.$$

A relevant question is how this argumentation can be generalised to the initial value problem

$$\begin{cases} \boldsymbol{x}'(t) &= A\boldsymbol{x}(t) \quad \text{for all } t > 0, \\ \boldsymbol{x}(0) &= \boldsymbol{x}_0 \end{cases}$$

where $\boldsymbol{x} : [0, \infty) \to \mathbb{C}^n$ is continuous and $A \in \mathbb{C}^{n\times n}$.

The answer is affirmative.

We **formally** define e^{tA} for $A \in \mathbb{C}^{n\times n}$ by setting

$$\mathrm{e}^{tA} := \sum_{j=0}^{\infty} \frac{(tA)^j}{j!} = \sum_{j=0}^{\infty} \frac{t^j A^j}{j!}, \tag{2.19}$$

where $A^0 = I$ as a convention.

It is not quite obvious whether the series on the right-hand side of (2.19) converges in any reasonable sense.

Let us treat the convergence next.

Define the partial sums

$$S_N(t) := \sum_{j=0}^{N} \frac{t^j A^j}{j!}.$$

For each $t \in \mathbb{R}$, the sequence $\{S_N(t)\}_N$ is a **Cauchy sequence** in (operator) norm $\|\cdot\| \equiv \|\cdot\|_2$.

Indeed, for $M < N$ we have

$$\begin{aligned}
\|S_N(t) - S_M(t)\| &= \Big\| \sum_{j=M+1}^{N} \frac{t^j A^j}{j!} \Big\| \leq \sum_{j=M+1}^{N} \Big\| \frac{t^j A^j}{j!} \Big\| \\
&\leq \sum_{j=M+1}^{N} \frac{(|t| \cdot \|A\|)^j}{j!} \\
&= \sum_{k=0}^{N-(M+1)} \frac{(|t| \cdot \|A\|)^{k+(M+1)}}{(k+(M+1))!} \\
&\leq \frac{(|t| \cdot \|A\|)^{M+1}}{(M+1)!} \sum_{k=0}^{N-(M+1)} \frac{(|t| \cdot \|A\|)^k}{k!} \\
&\leq \frac{(|t| \cdot \|A\|)^{M+1}}{(M+1)!} \sum_{k=0}^{\infty} \frac{(|t|\| \cdot A\|)^k}{k!} \\
&= \frac{(|t|\| \cdot A\|)^{M+1}}{(M+1)!} \mathrm{e}^{|t| \cdot \|A\|} \to 0
\end{aligned}$$

as $M = \min(M, N) \to \infty$ regardless of N.

So, the sequence is Cauchy for any $t \in \mathbb{R}$, and the reader is free to add the epsilons, if need be.

How did you get the estimate on the fourth line?

Why does RHS converge to zero?

As already mentioned, all Cauchy sequences converge to a unique limit in complete spaces.

Since $\mathbb{C}^{n\times n}$ equipped with the operator norm $\|\cdot\|$ is a complete normed vector (Banach) space, the definition

$$\mathrm{e}^{tA} := \lim_{N\to\infty} S_N(t)$$

is **good to go** for all $t \in \mathbb{R}$ and $A \in \mathbb{C}^{n\times n}$.

Observe that for fixed t and N, each partial sum $S_N(t)$ is a matrix polynomial of degree N, say

$$S_N(t) = p_{t,N}(A) = q_N(tA),$$

in the sense of Definition 2.3.2.

One way to learn about Cauchy sequences and completeness is to take the course on **Euclidean spaces**. It is shown there that any finite-dimensional vector space (with scalars either in $\mathbb{R}$ or $\mathbb{C}$), equipped with any (vector) norm, is complete.

Let us return to the system of ordinary differential equations, namely

$$\begin{cases} \boldsymbol{x}'(t) &= A\boldsymbol{x}(t) \quad \text{for all } t > 0, \\ \boldsymbol{x}(0) &= \boldsymbol{x}_0 \end{cases} \tag{2.20}$$

where $A \in \mathbb{C}^{n\times n}$ is the coefficient matrix, $\boldsymbol{x} : \mathbb{R}_+ \to \mathbb{C}^n$ is the solution, and $\boldsymbol{x}_0 \in \mathbb{C}^n$ carries the initial values.

In the series representation of e^{tA}, exactly **the same rules of computation hold** as in the scalar case, even with analogous arguments for their proofs.

Hence, the unique continuous solution of (2.20) is given by

$$\boldsymbol{x}(t) = e^{tA}\boldsymbol{x}_0.$$

That $\boldsymbol{x}(t)$ satisfies the differential equation part, follows because termwise differentiation of the matrix-valued **convergent** power series in t is allowed. The formal argument is exactly the same as in (2.17) with a replaced by A.

It is easy to see that $e^{0\cdot A} = I \in \mathbb{C}^{n\times n}$ for $A \in \mathbb{C}^{n\times n}$. Thus, the initial value condition $\boldsymbol{x}(0) = \boldsymbol{x}_0$ holds, too.

Similarity transformations can be used to simplify the computation of matrix exponentials.

To simplify notation, we use $t = 1$ in e^{tA} without loss of generality.

Let us start from easiest functions, polynomials.

We already saw that for any polynomial $p(s)$

$$A = SBS^{-1} \quad \Rightarrow \quad p(A) = Sp(B)S^{-1}$$

in terms of the matrix polynomials.

Now, we also saw that

$$\mathrm{e}^{A} = \lim_{N \to \infty} q_N(A)$$

for the sequence of polynomials

$$q_N(s) := s + s + \frac{s^2}{2!} + \frac{s^3}{3!} + \ldots + \frac{s^N}{N!}$$

applied on matrices as in Definition 2.3.2.

Now we can compute the limits in operator norm as follows:

$$\begin{aligned} \mathrm{e}^A &= \lim_{N\to\infty} q_N(A) = \lim_{N\to\infty} q_N(SBS^{-1}) \\ &= \lim_{N\to\infty} Sq_N(B)S^{-1} = S\left(\lim_{N\to\infty} q_N(B)\right)S^{-1} \\ &= S\mathrm{e}^B S^{-1}. \end{aligned} \tag{2.21}$$

What holds for matrix polynomials, holds also for the limits (computed in the operator norm) of matrix polynomials!

You can treat similarly **many other functions** than the exponential function, if you only can show that the function is a convergent limit of matrix polynomials.

If A is diagonalisable, then the computation becomes particularly easy.

Let us start with a diagonalisable $A \in \mathbb{C}^{n\times n}$ such that

$$A = S\Lambda S^{-1},$$

where $S \in \mathbb{C}^{n\times n}$ is invertible, and the diagonal matrix $\Lambda \in \mathbb{C}^{n\times n}$ has the n (possibly non-distict) eigenvalues of A, i.e. $\lambda_1, \ldots, \lambda_n$, on its diagonal.

Powers of diagonal matrices are easy to compute:

$$\begin{aligned}\operatorname{diag}(\lambda_1, \ldots, \lambda_n)^j &= \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}^j \\ &= \begin{bmatrix} \lambda_1^j & & & \\ & \lambda_2^j & & \\ & & \ddots & \\ & & & \lambda_n^j \end{bmatrix} = \operatorname{diag}(\lambda_1^j, \ldots, \lambda_n^j)\end{aligned}$$

where the omitted entries are all zeroes.

Further, for polynomials

$$p\left(\operatorname{diag}(\lambda_1, \ldots, \lambda_n)\right) = \operatorname{diag}(p\left(\lambda_1\right), \ldots, p\left(\lambda_n\right))$$

and finally the exponential function

$$\mathrm{e}^{\operatorname{diag}(\lambda_1,\ldots,\lambda_n)} = \operatorname{diag}(\mathrm{e}^{\lambda_1}, \ldots, \mathrm{e}^{\lambda_n}).$$

Example 2.4.1. *The symmetric matrix*

$$A := \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}^{-1}$$

is hereby diagonalised.

So, the matrix exponential of A can be written as

$$e^A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} e^{-1} & 0 \\ 0 & e^3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}^{-1}$$

in diagonalised form.

It would take some computations to carry out the matrix multiplications and inversion required.

The exponential of a non-diagonalisable, defective $A \in \mathbb{C}^{n\times n}$ can be transformed into a more tractable form by resorting to the Jordan normal form as in Theorem 2.3.2.

If A, S and J are as in Theorem 2.3.2, then

$$\mathrm{e}^A = S\mathrm{e}^J S^{-1},$$

as shown in (2.21).

Furthermore, it can be argued rather straightforwardly that the diagonal block decomposition behaves expectedly:

$$\mathrm{e}^J = \begin{bmatrix} \mathrm{e}^{J_{n_1}(\lambda_1)} & & & \\ & \mathrm{e}^{J_{n_2}(\lambda_2)} & & \\ & & \ddots & \\ & & & \mathrm{e}^{J_{n_l}(\lambda_l)}. \end{bmatrix},$$

where the Jordan blocks $J_{n_j}(\lambda_j)$, $j = 1, \ldots, l$, are as in (2.15) above. All omitted entries here are zeroes.

The exponentials $\mathrm{e}^{J_{n_j}(\lambda_j)}$ of the Jordan blocks as well, but we save our strength by not doing this here.

In order to understand the matrix exponential e^A, one simply exploits the similarity of A with the Jordan normal form J. If A is diagonalisable (i.e., non-defective), this reduces to diagonalisation $J = \Lambda$.

Diagonalisation or the Jordan form are theoretical tools that are usually not implemented numerically in practice.

Carrying out even diagonalisation requires solving the full matrix eigenvalue problem associated to A. This is very expensive, and may be even numerically inaccurate.

Numerical software, such as MATLAB, have other algorithms for computing the matrix exponential of a given square matrix, not addressed here.

In MATLAB, the command `xpm(a) }`produces the matrix exponential of `}. The comm`nd `xp(a) }`does something else — what?

Lecture 9.

2.4.1 Qualitative analysis of first-order multivariate differential equations

Even though the matrix exponential has little value in numerical simulations, it is valuable for qualitative understanding about the behavior of a solution to a system of non-linear **ordinary differential equations (ODE)** close to their **equilibrium points**.

Finding equilibrium points and understanding the solution behaviour there reveal much about the nature of the non-linear ODE.

Away from the equilibrium points the ODE trajectories behave in a more expected manner.

We do not consider here ODE's with **external control signals** but only the **autonomous** case

$$\begin{cases} \boldsymbol{x}'(t) &= \boldsymbol{F}(\boldsymbol{x}(t)) \quad \text{for} \;\; t > 0, \\ \boldsymbol{x}(0) &= \boldsymbol{x}_0 \in \mathbb{R}^n, \end{cases}$$

where $\boldsymbol{x} : [0, \infty) \to \mathbb{R}^n$ is the solution, and $\boldsymbol{F} : \mathbb{R}^n \to \mathbb{R}^n$ is a "smooth" function (all second partial derivatives continuous will do). Solutions are also called **trajectories.**)

We also require (But why?) from the solution that

$$\boldsymbol{x}(\cdot) \in C([0, \infty); \mathbb{R}^n) \cap C^1((0, \infty); \mathbb{R}^n).$$

Observe that for $A \in \mathbb{R}^{n\times n}$ the system of linear differential equations

$$\begin{cases} \boldsymbol{x}'(t) &= A\boldsymbol{x}(t) \quad \text{for all } t > 0, \\ \boldsymbol{x}(0) &= \boldsymbol{x}_0 \end{cases}$$

is a special case of

$$\begin{cases} \boldsymbol{x}'(t) &= \boldsymbol{F}(\boldsymbol{x}(t)) \quad \text{for } \ t > 0, \\ \boldsymbol{x}(0) &= \boldsymbol{x}_0 \in \mathbb{R}^n. \end{cases} \tag{2.22}$$

When we say "non-linear", we do not refer to any mathematical property rather than to a possible lack of a property — namely linearity.

We call $\tilde{\boldsymbol{x}} \in \mathbb{R}^n$ an **equilibrium (point)** of (2.22) if

$$\boldsymbol{F}(\tilde{\boldsymbol{x}}) = \mathbf{0}.$$

If a trajectory $\boldsymbol{x}(\cdot)$ satisfies $\boldsymbol{x}(t_0) = \tilde{x}$ for some $t_0 > 0$, then $\boldsymbol{x}'(t_0) = F(\tilde{x}) = \mathbf{0}$. So, the trajectory would never leave an equilibrium, unless someone "kicks it out" from it by some external control action or noise.

For non-linear systems, small (random) perturbations near unstable equilibria can have wildly different long-time consequences.

Take now $\tilde{\boldsymbol{x}} \in \mathbb{R}^n$ such that $\boldsymbol{F}(\tilde{\boldsymbol{x}}) = \boldsymbol{0}$.

We linearise $\boldsymbol{F}(\cdot)$ at the equilibrium point $\tilde{\boldsymbol{x}}$ to study the trajectories $\boldsymbol{x}' = \boldsymbol{F}(\boldsymbol{x})$ passing very near $\tilde{\boldsymbol{x}}$.

By the linear approximation and $\boldsymbol{F}(\tilde{\boldsymbol{x}}) = \boldsymbol{0}$,

$$\boldsymbol{F}(\boldsymbol{x}) \approx \boldsymbol{F}(\tilde{\boldsymbol{x}}) + D\boldsymbol{F}(\tilde{\boldsymbol{x}})(\boldsymbol{x} - \tilde{\boldsymbol{x}}) = D\boldsymbol{F}(\tilde{\boldsymbol{x}})(\boldsymbol{x} - \tilde{\boldsymbol{x}}),$$

where $D\boldsymbol{F}(\tilde{\boldsymbol{x}}) \in \mathbb{R}^{n \times n}$ is the Jacobian matrix of $\boldsymbol{F}$ evaluated at $\tilde{\boldsymbol{x}}$.

By plugging this approximation in (2.22), we get

$$\begin{aligned}
\boldsymbol{x}'(t) &\approx D\boldsymbol{F}(\tilde{\boldsymbol{x}})(\boldsymbol{x}(t) - \tilde{\boldsymbol{x}}) \\
&\Leftrightarrow \\
\frac{d}{dt}\left(\boldsymbol{x}(t) - \tilde{\boldsymbol{x}}\right) &\approx D\boldsymbol{F}(\tilde{\boldsymbol{x}})(\boldsymbol{x}(t) - \tilde{\boldsymbol{x}}) \\
&\Leftrightarrow \\
\boldsymbol{y}'(t) &\approx A\boldsymbol{y}(t) \text{ with } A := D\boldsymbol{F}(\tilde{\boldsymbol{x}})
\end{aligned}$$

by introducing the spatially shifted solution $\boldsymbol{y}(t) := \boldsymbol{x}(t) - \tilde{\boldsymbol{x}}$.

The system of linear ordinary differential equations on the last line is often **accurate enough** for studying the solutions of the original non-linear $\boldsymbol{x}' = \boldsymbol{F}(\boldsymbol{x})$ sufficiently near to the equilibrium $\tilde{\boldsymbol{x}}$.

The above linearisation process leads us to study

$$\boldsymbol{y}'(t) \approx A\boldsymbol{y}(t) \text{ with } A = D\boldsymbol{F}(\tilde{\boldsymbol{x}}) \tag{2.23}$$

for $\boldsymbol{y}(t)$ satisfying $\|\boldsymbol{y}(t)\| \ll 1$ for some intervals of time t.

The behaviour of (2.23) can be understood (even globally) in terms of the matrix exponential solutions

$$\boldsymbol{y}(t) = e^{tA}\boldsymbol{y}(0)$$

This behaviour can be understood in terms of the eigenvalues of $A = D\boldsymbol{F}(\tilde{\boldsymbol{x}})$.

The conclusion is particularly clear if A is diagonalisable.

Example 2.4.2. *Define* $\boldsymbol{F} : \mathbb{R}^2 \to \mathbb{R}^2$ *by*

$$\boldsymbol{F}([x_1, x_2]^T) := \begin{bmatrix} \frac{2}{3}x_1 - \frac{4}{3}x_1x_2 \\ x_1x_2 - x_2 \end{bmatrix}.$$

Consider a **Lotka–Volterra predator/prey model**

$$\begin{cases} \boldsymbol{x}'(t) &= \boldsymbol{F}(\boldsymbol{x}(t)) \quad \text{for } t > 0, \\ \boldsymbol{x}(0) &= \boldsymbol{x}_0 \in \mathbb{R}^n. \end{cases}$$

whose top (differential) equation is written in component form as

$$\begin{cases} x_1'(t) &= \frac{2}{3}x_1(t) - \frac{4}{3}x_1(t)x_2(t), \\ x_2'(t) &= x_1(t)x_2(t) - x_2(t). \end{cases}$$

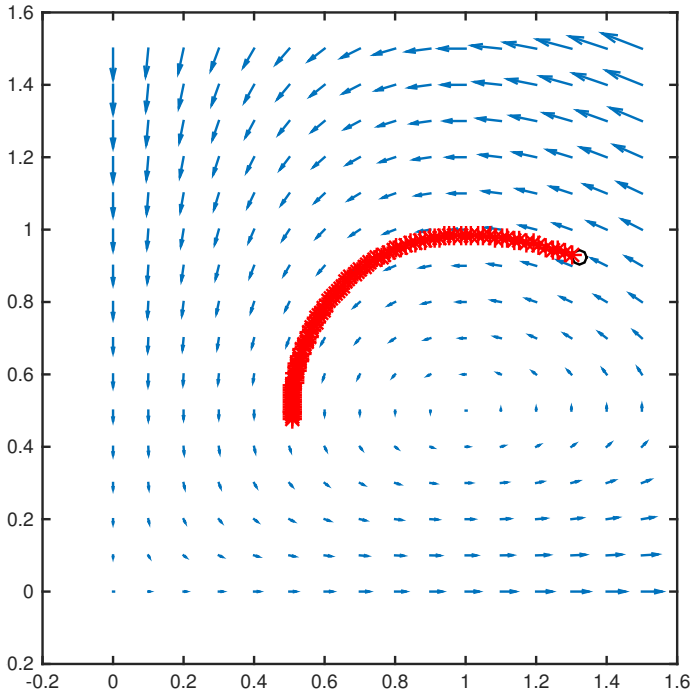
The unknowns x_1 *and* x_2 *are the number of preys and the number of predators, respectively.*

Suppose there are no predators: $x_2(t) = 0$ *for all* t*. Then* $x_1(t) = e^{\frac{2}{3}t}x_1(0)$*, and the prey population will explode.*

Suppose there is no prey: $x_1(t) = 0$ *for all* t*. Then predators will go extinct as* $x_2(t) = e^{-t}x_2(0)$*.*

However, predators eat prey at an intensity proportional to the product of the population sizes $x_1(t)x_2(t)$*. Eating will increase the predator population but decrease the prey population as modelled by the coupling terms.*

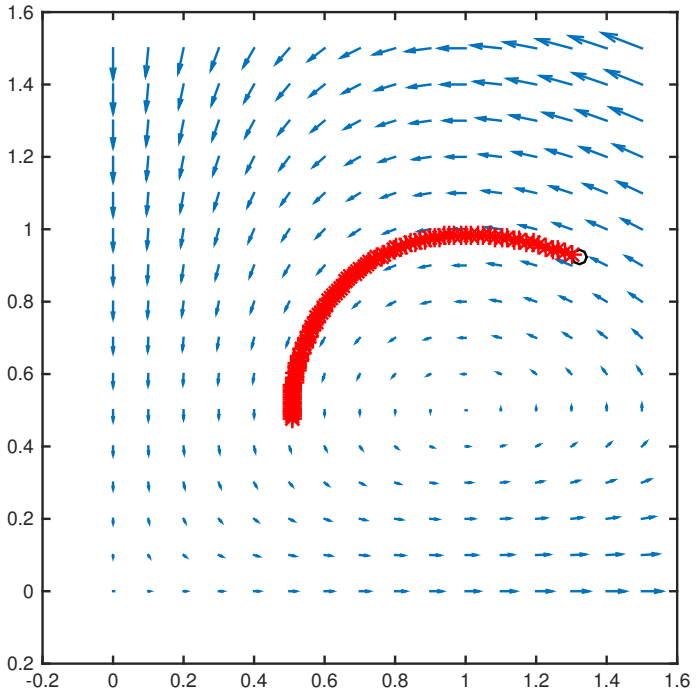
Some understanding about the behavior of the populations can be obtained by studying the Lotka–Volterra system in **phase space**.



Each point $(x_1, x_2) \in [-0.2, 1.6] \times [-0.2, 1.6]$ *represents the concurrent possible sizes of prey and predator populations. (Never mind the negative numbers... be positive for change.)*

Moreover, at each point (x_1, x_2)*, a vector is drawn to illustrate the vector field* $\boldsymbol{F} : \mathbb{R}^2 \to \mathbb{R}^2$.

Trajectories satisfying $\boldsymbol{x}'(t) = \boldsymbol{F}(\boldsymbol{x}(t))$ *have tangents in the direction of this vector field at all points.*



A trajectory starting from the black circle is visualised by the red thick line.

The system has two equilibria, namely $(1, 0.5)$ *and* $(0, 0)$.

At $(x_1, x_0) = (0, 0)$, *the predatory parrot is no more. Very stable. He has ceased to be. Expired and gone to meet its maker. However, the prey is now pushing up the daisies at an exponential rate.*

At $(x_1, x_0) = (1, 0.5)$ *we have reached all sustained development goals. Better not rock the boat, or we all get oscillatory again around the point* $(1, 0.5)$.

A more formal study of the equilibria $\tilde{\boldsymbol{x}}_1 := [1, 0.5]^T$ *and* $\tilde{\boldsymbol{x}}_2 := [0, 0]^T$ *now in column vector notation.*

A direct computation gives the Jacobian matrix we need

$$D\boldsymbol{F}(\boldsymbol{x}) = \begin{bmatrix} \frac{2}{3} - \frac{4}{3}x_2 & -\frac{4}{3}x_1 \\ x_2 & x_1 - 1 \end{bmatrix}.$$

At the equilibria we get

$$A_1 := D\boldsymbol{F}(\tilde{\boldsymbol{x}}_1) = \begin{bmatrix} 0 & -\frac{4}{3} \\ \frac{1}{2} & 0 \end{bmatrix} \quad \textit{and} \quad A_2 := D\boldsymbol{F}(\tilde{\boldsymbol{x}}_2) = \begin{bmatrix} \frac{2}{3} & 0 \\ 0 & -1 \end{bmatrix}.$$

We get for the spectra

$$\sigma(A_1) = \{i\sqrt{2/3}, -i\sqrt{2/3}\} \quad \textit{and} \quad \sigma(A_2) = \{-1, 2/3\}.$$

We **cannot** *conclude from the spectrum that* $\tilde{\boldsymbol{x}}_1$ *is a stable equilibrium. The linearised system has a* **centre or elliptic equilibrium** *here. (In fact, the point* $\tilde{\boldsymbol{x}}_1$ *is a stable equilibrium for the original non-linear system (that has closed trajectories) but we cannot prove it based on a linearisation argument.)*

The equilibrium $\tilde{\boldsymbol{x}}_2$ *is unstable because there is an eigenvalue with real part* > 0*. It is a* **saddle point** *because the real parts are of opposite sign. The predators have the tendency to die, and the prey population has the tendency to explode.*

Chapter 3

Least squares problems

To minimise the effect of measurement errors on the estimated parameters, the measurement is typically repeated several times.

This leads to an overdetermined problem that includes more observations than model parameters.

3.1 Normal equations

Example 3.1.1. *Determination of a spring constant from several noisy measurements.*

There are n weights of masses m_i, $i = 1, \dots, n$.

We measure the elongation x_i of the spring when the weight m_i is attached to it.

We assume an **ideal spring** *with the linear law*

$$x_i = kgm_i, \qquad i = 1, \dots, n, \tag{3.1}$$

where $k \in \mathbb{R}$ is the spring constant.

We assume that the acceleration g in the gravity field is known with practically perfect precision. However, there might be some measurement error in m_i's and in x_i's.

Further, the spring surely fails to be ideal to some extent but we want to model it as such, for simplicity.

So, all of the n equations in (3.1) cannot hold for any k simultaneously.

Instead, one determines (an estimate for) the parameter k by minimizing the 2-norm of the residual.

Let

$$A := \begin{bmatrix} gm_1 \\ gm_2 \\ \vdots \\ gm_n \end{bmatrix} \in \mathbb{R}^{n\times 1} \qquad \textit{and} \qquad \boldsymbol{b} := \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^{n}.$$

Solve k from the optimisation problem

$$k = \arg\min_{k\in\mathbb{R}} \|Ak - \boldsymbol{b}\|_2^2.$$

(The symbol $\arg\min$ *means: Find the argument (i.e., the* **variable** *value(s)) that minimises the following expression. The minimum value itsef is denoted by* $\min$ *as before.)*

It follows that

$$\begin{aligned}\|Ak - \boldsymbol{b}\|_2^2 &= (Ak - \boldsymbol{b})^T(Ak - \boldsymbol{b}) \\ &= (A^TA)k^2 - 2(A^T\boldsymbol{b})k + \|\boldsymbol{b}\|_2^2.\end{aligned}$$

Here all the coefficients A^TA, $2A^T\boldsymbol{b}$, *and* $\|\boldsymbol{b}\|_2^2$ *are scalars.*

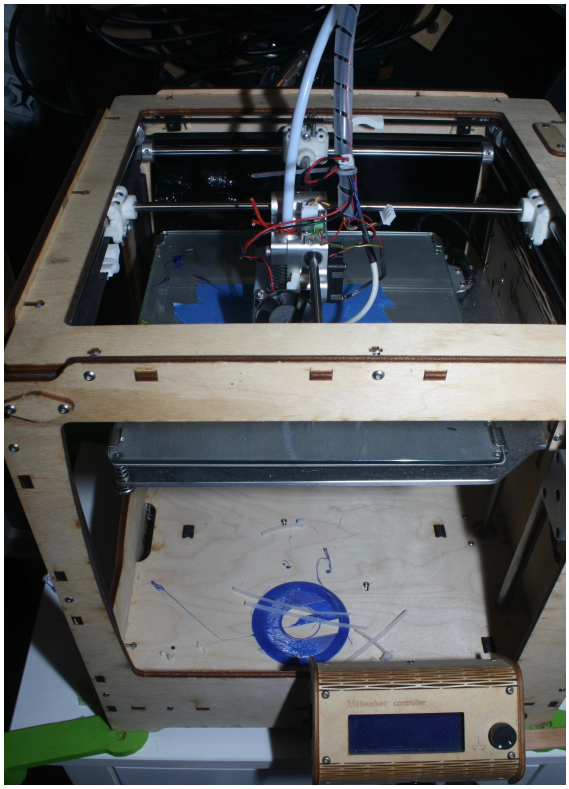
For the minimum, we have

$$\frac{d}{dk}\|Ak - \boldsymbol{b}\|_2^2 = 2(A^TA)k - 2(A^T\boldsymbol{b}) = 0,$$

giving

$$\begin{aligned}k = (A^TA)^{-1}A^T\boldsymbol{b} &= \left(\sum_{j=1}^{n} g^2m_j^2\right)^{-1}\left(\sum_{i=1}^{n} gm_ix_i\right) \\ &= \frac{1}{g}\frac{\sum_{i=1}^{n} m_ix_i}{\sum_{j=1}^{n} m_j^2}.\end{aligned}$$

The next example concerns the 3D printer "Ultimaker Original".



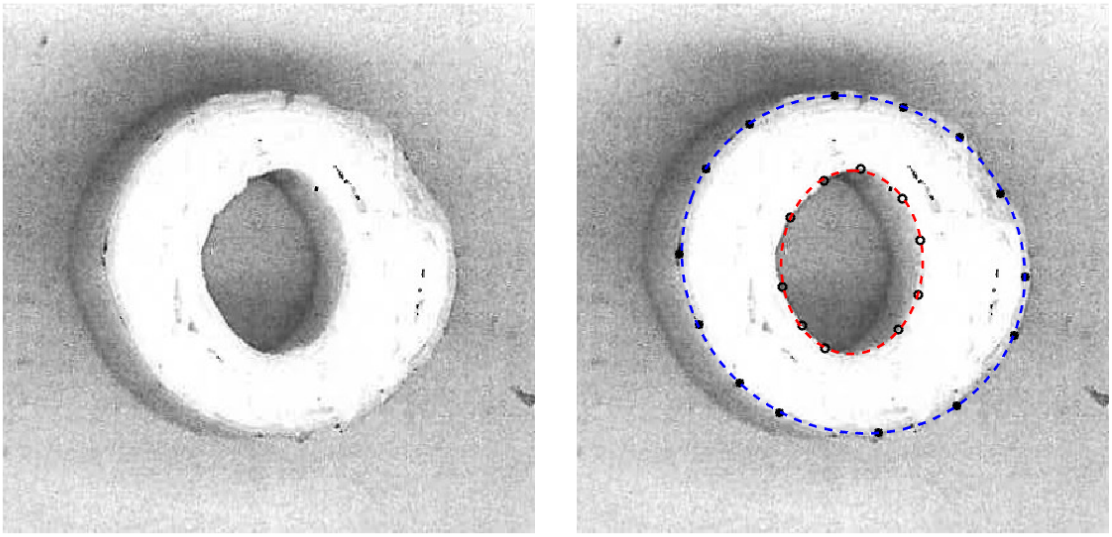
There are backlashes in the printer mechanics, some of then due to **elastic drive belts** from the stepper motors to print head axes.

These non-idealities produce characteristic geometry errors in prints.

The geometry error depends on many things. For example, higher printing speed leads to higher accelerations and forces in the mechanics.

We want to quantify that error and verify its main source.

Ultimaker Original is used to produce **spacers** shown here.



The lack of roundness of the spacers is measured from these images.

Some of the lack of roundness in images is due to the fact that the spacers are more ellipses than circles.

Unfortunately, the camera angle is not right above the spacers. Even if it were, there would be perspective in the images since the camera lense is not **telecentric**. Perspective images of circles are ellipses.

To measure how much the spacers are out of spec, we first match ellipses to a cloud of points that has been manually extracted from the image at the edges of the spacer.

Now we are ready to give our second example.

Example 3.1.2. *Find the ellipse in the photograph that is the* **best fit** *to the given set of points, say* $\boldsymbol{x}^{(i)} \in \mathbb{R}^2$, $i = 1, \ldots, n$.

There are many different ways to understand what "best fit" means.

We consider an ellipse through its implicit equations in $\mathbb{R}^2$*:*

$$\big\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + Ax_2^2 + Bx_1x_2 + Cx_1 + Dx_2 + E = 0, \\ A, B, C, D, E \in \mathbb{R}\big\},$$

In fact, this equation describes general conic sections (also parabolas and hyperbolas), and ellipses are obtained if $B^2 - 4A < 0$.

Surely our data is good enough so that the best fit will automatically be an ellipse. So, we need not consider $B^2 - 4A < 0$.

Because the x_1x_2 *term is present, the semiaxes need not be in the directions of coordinate axes.*

To find the best fit within conic sections, we look for real constants A, B, C, D and E that minimise the total fitting error in terms of square sums

$$\begin{aligned}&\mathcal{E}(A,B,C,D,E)\\&:=\sum_{i=1}^{n}\left(\left(x_1^{(i)}\right)^2+A\left(x_2^{(i)}\right)^2+Bx_1^{(i)}x_2^{(i)}+Cx_1^{(i)}+Dx_2^{(i)}+E\right)^2.\end{aligned}$$

Why do we use the square sum? Why not the sum of absolute values?

Writing

$$\boldsymbol{\alpha}=[A,B,C,D,E]^T\in\mathbb{R}^5$$

we can write the total fitting error $\mathcal{E}(A,B,C,D,E)$ in terms of matrices.

Indeed, for each point (x_i,y_i) in the point cloud, we have

$$\begin{aligned}&\left(x_1^{(i)}\right)^2+A\left(x_2^{(i)}\right)^2+Bx_1^{(i)}x_2^{(i)}+Cx_1^{(i)}+Dx_2^{(i)}+E\\&=\left[\left(x_2^{(i)}\right)^2,x_1^{(i)}x_2^{(i)},x_1^{(i)},x_2^{(i)},1\right]\boldsymbol{\alpha}+\left(x_1^{(i)}\right)^2.\end{aligned}$$

Now, we have one such equation for each point (x_i, y_i) in the point cloud.

We stack them to the matrix A and the vector $\boldsymbol{b}$ as follows:

$$A := \begin{bmatrix} \left(x_2^{(1)}\right)^2 & x_1^{(1)}x_2^{(1)} & x_1^{(1)} & x_2^{(1)} & 1 \\ \left(x_2^{(2)}\right)^2 & x_1^{(2)}x_2^{(2)} & x_1^{(2)} & x_2^{(2)} & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \left(x_2^{(n)}\right)^2 & x_1^{(n)}x_2^{(n)} & x_1^{(n)} & x_2^{(n)} & 1 \end{bmatrix} \quad \textit{and} \quad \boldsymbol{b} = - \begin{bmatrix} \left(x_1^{(1)}\right)^2 \\ \left(x_1^{(2)}\right)^2 \\ \vdots \\ \left(x_1^{(n)}\right)^2 \end{bmatrix}.$$

With this notation,

$$\mathcal{E}(A, B, C, D, E) = \|A\boldsymbol{\alpha} - \boldsymbol{b}\|_2^2.$$

We are now against the following **least squares optimisation problem***:*

$$\boldsymbol{\alpha} = \arg\min_{\boldsymbol{\alpha} \in \mathbb{R}^5} \|A\boldsymbol{\alpha} - \boldsymbol{b}\|_2^2.$$

There is a general recipe for all linear least squares problems, and the rest of this course is just about that.

Suppose that the system of linear equations

$$A\boldsymbol{x} = \boldsymbol{b} \quad \text{for} \quad A \in \mathbb{R}^{m\times n} \quad \text{and} \quad \boldsymbol{b} \in \mathbb{R}^m \tag{3.2}$$

is either "badly solvable" or not solvable at all.

For example, the matrix A and the vector $\boldsymbol{b}$ may be contaminated by measurement noise so that $\boldsymbol{b} \notin \mathrm{R}(A)$.

To deal with this situation, we need to consider "non-solutions" of a non-solvable system $A\boldsymbol{x} = \boldsymbol{b}$ which are — however — some kind of generalised solutions useful enough.

Motivated by the preceding two examples, we define a **least squares (LSQ) solution** of the linear system $A\boldsymbol{x} = \boldsymbol{b}$ as follows:

Definition 3.1.1. *A vector* $\boldsymbol{x} \in \mathbb{R}^n$ *is called a* **least squares solution** *of* (3.2) *if it (globally) minimises the square sum fitting error*

$$\mathcal{E}(\boldsymbol{x}) := \|A\boldsymbol{x} - \boldsymbol{b}\|_2^2. \tag{3.3}$$

It turns out that minimising (3.3) is equivalent to solving the **normal equation**

$$A^T A\boldsymbol{x} = A^T\boldsymbol{b}.$$

The normal equation has always at least one solution since $\mathrm{R}(A^TA) = \mathrm{R}(A^T)$. Why?

A least squares solution need not be unique. Why not?

Obviously $\mathrm{N}(A) \subset \mathrm{N}(A^TA)$, but also $\mathrm{N}(A^TA) \subset \mathrm{N}(A)$. For $\boldsymbol{z} \in \mathrm{N}(A^TA)$ we have$0 = \boldsymbol{z}^T(A^TA\boldsymbol{z}) = \|A\boldsymbol{z}\|_2^2$, i.e., $\boldsymbol{z} \in \mathrm{N}(A)$.

If (3.2) is solvable in the usual sense, then any of the solutions $\boldsymbol{x}$ is a least squares solution as well. And there are no others as we shall see next.

Theorem 3.1.1. *Let $A \in \mathbb{R}^{m\times n}$ and $\boldsymbol{b} \in \mathbb{R}^m$. Then $\boldsymbol{x} \in \mathbb{R}^n$ satisfies*

$$\boldsymbol{x} = \arg\min_{\boldsymbol{x}\in\mathbb{R}^m} \|A\boldsymbol{x} - \boldsymbol{b}\|_2^2.$$

if and only if

$$A^T A\boldsymbol{x} = A^T\boldsymbol{b}. \tag{3.4}$$

Proof. **If -part:**

Let $\boldsymbol{x} \in \mathbb{R}^n$ be a solution of the normal equations (3.4).

Write for any $\tilde{\boldsymbol{x}} \in \mathbb{R}^n$

$$\tilde{\boldsymbol{x}} = \boldsymbol{x} + \boldsymbol{z}.$$

Then

$$\begin{aligned}
&\|A\tilde{\boldsymbol{x}} - \boldsymbol{b}\|_2^2 \\
&= \|A(\boldsymbol{x} + \boldsymbol{z}) - \boldsymbol{b}\|_2^2 = \|(A\boldsymbol{x} - \boldsymbol{b}) + A\boldsymbol{z}\|_2^2 \\
&= \big((A\boldsymbol{x} - \boldsymbol{b}) + A\boldsymbol{z}\big)^T\big((A\boldsymbol{x} - \boldsymbol{b}) + A\boldsymbol{z}\big) \\
&= \|A\boldsymbol{x} - \boldsymbol{b}\|_2^2 + (A\boldsymbol{x} - \boldsymbol{b})^T A\boldsymbol{z} + (A\boldsymbol{z})^T(A\boldsymbol{x} - \boldsymbol{b}) + \|A\boldsymbol{z}\|_2^2 \\
&= \|A\boldsymbol{x} - \boldsymbol{b}\|_2^2 + 2\boldsymbol{z}^T(A^T A\boldsymbol{x} - A^T\boldsymbol{b}) + \|A\boldsymbol{z}\|_2^2 \\
&= \|A\boldsymbol{x} - \boldsymbol{b}\|_2^2 + \|A\boldsymbol{z}\|_2^2 \geq \|A\boldsymbol{x} - \boldsymbol{b}\|_2^2.
\end{aligned}$$

So, $\boldsymbol{x}$ is a (global) minimum of $\|A\boldsymbol{x} - \boldsymbol{b}\|_2^2$.

Only if -part:

Let $\boldsymbol{x} \in \mathbb{R}^n$ be some solution of the normal equations (3.4) (and we already know it exists!).

This time, let $\tilde{\boldsymbol{x}}$ be a **least squares solution** that minimises $\|A\boldsymbol{x} - \boldsymbol{b}\|_2^2$.

We again use this equation for defining $\boldsymbol{z}$:

$$\tilde{\boldsymbol{x}} = \boldsymbol{x} + \boldsymbol{z}.$$

This time, there is an equality $=$ in place of the inequality $\geq$ in what we just computed:

$$\|A\tilde{\boldsymbol{x}} - \boldsymbol{b}\|_2^2 = \|A\boldsymbol{x} - \boldsymbol{b}\|_2^2 + \|A\boldsymbol{z}\|_2^2 = \|A\boldsymbol{x} - \boldsymbol{b}\|_2^2.$$

So, $\|A\boldsymbol{z}\|_2 = 0 \quad \Rightarrow \quad A\boldsymbol{z} = \boldsymbol{0} \quad \Leftrightarrow \quad \boldsymbol{z} \in \mathrm{N}(A)$.

Hence, for $\tilde{\boldsymbol{x}} = \boldsymbol{x} + \boldsymbol{z}$ we have

$$A^T A\tilde{\boldsymbol{x}} = A^T A(\boldsymbol{x} + \boldsymbol{z}) = A^T A\boldsymbol{x} + A^T(A\boldsymbol{z}) = A^T\boldsymbol{b}$$

which completes the proof. □

Lecture 10.

(From 2018 version.)

3.2 Projection matrices

Projection matrices are related to decompositions of a given vector $\boldsymbol{x} \in \mathbb{R}^n$ into two parts. To this end, let $\mathcal{V}$ and $\mathcal{W}$ be subspaces of $\mathbb{R}^n$ and assume that

$$\mathcal{V} \cap \mathcal{W} = \{0\} \qquad \text{and} \qquad \mathcal{V} + \mathcal{W} = \mathbb{R}^n, \tag{3.5}$$

where

$$\mathcal{V} + \mathcal{W} = \{\boldsymbol{v} + \boldsymbol{w} \mid \boldsymbol{v} \in \mathcal{V} \text{ and } \boldsymbol{w} \in \mathcal{W}\}.$$

This means that $\mathbb{R}^n$ is the **direct sum** of $\mathcal{V}$ and $\mathcal{W}$, which is expressed as $\mathbb{R}^n = \mathcal{V} \oplus \mathcal{W}$.

Lemma 3.2.1. *If $\mathbb{R}^n = \mathcal{V} \oplus \mathcal{W}$, i.e. (3.5) is valid, then any $\boldsymbol{x}$ can be decomposed* **uniquely** *as*

$$\boldsymbol{x} = \boldsymbol{v} + \boldsymbol{w},$$

where $\boldsymbol{v} \in \mathcal{V}$ and $\boldsymbol{w} \in \mathcal{W}$.

Proof. Due to (3.5), it is obvious that any $\boldsymbol{x} \in \mathbb{R}^n$ has the required decomposition, and so we only need to worry about its uniqueness. Let $\boldsymbol{v}, \tilde{\boldsymbol{v}} \in \mathcal{V}$ and $\boldsymbol{w}, \tilde{\boldsymbol{w}} \in \mathcal{W}$ be such that

$$\boldsymbol{v} + \boldsymbol{w} = \boldsymbol{x} = \tilde{\boldsymbol{v}} + \tilde{\boldsymbol{w}}.$$

Hence,

$$\mathcal{V} \ni \boldsymbol{v} - \tilde{\boldsymbol{v}} = \tilde{\boldsymbol{w}} - \boldsymbol{w} \in \mathcal{W},$$

and by the first condition in (3.5), it must hold that

$$\boldsymbol{v} - \tilde{\boldsymbol{v}} = \tilde{\boldsymbol{w}} - \boldsymbol{w} = 0,$$

as the zero vector is the only common element of $\mathcal{V}$ and $\mathcal{W}$. In other words, $\boldsymbol{v} = \tilde{\boldsymbol{v}}$ and $\boldsymbol{w} = \tilde{\boldsymbol{w}}$, and thus the decomposition must be unique. □

The projections, or projection matrices $P_{\mathcal{V}}, P_{\mathcal{W}} \in \mathbb{R}^{n\times n}$ associated to a decomposition $\mathbb{R}^n = \mathcal{V} \oplus \mathcal{W}$ are defined via

$$P_{\mathcal{V}}\boldsymbol{x} = \boldsymbol{v} \qquad \text{and} \qquad P_{\mathcal{W}}\boldsymbol{x} = \boldsymbol{w},$$

where $\boldsymbol{v} \in \mathcal{V}$ and $\boldsymbol{w} \in \mathcal{W}$ are the unique vectors such that $\boldsymbol{x} = \boldsymbol{v} + \boldsymbol{w}$. It would be straightforward to prove that such matrices exist, i.e., that the functions sending $\boldsymbol{x}$ to its unique components in $\mathcal{V}$ and $\mathcal{W}$ are linear mappings. However, we take a more constructive path and find explicit representations for $P_{\mathcal{V}}$ and $P_{\mathcal{W}}$.

Let $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$ and $\{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_l\}$ be basis for $\mathcal{V}$ and $\mathcal{W}$, that is, these sets of vectors are linearly independent and

$$\mathcal{V} = \operatorname{span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\} \qquad \text{and} \qquad \mathcal{W} = \operatorname{span}\{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_l\}.$$

According to (3.5), any vector in $\mathbb{R}^n$ can be presented as a joint linear combination of $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$ and $\{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_l\}$. Let us prove that the joint set of vectors $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k, \boldsymbol{w}_1, \ldots, \boldsymbol{w}_l\}$ is also linearly

independent: Let $\boldsymbol{\alpha} \in \mathbb{R}^{k+l}$ be such that

$$\sum_{i=1}^{k} \alpha_i \boldsymbol{v}_i + \sum_{j=1}^{l} \alpha_{k+j} \boldsymbol{w}_j = 0$$

$$\Leftrightarrow \qquad \mathcal{V} \ni \sum_{i=1}^{k} \alpha_i \boldsymbol{v}_i = -\sum_{j=1}^{l} \alpha_{l+k} \boldsymbol{w}_l \in \mathcal{W},$$

where both sides must vanish by virtue of (3.5). Because $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$ and $\{\boldsymbol{w}_1, \ldots, \boldsymbol{w}_l\}$ are separately linearly independent, this means that the whole vector $\boldsymbol{\alpha} \in \mathbb{R}^{k+l}$ must vanish.

Hence, $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k, \boldsymbol{w}_1, \ldots, \boldsymbol{w}_l\}$ is a set of linearly independent vectors that span $\mathbb{R}^n$, i.e., they form a basis for $\mathbb{R}^n$. In particular, their total number $k + l$ must equal the dimension of $\mathbb{R}^n$; in the following, we write $l = n - k$.

We stack the two bases as columns of the matrices $W \in \mathbb{R}^{n \times k}$ and $V \in \mathbb{R}^{n \times (n-k)}$, respectively:

$$V = [\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k] \qquad \text{and} \qquad W = [\boldsymbol{w}_1, \ldots, \boldsymbol{w}_{n-k}].$$

With this notation, the two subspaces can be expressed as $\mathcal{V} = \mathrm{R}(V)$ and $\mathcal{W} = \mathrm{R}(W)$; see (1.5). The decomposition of $\mathbb{R}^n$ related to $\mathcal{V}$ and $\mathcal{W}$ is thus

$$\boldsymbol{x} = \boldsymbol{v} + \boldsymbol{w} = V\boldsymbol{z}_V + W\boldsymbol{z}_W = [V,\, W] \begin{bmatrix} \boldsymbol{z}_V \\ \boldsymbol{z}_W \end{bmatrix}$$

for some 'coordinate vectors' $\boldsymbol{z}_V \in \mathbb{R}^k$ and $\boldsymbol{z}_W \in \mathbb{R}^{n-k}$. Because the columns of $[W,\, V] \in \mathbb{R}^{n \times n}$ are a (linearly independent) basis for

$\mathbb{R}^n$, the matrix $[W,\, V]$ is invertible. Hence,

$$\begin{bmatrix} \boldsymbol{z}_V \\ \boldsymbol{z}_W \end{bmatrix} = [V,\, W]^{-1}\boldsymbol{x}.$$

On the other hand,

$$\boldsymbol{z}_V = [I,\, 0] \begin{bmatrix} \boldsymbol{z}_V \\ \boldsymbol{z}_W \end{bmatrix} \qquad \text{and} \qquad \boldsymbol{z}_W = [0,\, I] \begin{bmatrix} \boldsymbol{z}_V \\ \boldsymbol{z}_W \end{bmatrix},$$

where $I \in \mathbb{R}^{k\times k}$, $0 \in \mathbb{R}^{k\times(n-k)}$ in the first formula and $I \in \mathbb{R}^{(n-k)\times(n-k)}$, $0 \in \mathbb{R}^{(n-k)\times k}$ in the second one. Altogether, we have thus deduced that

$$P_{\mathcal{V}}\boldsymbol{x} = V\boldsymbol{z}_V = V[I,\, 0] \begin{bmatrix} \boldsymbol{z}_V \\ \boldsymbol{z}_W \end{bmatrix} = [V,\, 0][V,\, W]^{-1}\boldsymbol{x}, \tag{3.6}$$

where the second 0 belongs to $\mathbb{R}^{n\times(n-k)}$. Similarly,

$$P_{\mathcal{W}}\boldsymbol{x} = W\boldsymbol{z}_W = W[0,\, I] \begin{bmatrix} \boldsymbol{z}_V \\ \boldsymbol{z}_W \end{bmatrix} = [0,\, W][V,\, W]^{-1}\boldsymbol{x}, \tag{3.7}$$

where the latter 0 belongs to $\mathbb{R}^{n\times k}$. In other words, $P_{\mathcal{V}} = [V,\, 0][V,\, W]^{-1}$ and $P_{\mathcal{W}} = [0,\, W][V,\, W]^{-1}$, where the zeros denote zero matrices of appropriate sizes.

By construction, the matrices $P_{\mathcal{V}}$ and $P_{\mathcal{W}}$ must satisfy

$$P_{\mathcal{V}}^2 = P_{\mathcal{V}} \qquad \text{and} \qquad P_{\mathcal{W}}^2 = P_{\mathcal{W}}.$$

Indeed, e.g., $P_{\mathcal{V}}\boldsymbol{x}$ is already an element of $\mathcal{V}$, and so its unique decomposition of the form (3.5) must be $P_{\mathcal{V}}\boldsymbol{x} = P_{\mathcal{V}}\boldsymbol{x}+0$, i.e., $P_{\mathcal{V}}^2\boldsymbol{x} =$

$P_{\mathcal{V}}(P_{\mathcal{V}}\boldsymbol{x}) = P_{\mathcal{V}}\boldsymbol{x}$ for all $\boldsymbol{x} \in \mathbb{R}^n$. This could also be verified by a direct calculation based on (3.6). In other words, the projection matrices $P_{\mathcal{V}}$ and $P_{\mathcal{W}}$ are identity maps on their respective ranges. This is a fundamental property that can be taken as the definition of a projection (matrix).

Definition 3.2.1. *A matrix $P \in \mathbb{R}^{n\times n}$ is a projection if it satisfies*

$$P^2 = P. \tag{3.8}$$

Each projection matrix is related to a decomposition of the space $\mathbb{R}^n$ into two parts. Let $P \in \mathbb{R}^{n\times n}$ be a projection and consider the decomposition

$$\boldsymbol{x} = P\boldsymbol{x} + (I - P)\boldsymbol{x}, \tag{3.9}$$

that is, $\mathcal{V} = \mathrm{R}(P)$ and $\mathcal{W} = \mathrm{R}(I-P)$. Due to the property $P = P^2$ it is easy to see that these two subspaces satisfy the conditions (3.5): The second one is already proven by (3.9). On the other hand, if $\boldsymbol{x}$ belongs to both $\mathrm{R}(P)$ and $\mathrm{R}(I-P)$, it can be given as $\boldsymbol{x} = (I-P)\boldsymbol{z}$ for some $\boldsymbol{z}$, but as P is a projection, it also holds that $P\boldsymbol{x} = \boldsymbol{x}$. Putting these together,

$$\boldsymbol{x} = P\boldsymbol{x} = P(I - P)\boldsymbol{z} = (P - P^2)\boldsymbol{z} = (P - P)\boldsymbol{z} = 0,$$

i.e., the zero vector is the only common element of $\mathrm{R}(P)$ and $\mathrm{R}(I - P)$. Note that $I - P$ is also a projection:

$$(I - P)(I - P) = I - P - P + P^2 = I - 2P + P = I - P,$$

where we once again utilised the identity $P^2 = P$.

A special projection is related to subspaces $\mathcal{V}$ and $\mathcal{W}$ that are orthogonal.

Definition 3.2.2. *Let $\mathcal{V}$ and $\mathcal{W}$ be subspaces of $\mathbb{R}^n$ and $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ be some inner product. The subspaces $\mathcal{V}$ and $\mathcal{W}$ are orthogonal with respect to $\langle \cdot, \cdot \rangle$ if*

$$\langle \boldsymbol{v}, \boldsymbol{w} \rangle = 0$$

for all $\boldsymbol{v} \in \mathcal{V}$ and $\boldsymbol{w} \in \mathcal{W}$.

We also need the definition for the orthogonal complement of a subspace $\mathcal{V}$.

Definition 3.2.3. *Let $\mathcal{V}$ be subspace of $\mathbb{R}^n$ and $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ some innerpoduct. The orthogonal complement of $\mathcal{V}$ is the subspace*

$$\mathcal{V}^\perp := \{\boldsymbol{w} \in \mathbb{R}^n \mid \langle \boldsymbol{w}, \boldsymbol{v} \rangle = 0 \quad \text{for all } \boldsymbol{v} \in \mathcal{V}\}. \tag{3.10}$$

(Proving that $\mathcal{V}^\perp$ is actually a subspace is left as an exercise.)

It holds that

$$\mathcal{V} \cap \mathcal{V}^\perp = \{0\} \qquad \text{and} \qquad \mathcal{V} + \mathcal{V}^\perp = \mathbb{R}^n, \tag{3.11}$$

i.e., $\mathcal{V} \oplus \mathcal{V}^\perp = \mathbb{R}^n$. Indeed, if $\boldsymbol{v} \in \mathcal{V}$ also belongs to $\mathcal{V}^\perp$, then by definition

$$\langle \boldsymbol{v}, \boldsymbol{v} \rangle = 0, \qquad \text{i.e.,} \qquad \boldsymbol{v} = 0.$$

On the other hand, if $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$ is an orthonormal basis for $\mathcal{V}$,[1] i.e., the vectors are orthogonal (cf. Definition 1.2.3) and of unit length in the norm defined by the considered inner product, then any $\boldsymbol{x} \in \mathbb{R}^n$ can be written as

$$\boldsymbol{x} = \sum_{i=1}^{k} \langle \boldsymbol{x}, \boldsymbol{v}_i \rangle \boldsymbol{v}_i + \left(\boldsymbol{x} - \sum_{i=1}^{k} \langle \boldsymbol{x}, \boldsymbol{v}_i \rangle \boldsymbol{v}_i \right) =: \boldsymbol{v}_{\boldsymbol{x}} + \boldsymbol{w}_{\boldsymbol{x}}, \qquad (3.12)$$

which gives $\boldsymbol{x}$ as a sum of a vector in $\mathcal{V}$ and another in $\mathcal{V}^\perp$: Clearly, $\boldsymbol{v}_{\boldsymbol{x}} \in \mathcal{V}$ and proving that $\boldsymbol{w}_{\boldsymbol{x}} \in \mathcal{V}^\perp$ is left as a homework (write an arbitrary element of $\mathcal{V}$ in the orthonormal basis, take its inner product with $\boldsymbol{w}_{\boldsymbol{x}}$ and employ the orthonormality of the basis vectors). In fact, $\boldsymbol{v}_{\boldsymbol{x}}$ is the **orthogonal** projection of $\boldsymbol{x}$ onto $\mathcal{V}$ and $\boldsymbol{w}_{\boldsymbol{x}}$ the **orthogonal** projection of $\boldsymbol{x}$ onto $\mathcal{V}^\perp$, as will be defined in the following.

For the rest of this chapter, we only consider orthogonal complements with respect to the Euclidean inner product. A projection matrix related to an orthogonal decomposition $\mathbb{R}^n = \mathcal{V} \oplus \mathcal{V}^\perp$ is called an **orthogonal projection** matrix. Let $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$ and $\{\boldsymbol{v}_1^\perp, \ldots, \boldsymbol{v}_{n-k}^\perp\}$ be (not necessarily orthonormal) bases for $\mathcal{V}$ and $\mathcal{V}^\perp$, respectively, and define the matrices $V \in \mathbb{R}^{n \times k}$ and $V^\perp \in \mathbb{R}^{n \times (n-k)}$ in the standard manner, i.e.,

$$V = [\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k] \qquad \text{and} \qquad V^\perp = [\boldsymbol{v}_1^\perp, \ldots, \boldsymbol{v}_{n-k}^\perp].$$

[1] Such a basis can be constructed by the Gram–Schmidt ortogonalisation process as we will learn later.

Note once again that $\mathcal{V} = R(V) = \text{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k)$ and $\mathcal{V}^\perp = R(V^\perp) = \text{span}(\boldsymbol{v}_1^\perp, \ldots, \boldsymbol{v}_{n-k}^\perp)$. By definition (3.10), the columns of V are orthogonal to those of $V^\perp$ (in the Euclidean inner product), meaning that

$$V^T V^\perp = 0. \tag{3.13}$$

On the other hand, the decomposition of $\mathbb{R}^n$ related to $\mathcal{V}$ and $\mathcal{V}^\perp$ is

$$\boldsymbol{x} = V\boldsymbol{z}_V + V^\perp \boldsymbol{z}_{V^\perp} \tag{3.14}$$

for some yet-to-be-defined $\boldsymbol{z}_V \in \mathbb{R}^k$ and $\boldsymbol{z}_{V^\perp} \in \mathbb{R}^{n-k}$. Multiplying from the left by V^T and using the orthogonlity property (3.13), we get

$$V^T \boldsymbol{x} = V^T V \boldsymbol{z}_V.$$

Because $\text{N}(V^T V) = \text{N}(V)$ is trivial as the columns of $V \in \mathbb{R}^{n\times k}$ are linearly independent, $V^T V \in \mathbb{R}^{k\times k}$ is invertible and $\boldsymbol{z}_V$ can be solved as

$$\boldsymbol{z}_V = (V^T V)^{-1} V^T \boldsymbol{x}.$$

The projection matrix related to the decomposition $\mathbb{R}^n = \mathcal{V} \oplus \mathcal{V}^\perp$ is thus

$$P_{\mathcal{V}} = V(V^T V)^{-1} V^T \in \mathbb{R}^{n\times n}, \tag{3.15}$$

see (3.14). In particular, the orthogonal projection does not require one to form a basis for $\mathcal{V}^\perp$: the auxiliary basis $\{\boldsymbol{v}_1^\perp, \ldots, \boldsymbol{v}_{n-k}^\perp\}$ of $V^\perp$ does not anymore appear in (3.15). Moreover, if $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$ are orthonormal, then V is orthogonal/unitary, i.e. $V^T V = I$, and

so

$$P_{\mathcal{V}}\boldsymbol{x} = V I^{-1} V^T \boldsymbol{x} = V \begin{bmatrix} \boldsymbol{v}_1^T \boldsymbol{x} \\ \vdots \\ \boldsymbol{v}_k^T \boldsymbol{x} \end{bmatrix} = \sum_{i=1}^{k} (\boldsymbol{v}_i^T \boldsymbol{x}) \boldsymbol{v}_i, \tag{3.16}$$

which corresponds to the first term in (3.12).

An othonormal projection satisfies some useful identities. First of all, since inversion and transposition commute, we have

$$\begin{aligned} P_{\mathcal{V}}^T &= \big(V(V^TV)^{-1}V^T\big)^T = (V^T)^T\big((V^TV)^T\big)^{-1}V^T \\ &= V(V^TV)^{-1}V^T = P_{\mathcal{V}}, \end{aligned}$$

and hence also,

$$P_{\mathcal{V}}^T(I - P_{\mathcal{V}}) = P_{\mathcal{V}}(I - P_{\mathcal{V}}) = P_{\mathcal{V}} - P_{\mathcal{V}}^2 = P_{\mathcal{V}} - P_{\mathcal{V}} = 0. \tag{3.17}$$

After transposition, this yields

$$(I - P_{\mathcal{V}})^T P_{\mathcal{V}} = 0. \tag{3.18}$$

In fact, $I - P_{\mathcal{V}}$ is the orthogonal projection onto $\mathcal{V}^\perp$ (homework).

Example 3.2.1. *Let* $\boldsymbol{a}, \boldsymbol{x} \in \mathbb{R}^2$. *Consider a decomposition of the vector* $\boldsymbol{x}$ *into two components, one parallel and the other orthogonal to* $\boldsymbol{a}$. *Such a decomposition can be constructed as in* (3.12)*:*

$$\boldsymbol{x} = \frac{\boldsymbol{a}^T\boldsymbol{x}}{\|\boldsymbol{a}\|_2^2}\boldsymbol{a} + \left(\boldsymbol{x} - \frac{\boldsymbol{a}^T\boldsymbol{x}}{\|\boldsymbol{a}\|_2^2}\boldsymbol{a}\right) = \frac{1}{\|\boldsymbol{a}\|_2^2}\boldsymbol{a}\boldsymbol{a}^T\boldsymbol{x} + \left(I - \frac{1}{\|\boldsymbol{a}\|_2^2}\boldsymbol{a}\boldsymbol{a}^T\right)\boldsymbol{x},$$

where the division by $\|\boldsymbol{a}\|_2^2$ *corresponds to normalisation of* $\boldsymbol{a}$, *i.e.,* $\boldsymbol{a}/\|\boldsymbol{a}\|_2$ *is a unit vector. The two components of this decomposition are orthogonal (see Figure 3.1) and the related projection matrices are*

$$P = \frac{1}{\|\boldsymbol{a}\|_2^2}\boldsymbol{a}\boldsymbol{a}^T. \tag{3.19}$$

and $I - P$.

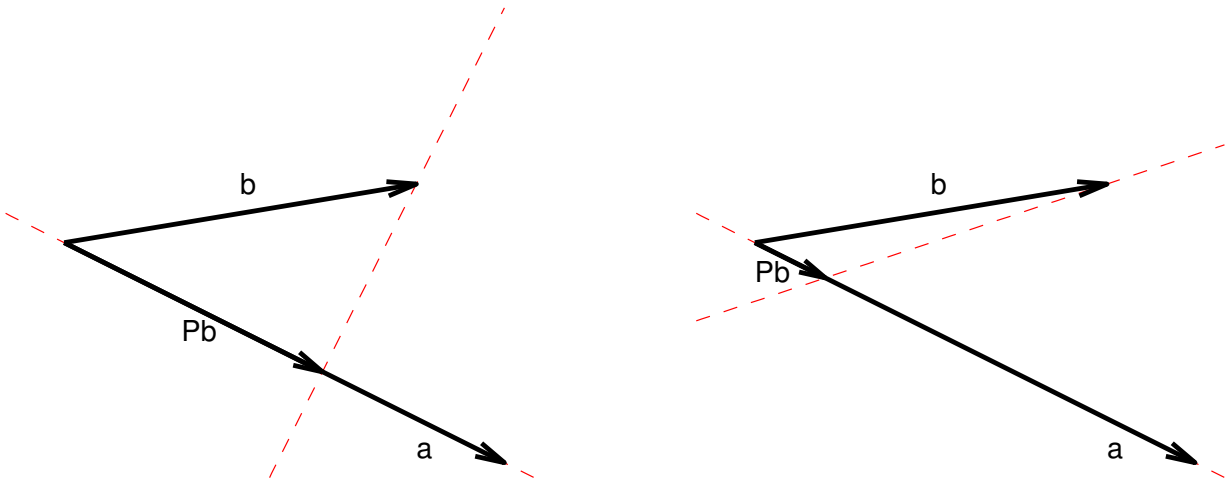


Figure 3.1: Left: Orthogonal projection of $\boldsymbol{a}$ onto $\boldsymbol{b}$. Right: One possible oblique projection of $\boldsymbol{a}$ onto $\boldsymbol{b}$. Both the projections are along the red dashed lines.

Example 3.2.2. *Let $\boldsymbol{a}, \boldsymbol{c}, \boldsymbol{x} \in \mathbb{R}^2$ be such that $[\boldsymbol{a}, \boldsymbol{c}] \in \mathbb{R}^{2\times 2}$ is invertible, i.e., $\boldsymbol{a}$ and $\boldsymbol{c}$ are linearly independent. Consider the decomposition of $\boldsymbol{x}$ in the basis defined by $\boldsymbol{a}$ and $\boldsymbol{c}$:*

$$\boldsymbol{x} = \alpha\boldsymbol{a} + \beta\boldsymbol{c}. \tag{3.20}$$

The coefficients α and β can be solved from the linear system

$$[\boldsymbol{a}, \boldsymbol{b}] \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \boldsymbol{x},$$

and the parameter α can then be extracted from the solution via

$$\alpha = [1, 0][\boldsymbol{a}, \boldsymbol{b}]^{-1}\boldsymbol{x},$$

where $[1, 0] \in \mathbb{R}^{1\times 2}$. Thus, one of the two projection matrices related to this decomposition is

$$P = [\boldsymbol{a}, 0][\boldsymbol{a}, \boldsymbol{b}]^{-1},$$

where $[\boldsymbol{a}, 0] \in \mathbb{R}^{2\times 2}$ (cf. (3.6)).

Lecture 11.

(From 2018 version.)

3.3 Geometric interpretation of a least squares solution

With the help of projection matrices, we can now understand the geometry behind the least squares problem (3.3) and the related normal equation (3.4). Let $P \in \mathbb{R}^m$ be an orthogonal projection from $\mathbb{R}^m$ to $\mathrm{R}(A)$. As P keeps elements in $\mathrm{R}(A)$ fixed, the least squares functional can be rewritten as

$$\begin{aligned}\|A\boldsymbol{x}-\boldsymbol{b}\|_2^2 &= \|PA\boldsymbol{x}-\boldsymbol{b}\|_2^2 = \|P(A\boldsymbol{x}-\boldsymbol{b})-(I-P)\boldsymbol{b}\|_2^2\\ &= \|PA\boldsymbol{x}-P\boldsymbol{b}\|^2 - \big(P(A\boldsymbol{x}-\boldsymbol{b})\big)^T(I-P)\boldsymbol{b}\\ &\quad - \big((I-P)\boldsymbol{b}\big)^T\big(P(A\boldsymbol{x}-\boldsymbol{b})\big) + \|(I-P)\boldsymbol{b}\|_2^2\\ &= \|A\boldsymbol{x}-P\boldsymbol{b}\| + \|(I-P)\boldsymbol{b}\|_2^2,\end{aligned}$$

where the last step follows from (3.17) or/and (3.18). Because the second term on the right-hand side of (3.21) does not depend on $\boldsymbol{x}$, to minimise the least squares functional, $\boldsymbol{x}$ should be chosen such that

$$A\boldsymbol{x} = P\boldsymbol{b}, \tag{3.21}$$

which has a (not-necessarily-unique) solution because $P\boldsymbol{b} \in \mathrm{R}(A)$ by definition. In other words, a minimiser of (3.3) is such $\boldsymbol{x}$ that $A\boldsymbol{x}$ equals the orthogonal projection of $\boldsymbol{b}$ onto $\mathrm{R}(A)$.

If $\mathrm{N}(A) = \{0\}$, i.e, $A\boldsymbol{\alpha} = 0$ if and only if $\boldsymbol{\alpha} = 0$, the columns of A are a (linearly independent) basis for $\mathrm{R}(A)$. Thus, the definition of an orthogonal projection matrix (3.15) gives $P = A(A^TA)^{-1}A^T$ and (3.21) turns into

$$A\boldsymbol{x} = A(A^TA)^{-1}A^T\boldsymbol{b}. \tag{3.22}$$

In particular, the unique solution of the normal equation (3.4), i.e.,

$$\boldsymbol{x} = (A^TA)^{-1}A^T\boldsymbol{b}$$

clearly satisfies (3.22) — as it should because it is the unique minimiser of (3.3) if $\mathrm{N}(A) = \mathrm{N}(A^TA) = \{0\}$ according to Theorem 3.1.1.

3.4 QR decomposition

The normal equation approach to solving the least squares problem has two drawbacks: First of all, if A does not have a trivial nullspace, the normal equation is not uniquely solvable. Secondly, even if $\mathrm{N}(A) = \{0\}$, the conditioning of the normal equation is much worse than that of the original problem (1.1): For example if $A \in \mathbb{R}^{n\times n}$, the condition numbers corresponding to the 2-norm satisfy

$$\kappa_2(A^TA) = \kappa_2(A)^2,$$

which follows from Lemma 1.2.5 of Chapter 1 and the fourth exercise sheet after noticing that A^TA is symmetric. According to the discussion in Section 1, if the condition number is squared, solving a linear system becomes much more sensitive to, e.g., floating point errors. In some extreme cases, resorting to the normal equation may even lead to loss of information, as demonstrated by the following example.

Example 3.4.1. *Let*

$$A = \begin{bmatrix} 1 & 1 \\ \epsilon & 0 \\ 0 & \epsilon \end{bmatrix} \quad \textit{so that} \quad A^T = \begin{bmatrix} 1 & \epsilon & 0 \\ 1 & 0 & \epsilon \end{bmatrix}.$$

Hence,

$$A^T A = \begin{bmatrix} 1+\epsilon^2 & 1 \\ 1 & 1+\epsilon^2 \end{bmatrix}.$$

In floating point arithmetics $fl(1+\epsilon^2) = 1$ *when* ϵ *is sufficiently small; note that for* $\epsilon \ll 1$*, the square* ϵ^2 *is much smaller that* ϵ *itself. When this happens,*

$$fl(A^T A) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix},$$

i.e., data is lost and, in this case, the normal equation is even no longer uniquely solvable. In Matlab, this happens if $\epsilon < 10^{-9}$*.*

To avoid the dramatic decrease in stability due to the normal equation approach and the related floating point errors, the least squares problem is typically solved using either the QR or the **singular value** (SVD) decomposition for the matrix A. Both methods can be understood as ways to compute an orthonormal basis for the subspace $\mathrm{R}(A)$.

The simplest way to compute the QR decomposition is via the **Gram–Schmidt orthogonalisation process** that is based on the following Lemma. In what follows, "orthogonality" refers to orthogonality in the sense of the Euclidean inner product.

Lemma 3.4.1. *Let* $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k\} \subset \mathbb{R}^m$, $k < m$, *be a set of orthonormal vectors, i.e.,* $\|\boldsymbol{q}_i\|_2 = 1$, $i = 1, \ldots, k$, *and*

$$\boldsymbol{q}_i^T \boldsymbol{q}_j = 0 \qquad \text{for } i \neq j.$$

In addition, assume that $\boldsymbol{a} \in \mathbb{R}^m$ *does not belong to* $\operatorname{span}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k)$ *and define*

$$\boldsymbol{q}_{k+1} = \frac{\tilde{\boldsymbol{q}}_{k+1}}{\|\tilde{\boldsymbol{q}}_{k+1}\|_2}, \qquad \text{where} \quad \tilde{\boldsymbol{q}}_{k+1} = \boldsymbol{a} - \sum_{i=1}^{k} (\boldsymbol{a}^T \boldsymbol{q}_i)\boldsymbol{q}_i. \tag{3.23}$$

Then $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_{k+1}\} \subset \mathbb{R}^n$ *is a set of orthonormal vectors and*

$$\operatorname{span}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_{k+1}) = \operatorname{span}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k, \boldsymbol{a}). \tag{3.24}$$

Proof. To begin with note that $\tilde{\boldsymbol{q}}_{k+1}$ defined in (3.23) is a nonzero vector because $\boldsymbol{a} \notin \operatorname{span}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k)$, i.e., $\boldsymbol{a}$ cannot be given as a linear combination of $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k$.

To prove the orthonormality of the set $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_{k+1}\}$ it is enough to prove that $\boldsymbol{q}_{k+1}$ is of unit Euclidean length and orthogonal to $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k$. From the first equation in (3.23), it is obvious that $\|\boldsymbol{q}_{k+1}\|_2 = 1$. Moreover, since $\boldsymbol{q}_{k+1}$ and $\tilde{\boldsymbol{q}}_{k+1}$ are parallel, it is actually enough to show that $\tilde{\boldsymbol{q}}_{k+1}$ is orthogonal to $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k$: for any

$j = 1, \ldots, k$, we have

$$\tilde{\boldsymbol{q}}_{k+1}^T \boldsymbol{q}_j = \Big(\boldsymbol{a} - \sum_{i=1}^{k} (\boldsymbol{a}^T \boldsymbol{q}_i)\boldsymbol{q}_i\Big)^T \boldsymbol{q}_j = \boldsymbol{a}^T \boldsymbol{q}_j - \sum_{i=1}^{k} (\boldsymbol{a}^T \boldsymbol{q}_i)(\boldsymbol{q}_i^T \boldsymbol{q}_j) = \boldsymbol{a}^T \boldsymbol{q}_j - \boldsymbol{a}^T \boldsymbol{q}_j = 0$$

due to the orthonormality of $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k\}$.

Although (3.24) follows straightforwardly from the definition of linear span, let us anyway carefully prove it for the sake of completeness. Assume first that $\boldsymbol{x} \in \text{span}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_{k+1})$, i.e.,

$$\boldsymbol{x} = \sum_{i=1}^{k+1} \alpha_i \boldsymbol{q}_i = \sum_{i=1}^{k} \alpha_i \boldsymbol{q}_i + \alpha_{k+1} \boldsymbol{q}_{k+1}$$

for some $\boldsymbol{\alpha} \in \mathbb{R}^{k+1}$. Note that (3.23) can be rewritten in the form

$$\boldsymbol{q}_{k+1} = \frac{1}{\|\tilde{\boldsymbol{q}}_{k+1}\|_2} \Big(\boldsymbol{a} - \sum_{i=1}^{k} (\boldsymbol{a}^T \boldsymbol{q}_i)\boldsymbol{q}_i\Big).$$

Hence,

$$\begin{aligned} \boldsymbol{x} &= \sum_{i=1}^{k} \alpha_i \boldsymbol{q}_i + \frac{\alpha_{k+1}}{\|\tilde{\boldsymbol{q}}_{k+1}\|_2} \Big(\boldsymbol{a} - \sum_{i=1}^{k} (\boldsymbol{a}^T \boldsymbol{q}_i)\boldsymbol{q}_i\Big) \\ &= \sum_{i=1}^{k} \Big(\alpha_i - \frac{\alpha_{k+1} \boldsymbol{a}^T \boldsymbol{q}_i}{\|\tilde{\boldsymbol{q}}_{k+1}\|_2}\Big) \boldsymbol{q}_i + \frac{\alpha_{k+1}}{\|\tilde{\boldsymbol{q}}_{k+1}\|_2} \boldsymbol{a}, \end{aligned}$$

which is obviously in $\text{span}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k, \boldsymbol{a})$, meaning that $\text{span}(\boldsymbol{q}_1, \ldots, , \boldsymbol{q}_{k+1}) \subset \text{span}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k, \boldsymbol{a})$.

On the other hand, if $\boldsymbol{x} \in \text{span}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k, \boldsymbol{a})$, then for some $\boldsymbol{\alpha} \in \mathbb{R}^{k+1}$,

$$\boldsymbol{x} = \sum_{i=1}^{k} \alpha_i \boldsymbol{q}_i + \alpha_{k+1}\boldsymbol{a} = \sum_{i=1}^{k} \alpha_i \boldsymbol{q}_i + \alpha_{k+1}\Big(\|\tilde{\boldsymbol{q}}_{k+1}\|_2 \, \boldsymbol{q}_{k+1} + \sum_{i=1}^{k} (\boldsymbol{a}^T \boldsymbol{q}_i)\boldsymbol{q}_i \Big)$$
$$= \sum_{i=1}^{k} (\alpha_i + \alpha_{k+1}\boldsymbol{a}^T\boldsymbol{q}_i)\boldsymbol{q}_i + \alpha_{k+1}\|\tilde{\boldsymbol{q}}_{k+1}\|_2 \, \boldsymbol{q}_{k+1},$$

which clearly belongs to $\text{span}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_{k+1})$.

Hence, also $\text{span}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k, \boldsymbol{a}) \subset \text{span}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_{k+1})$, which completes the proof. □

The intuitive idea of (3.23) is that one first subtracts from $\boldsymbol{a}$ its projections onto the one-dimensional subspaces defined by $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n$, leaving only the component of $\boldsymbol{a}$ orthogonal to $\text{span}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k)$, and then this component is normalised. In fact, one can write the second equation of (3.23) in the form

$$\tilde{\boldsymbol{q}}_{k+1} = (I - P_k)\boldsymbol{a},$$

where $P_k \in \mathbb{R}^{m \times m}$ is the orthogonal projection matrix onto the subspace $\text{span}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_k)$; see (3.16).

Using Lemma 3.4.1, it is straightforward to compute an orthonormal basis for the subspace

$$\text{R}(A) = \text{span}(\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n) \subset \mathbb{R}^m,$$

assuming the columns $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n$ of the matrix $A \in \mathbb{R}^{m\times n}$ are linearly independent, i.e., assuming $\mathrm{N}(A) = \{0\}$. Indeed, such basis $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n\}$ can be recursively obtained via

$$\boldsymbol{q}_j = \frac{\tilde{\boldsymbol{q}}_j}{\|\tilde{\boldsymbol{q}}_j\|_2}, \qquad \text{where} \quad \tilde{\boldsymbol{q}}_j = \boldsymbol{a}_j - \sum_{i=1}^{j-1} (\boldsymbol{a}_j^T \boldsymbol{q}_i)\boldsymbol{q}_i, \qquad \text{for } j = 1, \ldots, n.$$

In other words, one first defines $\boldsymbol{q}_1$ by simply normalizing $\boldsymbol{a}_1$, then one computes a unit vector $\boldsymbol{q}_2$ that is orthogonal to $\boldsymbol{q}_1$ and satisfies $\mathrm{span}(\boldsymbol{q}_1, \boldsymbol{q}_2) = \mathrm{span}(\boldsymbol{q}_1, \boldsymbol{a}_2) = \mathrm{span}(\boldsymbol{a}_1, \boldsymbol{a}_2)$, then one continues by computing a unit vector $\boldsymbol{q}_2$ that is orthogonal to both $\boldsymbol{q}_1$ and $\boldsymbol{q}_2$ and satisfies

$$\mathrm{span}(\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3) = \mathrm{span}(\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{a}_3) = \mathrm{span}(\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3),$$

and so on until $\boldsymbol{q}_n$ is computed and it holds that $\mathrm{span}(\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n) = \mathrm{span}(\boldsymbol{a}_1, \ldots, \boldsymbol{a}_n) = \mathrm{R}(A)$.

Take note that one can get the original columns of A back via

$$\boldsymbol{a}_j = \|\tilde{\boldsymbol{q}}_j\|_2 \, \boldsymbol{q}_j + \sum_{i=1}^{j-1} (\boldsymbol{a}_j^T \boldsymbol{q}_i)\boldsymbol{q}_i, \qquad j = 1, \ldots, n, \tag{3.25}$$

which demonstrates that, for any $j = 1, \ldots, n$, the jth column $\boldsymbol{a}_j$ of A can be given as a linear combination of $\boldsymbol{q}_1, \ldots, \boldsymbol{q}_j$, i.e., of (only) the first j orthonormal basis vectors of $\mathrm{R}(A)$ produced by the Gram–Schmidt process. Defining in the standard manner $Q = [\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n] \in \mathbb{R}^{m\times n}$ and collecting the coefficients in the linear combinations of (3.25) as columns of an upper triangular matrix

$R \in \mathbb{R}^{n\times n}$, the equations (3.25) can be written neatly in a matrix form

$$A = QR.$$

To be more precise, R can be given elementwise as

$$R_{i,j} = \begin{cases} \boldsymbol{a}_j^T \boldsymbol{q}_i & \text{if } i < j, \\ \|\tilde{\boldsymbol{q}}_i\|_2 & \text{if } i = j, \\ 0 & \text{if } i > j. \end{cases}$$

Note also that $Q^TQ = I \in \mathbb{R}^{n\times n}$ because the columns of Q are orthonormal.

Once a QR decomposition has been computed for A,[2] the least squares problem (3.3) can be formulated **equivalently** as the equation

$$R\boldsymbol{x} = Q^T\boldsymbol{b}.$$

There are (at least) two ways to deduce this. First of all, since $\mathrm{N}(A^TA) = \mathrm{N}(A) = 0$ by assumption, the unique solution of the least squares problem (3.3) is the unique solution of the normal equation (3.4). On the other hand, since

$$A^T = R^TQ^T \qquad \text{and} \qquad A^TA = R^TQ^TQR = R^TIR = R^TR,$$

the normal equation can be written as

$$R^TR\boldsymbol{x} = R^TQ^T\boldsymbol{b} \qquad \Longleftrightarrow \qquad R\boldsymbol{x} = Q^T\boldsymbol{b},$$

[2]Observe that when introducing the QR decomposition, we assumed that the columns of $A \in \mathbb{R}^{m\times n}$ are linearly independent, i.e., $\mathrm{N}(A) = \{0\}$, and so everything that follows is conditional on this assumption. In particular, this excludes the case $n > m$.

where the last step follows by multiplying from the left by the inverse of R^T.[3]

The second way is based on using the reformulation (3.21) of the least squares problem and writing the orthogonal projection matrix $P \in \mathbb{R}^{m\times m}$ onto $\mathrm{R}(A)$ with the help of the orthonormal basis $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_n\}$, that is,

$$P = Q\big(Q^TQ\big)^{-1}Q^T = QI^{-1}Q^T = QQ^T$$

according to (3.15). Hence, (3.21) reduces to

$$A\boldsymbol{x} = QR\boldsymbol{x} = QQ^T\boldsymbol{b} \qquad \Longleftrightarrow \qquad R\boldsymbol{x} = Q^T\boldsymbol{b},$$

where "$\Rightarrow$" follows by multiplying from left with Q^T and "$\Leftarrow$" by multiplying from left with Q.

[3]Observe that both R and R^T are invertible: As triangular square matrices, their eigenvalues are their diagonal elements $\|\tilde{\boldsymbol{q}}_i\|_2 > 0$, $i = 1, \ldots, n$. Hence, zero is not an eigenvalue for either of these square matrices, and thus they are both invertible.

There are two ways to numerically implement the Gram–Schmidt process, or the computation of a QR decomposition, namely classical and modified. In floating point arithmetics these two procedures have very different numerical stability properties. The more stable **modified Gram–Schmidt** procedure can be implemented in Matlab as follows:

```
function [Q,R] = my_gsmith(A)

Q = [];
for i=1:size(A,2)
    q = A(:,i);

    for k=1:size(Q,2)
        R(k,i) = q'*Q(:,k);
        q = q - R(k,i)*Q(:,k);
    end
    R(i,i) = norm(q);
    Q(:,i) = q/R(i,i);
end
```

And the **classical Gram-Schmidt** procedure as follows:

```
function [Q,R] = my_c_gsmith(A)

Q = [];
for i=1:size(A,2)
    q = A(:,i);

    for k=1:size(Q,2)
        R(k,i) = q'*Q(:,k);
    end

    for k=1:size(Q,2)
        q = q - R(k,i)*Q(:,k);
    end
    R(i,i) = norm(q);
    Q(:,i) = q/R(i,i);
end
```

In exact arithmetics, both these algorithms return $Q \in \mathbb{R}^{m \times n}$ such that $Q^T Q = I$ and $R(Q) = \mathrm{R}(A)$ as well as an upper triangular $R \in \mathbb{R}^{n \times n}$ such that $A = QR$. However, the two implementations of the Gram–Schmidt process have very different numerical stability properties. The exact definition of numerical stability is in this case rather complicated. One can, e.g., measure the orthogonality of Q, i.e. $\|I - Q^T Q\|$, or the accuracy of the decomposition, i.e. $\|A - QR\|$, in some suitable matrix norm. The loss of orthogonality is more prone to stability issues than the accuracy of the decomposition. For the modified Gram–Schmidt, one can prove numerical stability in both of these measures, whereas the classical Gram–Schmidt is not numerically stable.

Lecture 12.

3.5 Singular value decomposition

The **singular value decomposition** (SVD) is a matrix factorisation of the form

$$A = U\Sigma V^T (= U\Sigma V^*),$$

where $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ are orthogonal (resp., unitary) matrices. The matrix $\Sigma \in \mathbb{R}^{m\times n}$ is a **non-square** matrix with all off-diagonal entries zero, and **diagonal elements non-negative**.

For a symmetric (or Hermitian) and **positive** definite matrix A, the unitary diagonalisation

$$A = U\Lambda U^* = U\Lambda U^{-1}$$

is a SVD with $\Sigma = \Lambda$ and $V = U$.

If A is symmetric (Hermitian) but not non-negative, then some eigenvalue in Λ is negative. This sign problem can be easily dealt with to produce SVD. How?

In fact, SVD can be formed for **any** matrix A by allowing $V \neq U$ and non-square Σ. Before proving that, let us consider the nature and some consequences of the SVD.

The matrices U and V in SVD are not unique, given A.

From now on, we work in the real vector spaces. How can we?

If A is a $m \times n$ matrix, so is Σ in the SVD of A.

Depending whether $n > m$, $n = m$, or $n < m$, the matrix Σ takes different forms.

If $m < n$, Σ takes the form

$$\Sigma = \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_m \end{bmatrix},$$

where all omitted matrix elements are zeroes.

If $m > n$, Σ takes the form

$$\Sigma = \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \\ & & & \end{bmatrix},$$

where all omitted matrix elements are, again, zeroes.

If $n = m$, we have plainly $\Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_n) \in \mathbb{R}^{n \times n}$.

The non-negative numbers in the diagonal of Σ

$$\sigma_1 \geq \sigma_2 \geq \sigma_{\min\{m,n\}} \geq 0$$

are called **singular values** of A.

The intuitive idea behind the SVD of $A = U\Sigma V^T$?

In the beginning, there are two orthonormal (column vector) bases

$$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m\} \quad \text{for } \mathbb{R}^m \quad \text{and}$$

$$\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\} \quad \text{for } \mathbb{R}^n.$$

The bases relate to the **orthogonal** matrices U and V in the SVD, namely

$$U := [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m] \quad \text{and} \quad V := [\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n]\,.$$

Now for $\boldsymbol{x} \in \mathbb{R}^n$ we have

$$V^T\boldsymbol{x} = \begin{bmatrix} \boldsymbol{v}_1^T\boldsymbol{x} \\ \vdots \\ \boldsymbol{v}_n^T\boldsymbol{x} \end{bmatrix}$$

consisting of the cofficients of $\boldsymbol{x}$ in the representation in the basis $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$.

Thus, if $m \leq n$, we have

$$\Sigma V^T \boldsymbol{x} = \begin{bmatrix} \sigma_1 \boldsymbol{v}_1^T \boldsymbol{x} \\ \vdots \\ \sigma_m \boldsymbol{v}_m^T \boldsymbol{x} \end{bmatrix}$$

and if $m > n$ we have

$$\Sigma V^T \boldsymbol{x} = \begin{bmatrix} \sigma_1 \boldsymbol{v}_1^T \boldsymbol{x} \\ \vdots \\ \sigma_n \boldsymbol{v}_n^T \boldsymbol{x} \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

So, using the basis $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ makes it possible to operate by A as if we were operating with a diagonal matrix Σ — even if A was defective (i.e., non-diagonalisable).

Now it remains to come back to the original (natural) basis of $\mathbb{R}^m$ where the producet $A\boldsymbol{x}$ was formulated by using U:

$$A\boldsymbol{x} = \sum_{i=1}^{\min\{m,n\}} \sigma_i (\boldsymbol{v}_i^T \boldsymbol{x}) \boldsymbol{u}_i = \sum_{i=i}^{\min\{m,n\}} \sigma_i (\boldsymbol{u}_i \boldsymbol{v}_i^T) \boldsymbol{x}.$$

The last term in the previous equation is interesting enough to be given separately (without the column vector $\boldsymbol{x}$):

$$A = \sum_{i=i}^{\min\{m,n\}} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T$$

where some of the σ_i's could be zero.

Each of the matrices $\boldsymbol{u}_i \boldsymbol{v}_i^T \in \mathbb{R}^{m\times n}$ is rank 1 and also a **partial isometry**: Indeed,

$$\left(\boldsymbol{u}_i \boldsymbol{v}_i^T\right)^T \boldsymbol{u}_i \boldsymbol{v}_i^T = \boldsymbol{v}_i \boldsymbol{u}_i^T \boldsymbol{u}_i \boldsymbol{v}_i^T = \boldsymbol{v}_i \boldsymbol{v}_i^T$$

where $\left(\boldsymbol{v}_i \boldsymbol{v}_i^T\right)^2 = \boldsymbol{v}_i \boldsymbol{v}_i^T \boldsymbol{v}_i \boldsymbol{v}_i^T = \boldsymbol{v}_i \boldsymbol{v}_i^T$ is an orthogonal projection.

So, the SVD is a way of writing any matrix $A \in \mathbb{R}^{m\times n}$ as a sum of **at most** $\min\{m, n\}$ rank 1 matrices.

Obviously, because both U and V^T are invertible in $A = U\Sigma V^T$,

$$\operatorname{rank}(A) = \dim \mathrm{R}(A) = \dim R(\Sigma) = \max\{j \mid \sigma_j > 0\}$$

recalling that the singular values of A are given in non-increasing order $\sigma_1 \geq \sigma_2 \geq \sigma_{\min\{m,n\}} \geq 0$.

From now on, we denote $p := \operatorname{rank}(A)$, the number of **non-zero** singular values of A. Thus, $\sigma_p > 0$ and $\sigma_{p+1} = 0$.

Forming the SVD of $A \in \mathbb{R}^{m \times n}$ requires finding the orthonormal bases $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m\}$ and $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$.

The vectors $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m\}$ are the **left-singular vectors** of A in the SVD $A = U\Sigma V^T$.

Similarly, $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\}$ are the **right-singular vectors** of A in the SVD $A = U\Sigma V^T$.

We conclude that **if** A can be represented through its SVD

$$A = U\Sigma V^T$$

and $p = \text{rank}(A)$, then all information about A is encoded in

1. the left-singular vectors $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\}$,
2. the right-singular vectors $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$, and
3. the non-zero singular values $\sigma_1, \ldots, \sigma_p$.

Now there is this big **"if"** left that we have to deal with by proving a theorem.

We begin with a lemma that is related to the eigenvalues and eigenvectors of A^TA.

Lemma 3.5.1. *Let $A \in \mathbb{R}^{m\times n}$.*

Let $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\} \subset \mathbb{R}^n$ be an orthonormal set of eigenvectors of the symmetric matrix $A^TA \in \mathbb{R}^{n\times n}$

$$A^TA\boldsymbol{v}_j = \lambda_j \boldsymbol{v}_j$$

arranged so that the corresponding (real, non-negative) eigenvalues are given in the non-increasing order

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0$$

according to their multiplicities.

Define p so that

$$\lambda_p > 0 \quad \textit{but} \quad \lambda_{p+1} = 0.$$

Then, $A\boldsymbol{v}_i = 0$ for $i = p+1, \ldots, n$, and

$$\boldsymbol{u}_i := \frac{1}{\sqrt{\lambda_i}} A\boldsymbol{v}_i, \qquad i = 1, \ldots, p,$$

form an orthonormal set in $\mathbb{R}^m$.

In particular, $0 \leq p \leq \min\{m, n\}$, and in fact $p = \operatorname{rank}(A) = \operatorname{rank}(A^TA)$.

Proof. Because A^TA is a symmetric matrix, the eigenbasis $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n\} \subset \mathbb{R}^n$ and the real eigenvalues exists as required by Theorem 2.2.1 above.

Because A^TA is non-negative (i.e., positive semidefinite), all eigenvalues are non-negative since

$$0 \leq \|A\boldsymbol{v}_i\|_2^2 = \boldsymbol{v}_i^T A^T A \boldsymbol{v}_i = \lambda_i \|\boldsymbol{v}_i\|_2^2 = \lambda_i.$$

This computation also proves that $A\boldsymbol{v}_i = 0$ if and only if $\lambda_i = 0$ if and **only if** $i = p+1, \ldots, n$.

Hence, none of the vectors $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p$, indeed, well-defined by

$$\boldsymbol{u}_i = \frac{1}{\sqrt{\lambda_i}} A\boldsymbol{v}_i$$

is a zero vector.

Furthermore,

$$\begin{aligned}\boldsymbol{u}_i^T \boldsymbol{u}_j &= \frac{1}{\sqrt{\lambda_i \lambda_j}} \boldsymbol{v}_i^T A^T A \boldsymbol{v}_j = \frac{\lambda_j}{\sqrt{\lambda_i \lambda_j}} \boldsymbol{v}_i^T \boldsymbol{v}_j \\ &= \sqrt{\frac{\lambda_j}{\lambda_i}}\, \delta_{i,j} = \delta_{i,j}\end{aligned}$$

by the orthonormality of $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$

Surely you know "Kronecker's delta" by now?

We conlude that the set $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\}$ is orthonormal.

Since $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$ is an orthonormal set (hence, linearly free) in $\mathbb{R}^n$, we have $p \leq n$.

Since $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\}$ is an orthonormal set in $\mathbb{R}^m$, we have $p \leq m$.

We conclude that $p \leq \min\{m, n\}$. The claim about the rank is left as homework. □

Now we are in place to show that any matrix has a SVD.

Theorem 3.5.1. *Let $A \in \mathbb{R}^{m\times n}$. Then there exists an integer $0 \leq p \leq \min\{m,n\}$, positive coefficients $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_p > 0$ and sets of orthonormal vectors $\{\boldsymbol{v}_1,\ldots,\boldsymbol{v}_p\} \subset \mathbb{R}^n$ and $\{\boldsymbol{u}_1,\ldots,\boldsymbol{u}_p\} \subset \mathbb{R}^m$ such that*

$$A = \sum_{i=1}^{p} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T.$$

Proof. Let the the integer p and

$$\begin{aligned}
&\{\boldsymbol{v}_1,\ldots,\boldsymbol{v}_n\} \subset \mathbb{R}^n,\\
&\{\boldsymbol{u}_1,\ldots,\boldsymbol{u}_p\} \subset \mathbb{R}^m,\\
&\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p > 0
\end{aligned}$$

be as in Lemma 3.5.1.

Take $\boldsymbol{x} \in \mathbb{R}^n$, and write it in the eigenbasis of $A^TA \in \mathbb{R}^n$ as

$$\boldsymbol{x} = \sum_{i=1}^{n} (\boldsymbol{v}_i^T \boldsymbol{x})\boldsymbol{v}_i.$$

Because $A\boldsymbol{v}_i = 0$ for $i = p+1, \ldots, n$, we have

$$\begin{aligned} A\boldsymbol{x} &= \sum_{i=1}^{n} (\boldsymbol{v}_i^T \boldsymbol{x}) A\boldsymbol{v}_i = \sum_{i=1}^{p} (\boldsymbol{v}_i^T \boldsymbol{x}) A\boldsymbol{v}_i \\ &= \sum_{i=1}^{p} (\boldsymbol{v}_i^T \boldsymbol{x}) \sqrt{\lambda_i} \boldsymbol{u}_i = \sum_{i=1}^{p} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T \boldsymbol{x} \end{aligned}$$

by defining $\sigma_i := \sqrt{\lambda_i}$. □

We have now shown the existence of the SVD in the "economical" form

$$A = \sum_{i=1}^{p} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T$$

instead of the "wasteful" matrix form

$$A = U \Sigma V^T.$$

We have already shown how to get from the wasteful form to the economical form.

In contrast to what they call "real life", it is more difficult to get from economical to wasteful in linear algebra.

To get from economical to wasteful, we first introduce a number of **worthless zeroes** to the $m \times n$ matrix Σ.

If $p < \min(m, n)$, we must introduce additional singular values to produce the matrix Σ of sufficient size:

$$\sigma_{p+1} = \sigma_{p+1} = \ldots = \sigma_{\min(m,n)} = 0.$$

Now we can write $\Sigma \in \mathbb{R}^{m \times n}$ as required.

We are not quite done yet, since Lemma 3.5.1 (via Theorem 3.5.1) provides us only the left-singular vectors

$$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\} \subset \mathbb{R}^m$$

where $p \leq \min(m, n) \leq m$. If $p < m$, we need to get additional vectors $\boldsymbol{u}_{p+1}, \ldots, \boldsymbol{u}_m$ to get the $m \times m$ matrix $U = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p, \ldots, \boldsymbol{u}_m]$.

It does not really matter what these additional vectors are since they will get multiplied by zeroes in Σ in the product $U\Sigma V^T$.

However, they must be chosen so that the matrix U becomes orthonormal; i.e., so that

$$\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p, \ldots, \boldsymbol{u}_m\} \subset \mathbb{R}^m$$

is an orthonormal basis.

The orthonormality can always be achieved using the Gramm–Schmidt -process, starting from any extension to a basis of $\mathbb{R}^m$.

The arbitrarily generated 'extra left-singular vectors'

$$\boldsymbol{u}_{p+1}, \ldots, \boldsymbol{u}_m$$

and the 'extra right-singular vectors' – given by Lemma 3.5.1 but not communicated further by Theorem 3.5.1 – given by

$$\boldsymbol{v}_{p+1}, \ldots, \boldsymbol{v}_n$$

are not actually needed in constructing the economical form $A = \sum_{i=1}^{p} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T$ of the SVD.

These 'extra singular vectors' can, however, be used to determine $\mathrm{R}(A)$, $\mathrm{N}(A)$ and their orthogonal complements.

Theorem 3.5.2. *Let $A \in \mathbb{R}^{m \times n}$ with its SVD given by*

$$A = \sum_{i=1}^{p} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T = U \Sigma V^T$$

where $p \leq \min(m, n)$ is the number of positive singular values of A, and

$$U = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m] \qquad \text{and} \qquad V = [\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n] \,.$$

Then

(i) $\mathrm{R}(A) = \mathrm{span}(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p)$,

(ii) $\mathrm{R}(A)^{\perp} = \mathrm{span}(\boldsymbol{u}_{p+1}, \ldots, \boldsymbol{u}_m)$,

(iii) $\mathrm{N}(A) = \mathrm{span}(\boldsymbol{v}_{p+1}, \ldots, \boldsymbol{v}_n)$,

(iv) $\mathrm{N}(A)^{\perp} = \mathrm{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p)$.

Proof. **Claim (i):** Since $A\boldsymbol{x} = \sum_{i=1}^{p} \sigma_i \boldsymbol{u}_i \left(\boldsymbol{v}_i^T \boldsymbol{x}\right)$, we have $\mathrm{R}(A) \subset$ $\mathrm{span}(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p)$.

Conversely, for any

$$\boldsymbol{y} = \sum_{i=1}^{p} \alpha_i \boldsymbol{u}_i, \qquad \boldsymbol{\alpha} \in \mathbb{R}^p,$$

can be written as

$$A\Big(\sum_{j=1}^{p} \frac{\alpha_j}{\sigma_j} \boldsymbol{v}_j\Big) = \sum_{i=1}^{p} \sigma_i \boldsymbol{u}_i \Big(\sum_{j=1}^{p} \frac{\alpha_j}{\sigma_j} \boldsymbol{v}_i^T \boldsymbol{v}_j\Big) = \sum_{i=1}^{p} \sigma_i \frac{\alpha_i}{\sigma_i} \boldsymbol{u}_i = \boldsymbol{y}$$

due to the orthonormality of $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$. Hence $\mathrm{span}(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p) \subset$ $\mathrm{R}(A)$ and $\mathrm{rank}(A) = p$.

Claim (ii): Since the $m-p$ vectors $\boldsymbol{u}_{p+1}, \ldots, \boldsymbol{u}_m$ are pairwise orthonormal and also orthogonal to

$$\mathrm{R}(A) = \mathrm{span}(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p),$$

they must be a basis for the $m-p$ dimensional orthogonal complement

$$\mathrm{R}(A)^{\perp} := \{\boldsymbol{x} \in \mathbb{R}^m \,|\, \boldsymbol{x}^T\boldsymbol{y} = 0 \text{ for all } \boldsymbol{y} \in \mathrm{R}(A)\},$$

recalling the orthogonal direct sum decomposition $\mathbb{R}^m = V \oplus V^{\perp}$ for any subspace $V \subset \mathbb{R}^m$.

Claim (iii): Take $\boldsymbol{x} \in \mathrm{N}(A)$ and write it

$$\boldsymbol{x} = \sum_{j=1}^{n} \alpha_j \boldsymbol{v}_j, \qquad \boldsymbol{\alpha} \in \mathbb{R}^n.$$

Using $A = \sum_{i=1}^{p} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T$ we get

$$\begin{aligned}\boldsymbol{0} = A\boldsymbol{x} = A\Big(\sum_{j=1}^{n} \alpha_j \boldsymbol{v}_j\Big) &= \sum_{i=1}^{p} \sigma_i \boldsymbol{u}_i \Big(\sum_{j=1}^{n} \alpha_j \boldsymbol{v}_i^T \boldsymbol{v}_j\Big) \\ &= \sum_{i=1}^{p} \sigma_i \alpha_i \boldsymbol{u}_i\end{aligned}$$

since $\boldsymbol{v}_i^T\boldsymbol{v}_j = \delta_{i,j}$ and, hence, $\sum_{j=1}^{n} \alpha_j \boldsymbol{v}_i^T\boldsymbol{v}_j = \alpha_i$ by orthonormality.

Since $\sigma_1, \ldots, \sigma_p$ are positive and $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p$ are linearly independent, we get

$$\alpha_1 = \cdots = \alpha_p = 0.$$

Thus,

$$\boldsymbol{x} = \sum_{j=p+1}^{n} \alpha_j \boldsymbol{v}_j \in \mathrm{span}(\boldsymbol{v}_{p+1}, \ldots, \boldsymbol{v}_n)$$

and hence $\mathrm{N}(A) \subset \mathrm{span}(\boldsymbol{v}_{p+1}, \ldots, \boldsymbol{v}_n)$.

Conversely, if

$$\boldsymbol{x} = \sum_{j=p+1}^{n} \alpha_j \boldsymbol{v}_j \in \mathrm{span}(\boldsymbol{v}_{p+1}, \ldots, \boldsymbol{v}_n),$$

we get

$$A\boldsymbol{x} = \sum_{j=p+1}^{n} \alpha_j \left(\sum_{i=1}^{p} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T \boldsymbol{v}_j \right) = \sum_{j=p+1}^{n} \sum_{i=1}^{p} \alpha_j \sigma_i \boldsymbol{u}_i \delta_{i,j} = \boldsymbol{0}$$

since terms with $i = j$ do not appear in the above sum. Hence, $\mathrm{span}(\boldsymbol{v}_{p+1}, \ldots, \boldsymbol{v}_n) \subset \mathrm{N}(A)$ follows.

Claim (iv): The remaining orthonormal vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p$ must form a basis for the p dimensional orthogonal complement $\mathrm{N}(A)^\perp$ in $\mathbb{R}^n$. This is the same argument that derives **Claim (ii)** from **Claim (i)** above. □

We complete these lecture notes by demonstrating the SVD can be used for solving least squares problems even if $\mathrm{N}(A) = 0$.

Theorem 3.5.3. *Let $A \in \mathbb{R}^{m\times n}$ and $\boldsymbol{b} \in \mathbb{R}^m$. Assume that the SVD of A is given by*

$$A = \sum_{i=1}^{p} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T \tag{3.26}$$

where $p = \mathrm{rank}(A)$. Then

$$\boldsymbol{x}^\dagger := \sum_{j=1}^{p} \frac{1}{\sigma_j} (\boldsymbol{u}_j^T \boldsymbol{b}) \boldsymbol{v}_j$$

is a solution of the normal equation

$$A^T A \boldsymbol{x} = A^T \boldsymbol{b}.$$

Proof. This is a good summing exercise on orthonormal expansions.

The proof is a computation based on (3.26) and

$$A^T = \Big(\sum_{j=1}^{p} \sigma_j \boldsymbol{u}_j \boldsymbol{v}_j^T \Big)^T = \sum_{j=1}^{p} \sigma_j \big(\boldsymbol{u}_j \boldsymbol{v}_j^T \big)^T = \sum_{j=1}^{p} \sigma_j \boldsymbol{v}_j \boldsymbol{u}_j^T .$$

Indeed,

$$\begin{aligned} A\boldsymbol{x}^\dagger &= \sum_{i=1}^{p} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^T \Big(\sum_{j=1}^{p} \frac{1}{\sigma_j} (\boldsymbol{u}_j^T \boldsymbol{b}) \boldsymbol{v}_j \Big) \\ &= \sum_{i=1}^{p} \sum_{j=1}^{p} \frac{\sigma_i \boldsymbol{u}_i}{\sigma_j} (\boldsymbol{u}_j^T \boldsymbol{b})(\boldsymbol{v}_i^T \boldsymbol{v}_j) \\ &= \sum_{i=1}^{p} \sum_{j=1}^{p} \frac{\sigma_i \boldsymbol{u}_i}{\sigma_j} (\boldsymbol{u}_j^T \boldsymbol{b}) \delta_{i,j} \\ &= \sum_{i=1}^{p} \frac{\sigma_i \boldsymbol{u}_i}{\sigma_i} (\boldsymbol{u}_i^T \boldsymbol{b}) = \sum_{i=1}^{p} (\boldsymbol{u}_i^T \boldsymbol{b}) \boldsymbol{u}_i, \end{aligned}$$

where we used the orthonormality of $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$.

In the same manner,

$$\begin{aligned} A^T(A\boldsymbol{x}^\dagger) &= \sum_{j=1}^{p} \sigma_j \boldsymbol{v}_j \boldsymbol{u}_j^T \Big(\sum_{i=1}^{p} (\boldsymbol{u}_i^T \boldsymbol{b}) \boldsymbol{u}_i \Big) \\ &= \sum_{j=1}^{p} \sum_{i=1}^{p} \sigma_j \boldsymbol{v}_j (\boldsymbol{u}_i^T \boldsymbol{b})(\boldsymbol{u}_j^T \boldsymbol{u}_i) \\ &= \sum_{j=1}^{p} \sum_{i=1}^{p} \sigma_j \boldsymbol{v}_j (\boldsymbol{u}_i^T \boldsymbol{b}) \delta_{i,j} \\ &= \sum_{j=1}^{p} \sigma_j (\boldsymbol{u}_j^T \boldsymbol{b}) \boldsymbol{v}_j \end{aligned}$$

where we used the orthonormality of $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_p\}$.

On the other hand

$$A^T \boldsymbol{b} = \sum_{j=1}^{p} \sigma_j \boldsymbol{v}_j \boldsymbol{u}_j^T \boldsymbol{b} = \sum_{j=1}^{p} \sigma_j (\boldsymbol{u}_j^T \boldsymbol{b}) \boldsymbol{v}_j,$$

and $A^T A \boldsymbol{x}^\dagger = A^T \boldsymbol{b}$ follows. □

Corollary 3.5.1. *Make the same assumptions and use the same notation as in Theorem 3.5.3.*

All solutions of the least squares problem of Definition 3.1.1 are of the form

$$\boldsymbol{x} = \boldsymbol{x}^\dagger + \boldsymbol{z} \tag{3.27}$$

for some $\boldsymbol{z} \in \mathrm{N}(A)$.

Proof. The parameterisation of the solution set of $A\boldsymbol{x} = \boldsymbol{b}$ in terms of $\mathrm{N}(A)$ and a particular solution $\boldsymbol{x}_p$. was given in Lemma 1.1.2.

Now we use that result with $A^T A$ in place of A and $\boldsymbol{x}^\dagger$ given by Theorem 3.5.3 in place of $\boldsymbol{x}_p$.

We need to observe that $\mathrm{N}(A^T A) = \mathrm{N}(A)$ as we have seen.

The claim follows since all solutions of the least squares problem are solutions of the normal equation, and *vice versa*. □

According to Theorems 3.5.2 and 3.5.3, the special least squares solution satisfies

$$\boldsymbol{x}^\dagger = \sum_{j=1}^{p} \frac{1}{\sigma_j}(\boldsymbol{u}_j^T \boldsymbol{b})\boldsymbol{v}_j \in \text{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p) = \text{N}(A)^\perp. \tag{3.28}$$

Hence, any (other) solution $\boldsymbol{x}^\dagger + \boldsymbol{z}$, $\boldsymbol{x}^\dagger \in \text{N}(A)^T$ and $\boldsymbol{z} \in \text{N}(A)$, of the least squares problem satisfies

$$\begin{aligned} \|\boldsymbol{x}^\dagger + \boldsymbol{z}\|_2^2 &= \|\boldsymbol{x}^\dagger\|_2^2 + 2\boldsymbol{z}^T\boldsymbol{x}^\dagger + \|\boldsymbol{z}\|_2^2 \\ &= \|\boldsymbol{x}^\dagger\|_2^2 + \|\boldsymbol{z}\|_2^2 \geq \|\boldsymbol{x}^\dagger\|_2^2. \end{aligned}$$

Thus, $\boldsymbol{x}^\dagger$ in (3.28) is the solution of the least squares problem

$$\arg\min_{\boldsymbol{x}\in\mathbb{R}^m} \|A\boldsymbol{x} - \boldsymbol{b}\|_2^2$$

that has the smallest Euclidean norm.

We may write

$$\boldsymbol{x}^\dagger = A^\dagger \boldsymbol{b}, \qquad \text{where} \quad A^\dagger = V \Sigma^\dagger U^T \in \mathbb{R}^{n \times m}$$

and $\Sigma^\dagger \in \mathbb{R}^{n \times m}$ is of the form

$$\Sigma^\dagger = \begin{bmatrix} \sigma_1^{-1} & & & & & & \\ & \sigma_2^{-1} & & & & & \\ & & \ddots & & & & \\ & & & \sigma_p^{-1} & & & \\ & & & & 0 & & \\ & & & & & \ddots & \\ & & & & & & 0 \end{bmatrix} \text{ if } m > n,$$

and of the form

$$\Sigma^\dagger = \begin{bmatrix} \sigma_1^{-1} & & & & & & \\ & \sigma_2^{-1} & & & & & \\ & & \ddots & & & & \\ & & & \sigma_p^{-1} & & & \\ & & & & 0 & & \\ & & & & & \ddots & \\ & & & & & & 0 \\ & & & & & & \\ & & & & & & \end{bmatrix} \text{ if } m < n.$$

The 'almost-an-inverse-matrix' $A^\dagger \in \mathbb{R}^{n \times m}$ is called the **Moore–Penrose pseudoinverse** of $A \in \mathbb{R}^{m \times n}$.

THE END

is nigh.